BIMopen user manual



BIMopen
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BIMopen user manual

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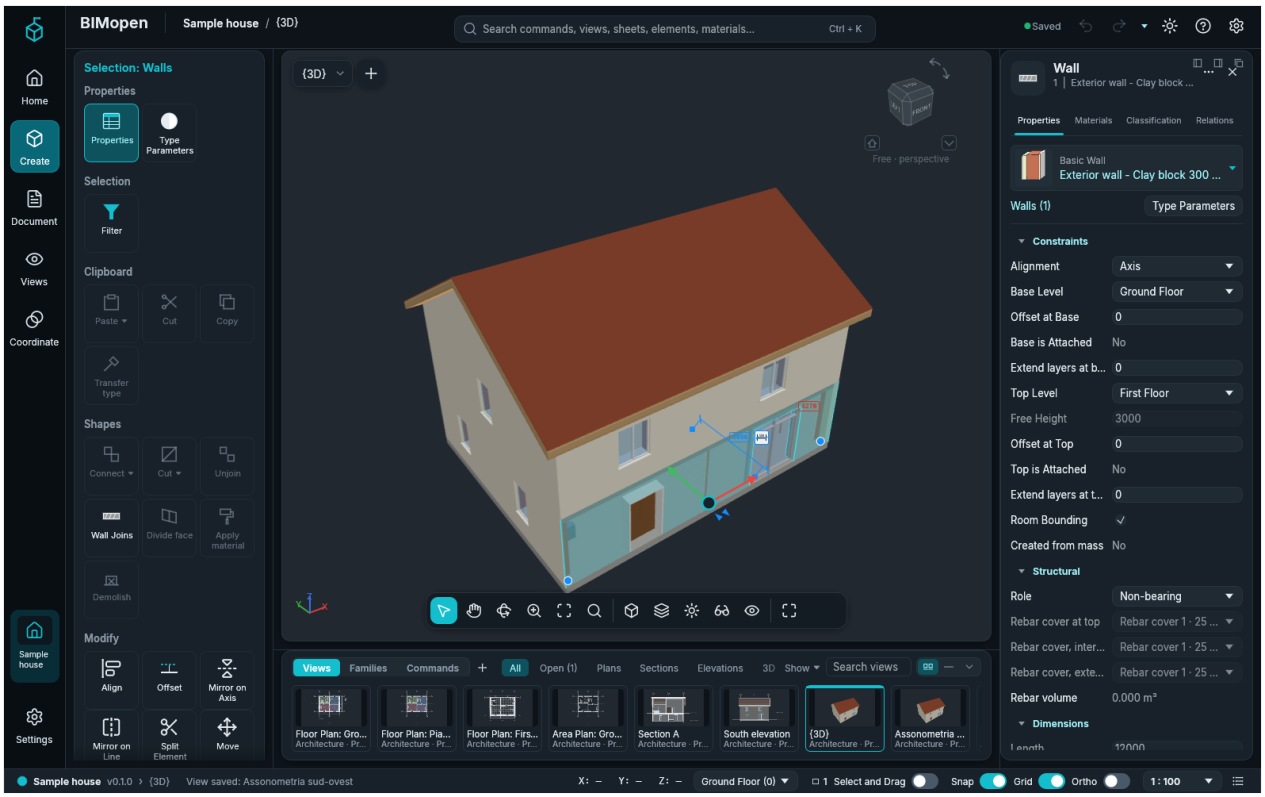
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BIMopen user manual

BIMopen is an open-source application for building information modelling (BIM). You model the building with walls, doors, windows, floors, roofs, stairs and rooms, and the drawings come from the model: plans, sections, elevations, 3D views, schedules and sheets are views of the same model and change together when you edit it.

BIMopen runs on the desktop (Linux, Windows, macOS) and in a web browser, and it can be driven from a command line and by AI agents over MCP. Its interface speaks six languages: Italian, English, German, French, Spanish and Arabic, the last one written from right to left.



How this manual is organised

Chapter	What it covers
Getting started	Installing and starting BIMopen, the sample project, and a first small house from an empty project to a sheet
The interface	The BIMopen style: the rail and the tool panel, the top bar, the command search, the options strip, the element panel, the Project Index, the views strip, the drawing area, the status bar and how to move around in the views
Interface styles	The Classic style, and touch mode on touch monitors, tablets and phones
Projects and files	Opening, saving, exporting and importing projects
Selecting and modifying	Selecting elements and the Modify tools
Walls, doors and windows	The building envelope and its openings
Floors, roofs, stairs and components	Floors, roofs, stairs, railings, curtain walls and placed components
Views and graphics	Plans, sections, elevations, 3D views and how they are drawn
Annotation and 2D details	Dimensions, text, tags, rooms, areas and 2D details
Sheets, schedules and printing	Sheets, schedules, printing and project settings
Command line and MCP	Typing commands, `bimopen-cli` and the MCP server for agents

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Chapter	What it covers
Keyboard shortcuts	Every key and typed shortcut
Troubleshooting	What to do when something does not work as expected

The chapters from *Projects and files* to *Sheets, schedules and printing* are being written area by area: each one is completed when the features it describes have been checked one by one through the interface. Until then they list what they will contain.

Conventions

- The manual describes and pictures the **BIMopen** interface style, the one BIMopen starts in. Interface styles shows the **Classic** style and touch mode, which run the same commands.
- Names of the interface are written in **bold**, as they appear on the screen in English: **Create** on the rail, **Wall** tile, **Properties** tab.
- A path through the interface is written with arrows: **Create** ▸ **Build** ▸ **Wall** means **Create** on the rail, then the **Build** group of the tool panel, then its **Wall** tile.
- Keys are written like `Ctrl+Z`; a shortcut typed at the keyboard, letter by letter, is written in capitals, like `WA`.
- Commands typed at the command line, and their output, are written `like this`. Points are `x,y` in metres.
- Notes, tips and warnings are set apart:

> **Note**
>
> A note adds something worth knowing.

> **Tip**
>
> A tip shows a quicker or better way.

> **Warning**
>
> A warning tells you about something that can lose work or give a wrong result.

What this manual describes

This manual describes BIMopen as it is today, and only what has been tried in the program. When a feature is not described yet, it may still exist: the roadmap *(https://atsanna.github.io/bimopen-site/roadmap.html)* lists what is available and what is planned. The whole manual is also a PDF, one per language: the last entry of the navigation, **PDF**, downloads it.

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Getting started

This chapter shows how to start BIMopen on the desktop and in the browser, what the sample project contains, and how to model a first small house: four walls, a door, a window, a floor, a roof, a 3D view and a sheet with the plan on it. It takes about ten minutes.

Running BIMopen on the desktop

BIMopen is built from its source code with Rust, version 1.90 or later. On Linux the graphical interface also needs the development files of xkbcommon, Wayland and GTK 3.

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev   # Debian, Ubuntu
cargo run --release -p bimopen -- --sample                             # opens the sample house
```

The program is `bimopen`. It accepts a file to open and one option:

Command	What it opens
`bimopen`	A new, empty project
`bimopen --sample`	The sample house
`bimopen house.bimopen`	The project saved in `house.bimopen`
`bimopen --no-splash`	As `bimopen`, without the start screen

You can also open a project by dragging a `.bimopen` file from the file manager into the window.

Start screen

While BIMopen loads, the start screen appears: a window without a frame, centred on the screen, with the name, the version, the loading stage under way with its percentage and a turquoise line growing along the bottom edge. The stages are the real ones (preferences, fonts, graphics card, families, opening the project); the screen stays at least half a second, so it does not flash, and fades out when the main window is ready. It speaks the language chosen in the app, otherwise the system's when it is one of the six, otherwise English.

To leave it out, clear **Options ▸ General ▸ Show the start screen** (BIMopen remembers it), or start `bimopen --no-splash`. On Wayland the compositor decides where the window appears (usually in the middle).

Language and theme

The first time, BIMopen speaks the language of the operating system when it is one of its six languages (Italian, English, German, French, Spanish, Arabic), and English otherwise. To change it, click the gear at the right end of the top bar (**Options**) and choose the **Language** under **User Interface**. The interface changes at once.

In **Options** you also choose the **Theme**, **Light** or **Dark** (the sun or the moon in the top bar switches it too), and whether the background of the views is inverted (**Invert background color**: dark views with light lines). Choosing the dark theme also inverts the background; you can turn it back off. At the first start BIMopen uses the dark theme, with the background inverted. It remembers the language, the theme and the background for the next time.

Running BIMopen in the browser

The web version is the same program, compiled for the browser. Open atsanna.github.io/bimopen-site/app/ *(https://atsanna.github.io/bimopen-site/app/)*: it starts with the sample house. Nothing needs to be installed, and the project stays in your browser: it is not sent anywhere.

The address accepts these options, which can be combined with `&`:

Address	Effect
`app/?lang=de`	Opens in German for this visit (`it`, `en`, `de`, `fr`, `es`, `ar`)
`app/?empty`	Starts with a new, empty project instead of the sample
`app/?webgl`	Draws with WebGL2 even where WebGPU is available

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The browser must support WebGPU or WebGL2. In the browser some things work differently:

- **Open…** cannot show a file dialog: drag a `.bimopen` file into the page to open it;
- **Save** and the exports download the file, with the name of the project.

To build the web version yourself:

```
cargo xtask web                          # builds dist/web
python3 -m http.server 8080 -d dist/web  # then open http://localhost:8080
```

`cargo xtask web` needs the `wasm32-unknown-unknown` target of Rust and the `wasm-bindgen` command of the version written in `Cargo.lock`; with binaryen's `wasm-opt` installed, the result is smaller.

The sample project

The sample house shows most of what BIMopen can do. Open it with **Application menu ▸ Sample project** (the application menu is BIMopen's mark, at the top of the rail), with `bimopen --sample`, or by typing `sample` at the command line.

It contains:

- two levels, each with its floor plan, and five grid lines;
- fourteen walls of four types, with layers, joined at the corners;
- six doors and twelve windows, tagged in the plans;
- two floors, a pitched roof, a stair with its railings and a storefront curtain wall;
- furniture and plumbing fixtures, eight rooms coloured by use, with a legend;
- a section, an elevation, an area plan, a saved 3D view and a free 2D view with a wall detail;
- dimensions, text notes and the schedules of walls, doors, windows, rooms and materials;
- its place in **Cagliari**, where its address is: the sun and the shadows are those of that city.

> Tip
>
> The sample is a good place to try a tool before using it on your own project: a new sample is always one click away, and nothing you change in it is saved unless you save it.

Your first house

The steps below start from an empty project and end with a sheet. Every step names what to click in the **BIMopen** interface style, the one BIMopen starts in; the interface chapter explains each part of the window.

1. Start a new project

1. Click BIMopen's mark at the top of the rail: the **application menu** opens.
2. Click **New ▸ Project…**. The **New project** dialog proposes BIMopen's default template: click **OK**.

The new project has two levels, **Ground Floor** and **First Floor**, a floor plan for each, and the default 3D view. It has no walls yet. Its wall, door, window, floor and roof types are ready to use.

2. Open the ground floor plan

In the views strip under the drawing area, click the thumbnail **Floor Plan: Ground Floor**. The plan opens in the drawing area, and the top bar and the view selector show **Floor Plan: Ground Floor**.

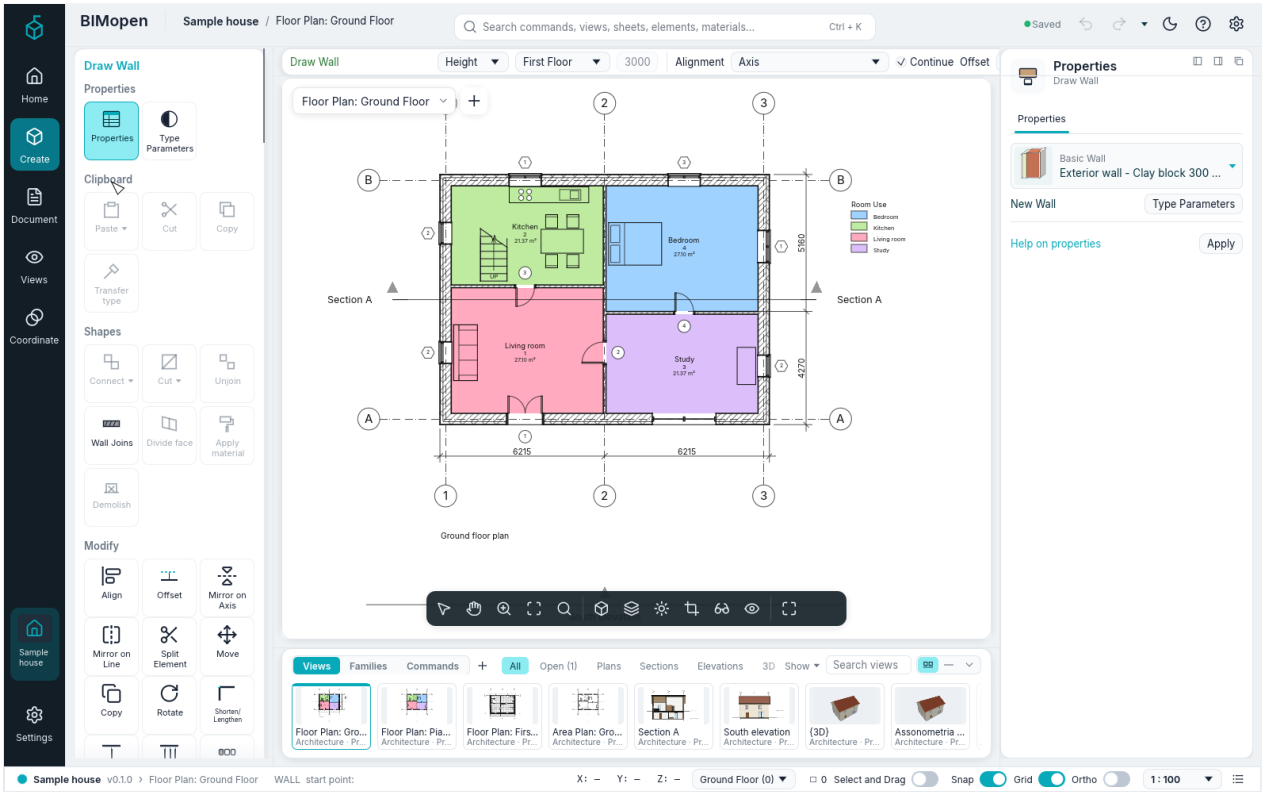
> Note
>
> **Views** on the rail also shows the **Project Index**, the tree of every view: there, a single click on a view shows its properties in the element panel without opening it, and a double-click opens it.

3. Draw four walls

1. In the tools of **Create**, the rail entry BIMopen opens with, click **Wall** in the **Build** group (if the tool panel is folded, click **Create** on the rail first). The contextual card **Draw Wall** comes on top of the tool panel, the options strip over the view shows the settings of the wall, and the element panel shows the type of the next wall, **Exterior wall - Clay block 300 + ETICS**.
2. In the status bar, turn on **Ortho**: the walls stay horizontal or vertical.

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3. Click in the plan where the first corner goes. The status bar asks for the next point: `WALL next point [Esc to finish]:`.



1. Click the second, third and fourth corners. While you move the mouse, the length and the angle of the wall being drawn are shown next to it.
2. Click the first corner again: the four walls are joined at the corners.
3. Press `Esc` to stop the chain, and `Esc` again to end the tool.

> **Tip**
>
> You can draw the same walls by typing their corners in metres: `wall 0,0 10,0 10,6 0,6 c`. The final `c` closes the chain. See Command line and MCP.

4. Add a door and a window

1. Click **Create ▸ Build ▸ Door**, then click on a wall where the door goes. The door is placed in the wall, which is cut around it. The command log confirms it: `Door «Single swing door 80x210» in wall …`.
2. Press `Esc`: the tool ends and the contextual card goes away.
3. Click **Create ▸ Build ▸ Window**, click on another wall, and press `Esc`.

A door or a window can only be placed in a wall: after a click away from the walls, the command line answers `no wall at that point` and nothing is placed.

5. Add a floor and a roof

1. Click **Create ▸ Build ▸ Flooring**. Sketch mode opens: the contextual card **Edit | Create Floor Boundary**, with **Pick Walls** active.
2. Move the cursor over a wall and press `Tab`: the four walls are highlighted. Click: their lines go on the inner faces of the walls.
3. Click **Finish Edit Mode**: the floor is made and the command log gives its area.
4. Click **Create ▸ Build ▸ Roof**. **Pick Walls** is active again; the options strip has **Defines Slope** (ticked) and the **Overhang** (500 mm by default).
5. Press `Tab` over a wall, click, then **Finish Edit Mode**: a hip roof sloping at 30° covers the walls, with the overhang all around.

Floors, roofs, stairs and components tells more about sketch mode.

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6. Look at it in 3D

Click the **{3D}** thumbnail in the views strip. The default 3D view, **{3D}**, opens. Drag with the right mouse button (or with `Shift` and the middle button) to orbit, drag with the middle button to pan, and turn the wheel to zoom; the left button selects, as in plans. `Home` fits the whole model in the view.

7. Put the plan on a sheet

1. Click **Document** on the rail, then **Sheet** in the **Sheets** group. The **New Sheet** dialog lists the title blocks: none, A0, A1, A2 or A3, with their size in millimetres.
2. Click **OK**. The sheet is created with the next free number (**A101** for the first one, then **A102**...) and opens.
3. Click **Document ▸ Sheets ▸ View**. The **Views** dialog lists the views that are not on a sheet yet.
4. Choose **Floor Plan: Ground Floor** and click **Add View to Sheet**. A rectangle of the plan's size follows the cursor: click on the sheet where the plan goes. It is placed at its own scale, 1 : 100, as a viewport, which stays selected.

8. Undo, redo and save

- `Ctrl+Z` undoes the last change (here, the viewport goes away); `Ctrl+Y` redoes it. The arrows of the top bar, **Undo** and **Redo**, do the same.
- `Ctrl+S` saves the project, as a click on the save state in the top bar. A project that has never been saved asks for a file name first; the file has the extension `.bimopen`. In the browser, the file is downloaded.

The top bar says **Unsaved changes** while there are changes to save, and **Saved** after saving.

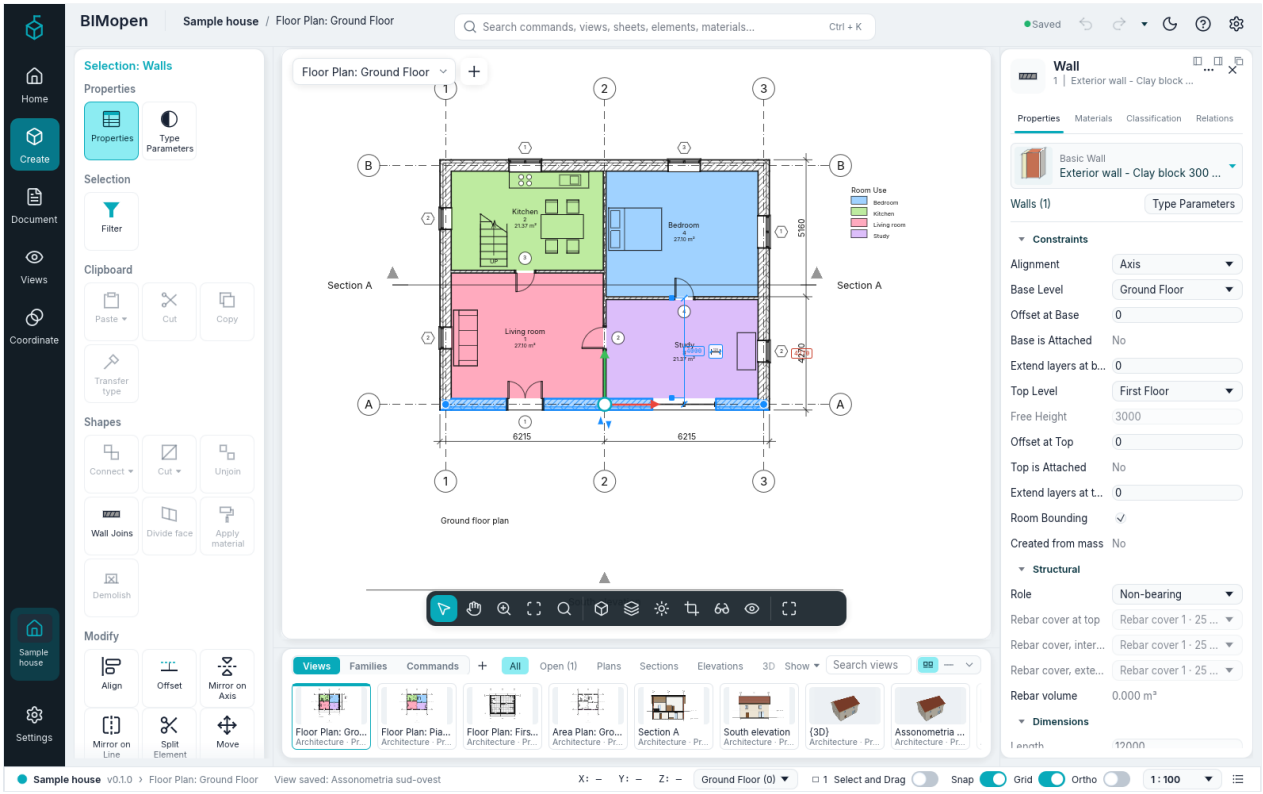
What next

- The interface describes every part of the window.
- Keyboard shortcuts lists the keys and the two-letter shortcuts, such as `WA` for a wall.
- Command line and MCP shows how to type commands, run scripts and let an agent model for you.

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The interface

This chapter describes the **BIMopen** interface style, the one BIMopen starts in, on the desktop and in the browser. The other looks, the **Classic** style and touch mode for tablets and phones, run the same commands with the same names; Interface styles shows them and tells how to switch.



The window has these parts:

1. the **rail**, on the left edge: BIMopen's mark, which opens the application menu, then **Home**, **Create**, **Document**, **Views** and **Coordinate**, and at the bottom the open project and **Settings**;
2. the **tool panel**, beside the rail: the tools of the rail entry chosen, as cards of tiles;
3. the **top bar**: the project and the active view, the command search, which is also the command line, the save state, **Undo** and **Redo**, the quick access tools, the light or dark theme, the **?** menu and **Options**;
4. the **drawing area**, in the middle: the views, each with its view selector at the top and the view bar at the bottom; while a tool works, the **options strip** runs over it;
5. the **element panel**, on the right: the properties of the selection, or of the active view;
6. the **views and families strip**, under the drawing area;
7. the **status bar**, along the bottom.

The panels keep their width whatever they show: a long name is cut with «...» and its tooltip shows it whole. Drag the edge of the tool panel or of the element panel to make it wider or narrower; BIMopen remembers it for the next session. A click on the rail entry already chosen folds the tool panel, and a second click opens it again.

In Arabic the whole window is mirrored: the rail and the tool panel are on the right, the element panel on the left, and the strips read from right to left.

Application menu

BIMopen's mark, at the top of the rail, opens the application menu: the commands that work on the whole project. The **Home** entry of the rail shows the project with its levels, views, sheets and schedules, and reminds you that the file commands are in this menu.

Item	What it does
New	Closes the current project and starts a new one from the default template: two levels with their plans, and the default types

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Item	What it does
Open…	Opens a `.bimopen` project, or imports an `.rvt` file (experimental: only the project name and the level names), chosen in a file dialog. In the browser, drag the file into the page instead
Sample project	Opens the sample house
Save `Ctrl+S`	Saves the project in its file; a project never saved asks for a name, like **Save As…**
Save As…	Saves the project in a new `.bimopen` file and goes on working on that file
Print… `Ctrl+P`	Prints sheets to PDF
Export ▸ Export IFC4…	Writes the model as an IFC4 file
Export ▸ Export plan SVG…	Writes the active plan as an SVG drawing
Export ▸ Export view PNG…	Writes the active view as a PNG image
Open Documents	Lists the open view windows; click one to bring it to the front
Options	Opens the **Options** dialog
Exit	Closes BIMopen (desktop only)

New, **Open…** and **Sample project** replace the current project without asking: save it first if you want to keep it.

> Warning
>
> Unsaved changes are lost when you choose **New**, **Open…** or **Sample project**. The top bar says **Unsaved changes** while there are changes to save.

Options

The gear at the right end of the top bar, or **Application menu ▸ Options**, opens **Options**. It has, among others, these settings:

Setting	Values	Effect
User Interface ▸ Language	Italiano, English, Deutsch, Français, Español, العربية	The language of the whole interface, at once
User Interface ▸ Theme	**Light**, **Dark**	The colours of the window. **Dark** also turns on **Invert background color**. The sun or moon in the top bar switches it too
User Interface ▸ Interface	**BIMopen**, **Classic**	The interface style, see Interface styles
User Interface ▸ Touch mode	**Automatic**, **Always**, **Never**	The layout for touch screens, see Interface styles
Graphics ▸ Invert background color	on, off	Views with a dark background and light lines
Orientation cube ▸ Orientation cube size, **Inactive opacity**	**Small**, **Medium**, **Large**; 0 % to 100 %	The size of the cube, and how much it shows while the pointer is elsewhere
Orientation cube ▸ Keep scene upright	on, off	Dragging the cube never turns the model upside down; off, the view can pass over the top
Orientation cube ▸ Fit to view	on, off	A click on the cube also fits the model in the window

BIMopen remembers the language, the theme and the style the next time it starts.

Top bar and quick access tools

The top bar shows, from the left, **BIMopen**, the name of the project (a click opens **Project information**) and the active view; in the middle the command search; on the right:

Control	What it does
Saved / **Unsaved changes**	The save state; a click saves, `Ctrl+S`

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Control	What it does
Undo	Undoes the last change, `Ctrl+Z`. Greyed when there is nothing to undo
Redo	Redoes the change just undone, `Ctrl+Y`. Greyed when there is nothing to redo
The quick access tools	The tools you added, then the arrow of the quick access toolbar
The sun or the moon	Switches to the light or to the dark theme
?	**Help** `F1`, **Commands** (every command of the command line, as typing `help`) and **About** (the version, the licence and the credits)
The gear	**Options**

Right-click a tile of the tool panel and choose **Add to Quick Access Toolbar**: the tool shows in the top bar and starts from there whatever rail entry is open. Right-click it there for **Remove from Quick Access Toolbar** or **Add Separator** (a line to its right). The small arrow after the tools, **Customize Quick Access Toolbar**, lists the default tools (**Open…**, **Save**, **Aligned dimension**, **Tag by Category**, **Text**, **3D View**, **Section**, **Close Hidden**, **Switch View**): tick one to show it, untick it to remove it. In the same menu, **Customize Quick Access Toolbar…** opens a window with the tools in order: **Move Up** and **Move Down** move the one selected, **Add Separator** puts a line after it, **Remove** takes it (or a line) away, **OK** applies. The tools are kept for the next sessions.

? ▸ **Help** (`F1`) opens this manual, in the language of the interface, in the browser. `F1` opens the page of what you are doing: the open dialog, the tile under the mouse (its tip ends with **Press F1 for more information**) or the active tool.

Rail and tool panel

Each entry of the rail shows its tools in the tool panel, as cards: a card has a title, its groups, and in each group big tiles with an icon and a name. A tile with a small ▾ mark has a menu: a click on the tile starts the command, a click on the mark opens the menu with its variants. Resting the mouse on a tile shows what it does and, in brackets, the shortcut you can type to start it. The commands BIMopen does not run yet are greyed, and their tip says «Not available yet»; **Level**, **Grid** and **Guide plane** are greyed in a 3D view («Not available in the 3D view»).

Rail	Card	Group	Commands
Create	**Modelling** ▸ **Architecture**	**Build**	**Wall** ▾, **Door**, **Window**, **Object** ▾, *Column* ▾, **Roof** ▾, *Ceiling*, **Flooring** ▾, **Curtain Wall**, **Curtain grid**, **Mullion**
		Vertical Links	**Railing** ▾, **Ramp**, **Stair**
		3D elements	*3D text*, *3D line*, *Group* ▾
		Rooms and Areas	**Room**, **Room Separator**, **Room Label** ▾, **Area** ▾, **Area Outline**, **Area Label** ▾
		Openings	*By face*, *Shaft*, *In a wall*, *Vertical opening*, *Dormer*
		References	**Level**, **Grid**
		Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
	Modelling ▸ **Structure**	**Structural elements**	*Beam*, *Structural wall* ▾, *Structural column*, *Structural floor* ▾, *Truss*, *Brace*, *Beam layout*
		Connections	*Connection* ▾, *Automatic joints*
		Foundations	*Pad footing*, *Strip footing*, *Raft foundation* ▾
		3D elements	**Object** ▾, *3D text*, *3D line*, *Group* ▾
		Openings	*Normal to a face*, *Shaft*, *In a wall*, *Vertical opening*, *Dormer*
		References	**Level**, **Grid**
		Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
	Modelling ▸ **MEP**	**Air**	*Duct*, *Duct from a path*, *Duct fitting*, *Duct accessory*, *Duct insulation*, *Flexible duct*, *Air outlet*
		Prefabricated parts	*Prefabricated part*, *Multi-point route*
		System details	*Duct bracing*
		P&ID diagrams	*P&ID editor*

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Rail	Card	Group	Commands
		Mechanical	*Mechanical equipment*, *Control device*
		Other systems	*Plumbing* ▾, *Electrical* ▾, *Modelling* ▾, **Work Plane** ▾
	Reinforced concrete	**Reinforcement**	*Rebar*, *Reinforcement along a path*, *Adaptive rebar*, *Reinforcement over an area*, *Mesh over an area*, *Single mesh*, *Concrete cover*, *Coupler*
		Settings	*Cover options*, *Rebar options*
		Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
	Metal	**Connections**	*Connection* ▾, *Automatic joints*
		Steelwork parts	*Plate*, *Bolts* ▾, *Weld*
		Fabrication	*Cope*, *Skewed cut*, *Shorten*, *Shaped cut*
		Automatic cuts	*End cut*, *Corner joint* ▾, *Intersection*, *Cut with*
		Work Plane	**Define**, **Display**, **Guide plane**, *Plane viewer*
	Prefabrication	**Division**	*Divide*
		Joints	*Assembly parts*
		Reinforcement	*Rebar*, *Custom mesh*
		Production	*Workshop drawings*, *Export for CAM*
		Settings	*Prefabrication settings*, *Cold-formed profiles*
		Automatic update	*Suspend* ▾
	Link and import	**Link**	*Link project*, *Link IFC*, *Link CAD*, *Link topography*, *Decal* ▾, *Point cloud*, *Manage links*, *Link PDF*, *Link image*, *Federated model* ▾
		Import	*Import CAD*, *Import gbXML*, *Import PDF*, *Import image*
		Load from library	**Load Family**, *Load as group*, **Insert from File** ▾
	Massing & site	**Massing**	*Display masses* ▾, *New mass*, *Place a mass*, *Draw in 3D*
		From mass faces	*Curtain Wall*, *Roof*, *Wall*, *Slab*, *Terrain from face*
		Site	*Terrain*, *Site element*, *Parking*, *Building pad*, *Surrounding context*
		Edit site	*Divide terrain*, *Merge terrains*, *Sub-area*, *Property line*, *Grading*, *Contour labels*
Document	**Document**	**Sheets**	**Sheet**, **View**, **Title Block**, **Revisions**, **Guide Grid**, **Deactivate View**
		Dimension	**Aligned**, **Linear**, **Angular**, **Radial**, *Diameter*, **Arc length**, **Spot dimension** ▾
		Detail	**Detail line**, **Filled region** ▾, **Detail symbol** ▾, **Revision Cloud**, *2D group* ▾
		Text	**Text**, *Spell checker*
		Tag	**Tag**, **Tag all**, *Material*, **Color legend**, *Beam tags*, *Multi-category tag*, *View reference*, *Tread numbers*, *Rebar tag* ▾, *Coded note* ▾, *Duct legend*, *Pipe legend*
		Symbol	**Symbol**, *Span direction*, *Rebar symbol*, *Stair path*
Views	**Views**	**Graphics**	**View presets** ▾, **Visibility and Graphics**, **Filters**, **Thin lines**, *Show hidden lines*, *Remove hidden lines*, *Cut outline* ▾
		Presentation	*Realistic image*
		Create	**3D View** ▾, **Section**, **Enlarged detail** ▾, **Plan** ▾, **Elevation** ▾, **Free 2D View**, **Duplicate View** ▾, **Legends** ▾, **Schedules** ▾, *Extent box*
		Windows	**Switch View** ▾, **Close others**, **Tile**, **Cascade**, **Replicate**, **Maximize**, **Panels** ▾
	Project Index		The tree of the project's views, schedules, sheets and families
Coordinate	**Coordinate**	**Energy**	*Energy analysis*
		Check	*Check clashes*, *Show loads*, *Disconnected connections*, *Electrical loads*, *Escape route*

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Rail	Card	Group	Commands
		Analytical model	*Analytical member*, *Analytical panel*, *Analytical node*, *Supports and restraints*, *Automatic analytical model*, *Analysis results*
		Spaces and zones	*Space*, *Space separator*, *Space tag*, *Space names*, *Drawn zone*, *Zone from spaces*
		Shared work	*Worksets*, *Synchronize*, *Show history*, *Enable sharing*, *Active workset*, *Dim other worksets*, *Reload changes*, *Release elements*, *Permission requests*, *Sharing status*
		Archive	*Restore backup*, *Publishing options*
		Coordination	*Copy and monitor* ▾, *Coordination review* ▾, *Coordination options*, *Realign hosts*, *Federated model changes*
Settings	**Settings**	**Settings**	**Materials**, **Styles by Category**, **Snap**, **Project information**, **Project parameters**, **Project units**, **Copy Settings from Project**, *General parameters*
		Phases	*Phases*
		Options	*Design options*, *Main version*
		Numbering	*Renumber*
		Position	*Shared coordinates* ▾, *Move project* ▾
		Project management	*Project images*, *Opening view*
		Selections	*Save selection*, *Load selection*, *Edit selection*
		Inquiry	*Select by number*, *Selection numbers*, *Warnings*

In italics the commands BIMopen does not run yet; ▾ marks a menu.

Press and release `Alt`, or press `F10`, to show the key tips: a letter on each rail entry and, on the tiles of the open card, their tool's shortcut. Type an entry's letter to open it, then a tile's letters to start its tool; `Esc` hides the tips. While they show, what you type does not go to the command line.

The Create tools

The tools you use most to model are in **Create ▸ Modelling ▸ Architecture**:

Group	Tiles
Build	**Wall** `WA`, **Door** `DR`, **Window** `WN`, **Roof** `RF`, **Flooring** `SB`, **Object** `CM`, **Curtain Grid** `CG`, **Mullion** `MU`
Vertical Links	**Railing** `RA`, **Stair** `ST`
Rooms and Areas	**Room** `RM`, **Room Separator** `RS`, **Area** ▾ (**Area Plan**, **Area Outline**, **Area**), **Legend** `LG`, **Settings** ▾ (**Area and Volume Computations**, **Color Schemes**)
References	**Level**: drawn in an elevation or a section, two clicks at the new level's height (from a plan it opens an elevation); **Grid** `GR`

Area Outline and **Area** work in an area plan: until one is open they are greyed, and their tip says to open an area plan first.

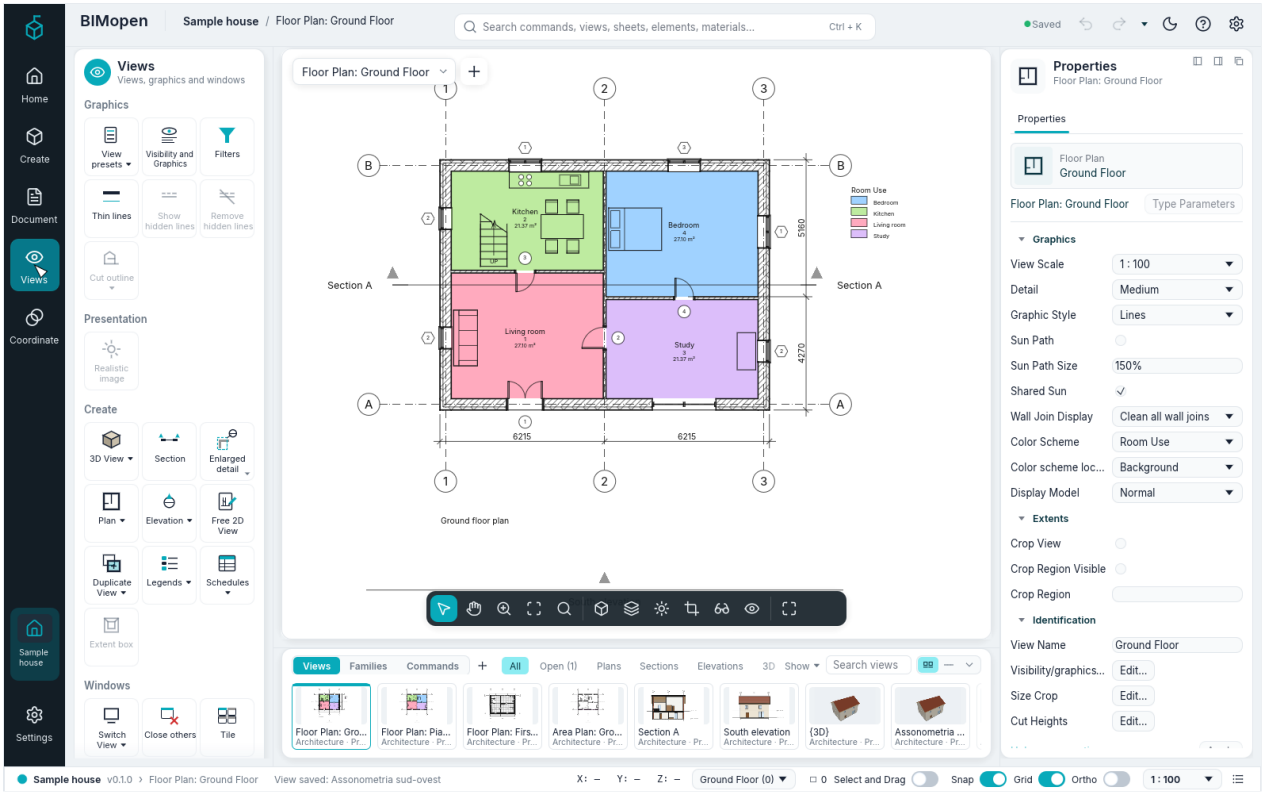
The Document tools

Group	Tiles
Sheets	**Sheet**: a new sheet with the title block you choose; **View**: places a view on the open sheet
Dimension	**Aligned** `DI`, **Linear** `DL`, **Angular** `DA`
Text	**Text** `TX`
Detail	**Detail line**, **Filled region**, **Masking region**, **Detail component** `CD`, **Repeating detail** `RD`, **Insulation** `IS`
Tag	**Tag All** `TG`

Sheets ▸ View works only while a sheet is open.

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The Views tools



Group	Tiles
Create	**3D View** ▾ (**Default 3D View**, **Save View**), **Plan** ▾ (**Floor Plan**, **Area Plan**), **Duplicate View** ▾ (**Duplicate**, **Duplicate with Detailing**), **Section** `SE`, **Elevation** ▾ (**Elevation** `EL`, **Framing Elevation**), **Enlarged detail** `DT` (a callout), **Free 2D View** `VD`, **Schedules** ▾ (**Schedule/Quantities**, **Material Takeoff**, **Sheet List**, **Note Block**, **Predefined schedules**)
Graphics	**Visibility and Graphics** `VG`, **Filters**, **View templates**, **Apply template to current view**, **Create template from current view**
Windows	**Switch View**, **Replicate**, **Cascade**, **Tile**, **Maximize**, **Close others**, **Panels**

Save View works only while a 3D view is open, and **Duplicate with Detailing** only for views that are not 3D. Under the tools, the **Project Index** card holds the whole tree of the project.

Every ▾ menu also lists, greyed, the entries not available yet; the command search finds them all.

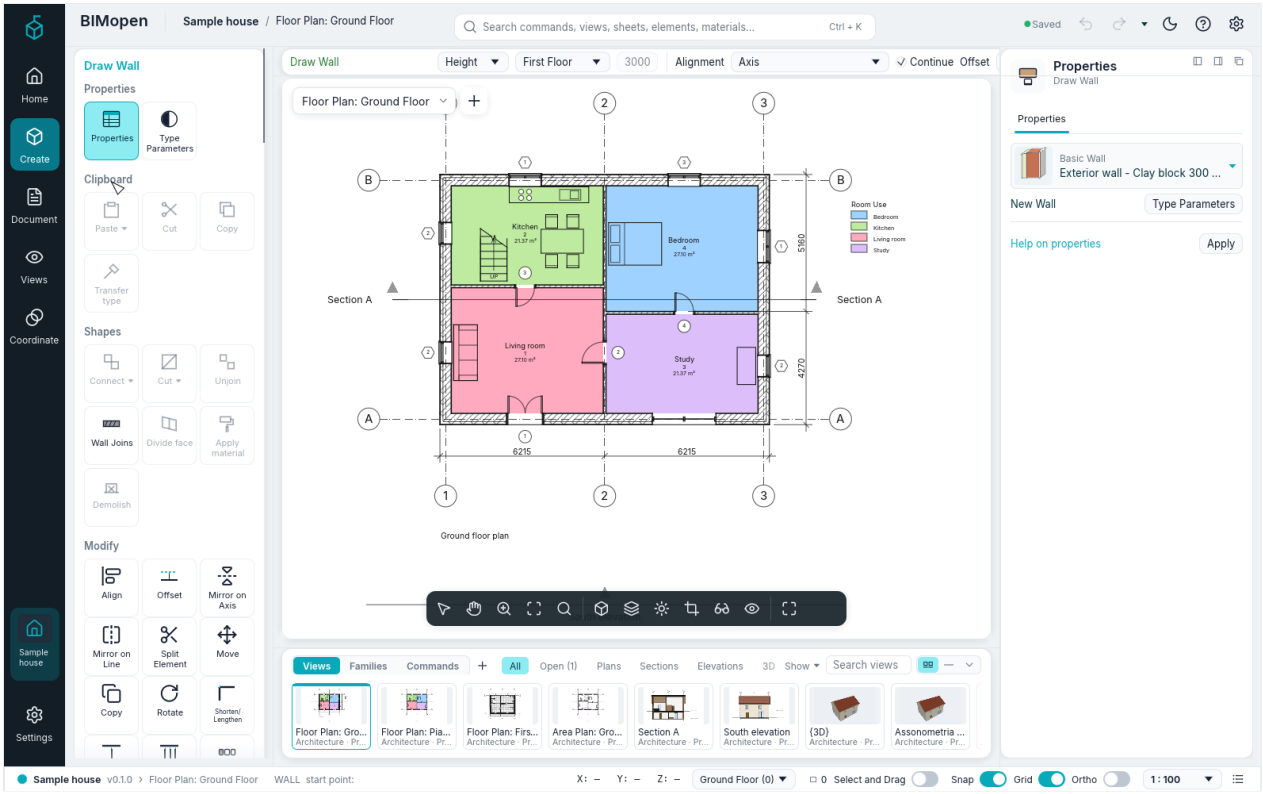
The Settings tools

Group	Tiles
Settings	**Materials**, **Project information**, **Styles by Category**, **Line styles**, **Purge unused**, **Project units** `UN`, **Snap**, and **Additional settings** (**Line styles**, **Fill patterns**)
Inquiry	**Check**: checks the openings and references of the model and reports at the command line

Snap turns snapping on and off for the whole program: it is highlighted while on, as the **Snap** switch of the status bar. **Additional settings** ▸ **Snaps...** opens the **Snap** window: the length and angular increments, the object snaps and the overrides (see Snap).

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The contextual card



When you start a tool or select something, a card comes on top of the tool panel, whatever rail entry is open, with the name of what you are doing in the accent colour: **Draw Wall** while drawing walls, **Selection: Walls** with walls selected, **Multi-Select** with elements of different categories, **Edit | ...** in the other modes. When the tool ends, or the selection is cleared, the card goes away; `Esc` or a click on empty space does the same.

It always has these groups:

Group	Tiles
Properties	**Properties**: shows or hides the element panel; **Type Parameters**: the properties of the type of the selection, or of the next element the tool will place
Modify	**Align** `AL`, **Offset** `OF`, **Mirror on Axis** `MM`, **Mirror on Line** `DM`, **Split Element** `SL`, **Split with Joint** `SG`, **Move** `MV`, **Copy** `CO`, **Rotate** `RO`, **Shorten/Lengthen** `TR`, **Shorten/Lengthen one element**, **Shorten/Lengthen several elements**, **Repeat in Series** `AR`, **Scale** `RE`, **Fix** `PN`, **Unpin** `UP`, **Delete** (`Del`)

Delete works only when something is selected. The selecting and modifying chapter describes the Modify tools.

Other groups appear for what is selected or for the tool:

When	Group and tiles
Anything selected	**Selection**: **Filter** (see Selecting and modifying)
Doors or windows selected	**Opening**: **Flip facing**, **Flip hand**
Walls selected	**Modify Wall**: **Link Top/Base** (the walls rise to the roof above them), **Unlink Top/Base**
A level selected	**View**: **Open plan**
Sections or elevations selected	**Section**: **Split Segment** (sections only), **Flip**, **Go to View**
A callout selected	**Detail**: **Go to View**
Curtain walls selected	**Curtain Wall**: **Curtain Grid**, **Mullion**, **Panel**
Flooring or **Roof** tool	**Draw**: **Boundary Line**, **Pick Walls**
Stair or **Railing** tool	**Mode**: **Finish Edit Mode**; **Draw**: **Run** or **Lines**
Mullion tool	**Placement**: **Grid Line**, **Grid Line Segment**, **All Grid Lines**
A detail tool	The drawing shapes of the tool

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Options strip

While a tool works, a strip runs over the top of the view: the name of what you are doing, then the settings of the tool. They apply to the next elements you place.

Tool	Settings on the options strip
Wall	**Height** or **Depth**; the level the wall reaches, or **Unconnected** with its value; **Alignment**; **Continue**; **Offset**; **Radius** (greyed: curved walls are not available yet); **Join at the ends** and **Automatic joining** (greyed); **Exterior on the left** or **Exterior on the right** (the space bar swaps them)
Door, **Window**	**Load Family**; **Model on site** (greyed); **Tag on Placement** with **Horizontal** or **Vertical**, **Leader** and its length; **Loaded tags…** (greyed)
Object	**Load Family**; **Model on site** (greyed); **Rotate after inserting**: the object placed stays selected and **Rotate** starts
Roof	The shape: **Gable**, **Hip** or **Flat**; **Slope**, from 5° to 60°; **Overhang**, up to 2 m
Stair	**Width**, in millimetres; **Railing Type**, or **None**
Dimensions	What the cursor picks first on a wall: **Wall centerlines** or **Wall faces**
Enlarged detail (callout)	The scale of the new detail view, or **Reference other view** and the detail or free 2D view the callout points at (always on in a free 2D view)
Area Outline	**Line** or **Pick lines**; with **Pick lines**, **Apply area rules** or an **Offset**

The **Ortho** and **Snap** switches of the status bar work for every tool that draws lines: **Ortho** keeps the segments horizontal or vertical, **Snap** catches endpoints, midpoints and intersections. With **Ortho** off, holding `Shift` while drawing gives the same constraint; with it on, `Shift` releases it.

Element panel

The element panel, on the right, shows and changes the parameters of what you are working on. Its header has the category and the name of what it shows, **Close** (which clears the selection) and **More actions** (⋯: **Type Parameters** and the actions of the contextual card). Under the header, its tabs:

Tab	What it holds
Properties	The type selector, then the parameters, in groups (**Constraints**, **Graphics**, **Identification**…). Click the name of a group to fold it. Values in grey are read-only: they are computed from the model. **Apply**, or `Enter`, confirms a value you are typing
Materials	For walls, floors and roofs: the layers of the type with their colours and thicknesses, and a preview of the layers
Classification	The category, the type and the IFC class
Relations	The level, the host, the doors and windows it hosts, the views and the schedules it appears in

The **type selector** shows the family in small letters, the type underneath, and a preview of the type: walls, floors and roofs as their layers, doors, windows and components as a small picture. Click it to choose another type: the selected elements change type, or, while a tool is active, the next element gets that type. Open it to search the types by name; they are grouped by family, each with its preview. The arrow keys move, `Enter` picks, `Esc` closes the list.

What the panel shows depends on the situation:

Situation	The panel shows
Nothing selected, no tool	The properties of the active view (scale, detail, graphic style…), or of the view clicked in the Project Index
A tool active	The type of the next element, as **New Wall**, **New Door**…
One element selected	Its type and its instance parameters
Several elements of one category	Their number, as **Walls (3)**
Elements of different categories	**Multi-Select**, **Common (5)**, and how many of each category, as **3 × Walls**

Lengths are written and read in the project's units (millimetres in a new project). You can type a value with its unit, for example `2.7m`.

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Project Index

Views on the rail shows, under its tools, the **Project Index**: the views, schedules, sheets and families of the project in a tree. Click ⊟ or ⊞ to fold or unfold a branch.

Branch	What it contains
Views (all) ▸ Floor Plans	The floor plans of every level, from the lowest, and the callouts drawn in plans
Views (all) ▸ Area Plans (...)	The area plans, one branch per area scheme
Views (all) ▸ Sections , Elevations	The sections and the elevations
Views (all) ▸ Detail Views	The callouts drawn in sections
Views (all) ▸ 3D Views	**{3D}**, the default 3D view, and the saved 3D views
Views (all) ▸ Free 2D Views	The free 2D views
Schedules/Quantities	The project's schedules, then the predefined schedules
Sheets (all)	The sheets, as number - name
Families	The types of the project, by category: walls, doors, windows, floors, roofs, components and the others

The branches of area plans, sections, elevations, detail views and free 2D views appear only when they have something in them. The active view is written in bold.

With a view:

- a **click** shows its properties in the element panel, without opening it;
- a **double-click** opens it;
- a **right-click** opens its menu: **Open**, **Close** (closes its windows), **Duplicate View ▸** (**Duplicate**, **Duplicate with Detailing**), **Apply view template...**, **Create view template from view...**, **Delete**, **Rename...**, **Properties**.

The menu of a sheet has **Apply view template to all views...** instead of the two template items, and no **Duplicate View**. **{3D}** cannot be deleted or renamed.

With a schedule of the project, a click selects it, a double-click opens it, and the right-click menu has **Open**, **Schedule Properties...** and **Delete**. A predefined schedule opens with a click.

With a type, under **Families**, a double-click opens its type parameters; the right-click menu has **Type Parameters**, **Duplicate...**, **Create Instance** (starts the tool that places it, with that type) and **Delete**.

After a click in the Project Index, the keyboard moves in it: `Up` and `Down` go from row to row (the view's properties show in the element panel), `Right` opens a branch, `Left` closes it or goes up to the branch the row is in, `Enter` opens the view.

Views go on sheets from here too. Drag a plan, section, elevation, free 2D view, legend, saved 3D view or schedule onto the open sheet: its viewport follows the cursor and stays where you drop it. Or drop it on a sheet's name in the tree: the sheet opens and the viewport follows the cursor until you click. While you drag, the tree scrolls when the cursor nears its edge. The right-click menu of a sheet has **Add View...**. The views on a sheet are listed under it; the right-click menu of one has **Remove from Sheet**, and the menu of **Sheets (all)** has **New Sheet...**. In a 3D view the right-click menu also has **Zoom to 3D Extents**: the whole model, the elements hidden in the view too. A view already on a sheet is refused, with the reason on the command line; legends and schedules can go on several sheets. A saved 3D view is drawn on the sheet with the edges seen from its camera.

The right-click menu of a plan, section or free 2D view has **Find Referring Views**: the views that show its symbol and the sheets it is on; **Go to View** opens the one chosen.

Views and families strip

The strip under the drawing area has three modes, chosen with the buttons on its left:

- **Views**: live thumbnails of the views, grouped by **All**, **Open (n)**, **Plans**, **Sections**, **Elevations**, **3D**, **Legends**, **Schedules** and **Sheets**. The current view is framed, open views have an accent line. Click a thumbnail to open the view; right-click for the view's menu, with **Open in the active window**; drag it onto the open sheet, or onto a sheet's thumbnail, to place it there. **+** makes a new view: **Default 3D View**, **Save View**, **Duplicate**, **Sheet** or **Schedule/Quantities**.
- **Families**: one preview for each type, grouped by family category. Click a preview to place that type; drag it into a plan to insert it where you let go. A door or a window dropped near a wall goes into it, along it.

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• **Commands**: the history of the command line. Click a line you typed to run it again.

On the right, **Show** filters the thumbnails by discipline and phase, the search box by name, and three buttons set the size: **Previews**, **Compact** and **Hide**.

Panels

The element panel can move. Drag it by its header: it docks on the left or on the right of the drawing area, or floats where you let it go. The buttons at the top right of the panel dock it left, dock it right or make it float; **Views ▸ Windows ▸ Panels** shows or hides it and the other panels. Each panel stays where you put it in the next sessions.

While you drag a panel, only the places where it really docks are drawn, each as the panel it will become. The place under the pointer is drawn strongly, the others faintly. Elsewhere nothing is drawn, and the panel floats where you let it go.

Dialogs

Every dialog opens in the middle of the screen, each time it opens, even if it was moved the last time or the window has changed size. Once open, it stays where it is: drag it by its title bar to move it. A change inside it, such as the **Preview** switch, does not move it.

Every dialog has a fixed size, which does not change with its content. The exception is the dialogs built around a long list or table (materials, visibility/graphics, object styles, filters, line styles and patterns, fill patterns, line weights, schedule properties, project and shared parameters, keyboard shortcuts, purge unused, IFC export classes, view templates): dragging one of their edges makes them larger, never smaller than they first open nor larger than the screen, and the list grows with them. They open again in the middle with the last size given.

Drawing area and view windows

Every view you open is a window in the drawing area. A view opened again comes to the front; at least one window always stays open. At the top left of the view, the **view selector** shows its name: its menu lists the views and arranges the windows, and **+** beside it makes a new view. **Views ▸ Windows** arranges them too:

Tile	What it does
Switch View	Lists the open windows; the active one is ticked
Replicate	Opens a second window on the active view, with its own zoom
Cascade	Overlapping windows, each with its title bar: drag a title to move a window, its edges to resize it
Tile	All the windows side by side, each with a title bar; the active one has a coloured title and frame
Maximize	The active window fills the drawing area; the others wait underneath
Close others	Closes every window except the active one
Panels	Shows or hides the **Properties**, the **Project Index**, the **Command line**, the **Navigation bar**, **Keyboard Shortcuts** and **Recent Files**

Click in a window to make it active: the tools work in the active window, the others only show the model. In **Tile** and **Cascade**, the **×** on a window's title bar closes it. `Ctrl+Tab` goes to the window you used before.

Tile puts the active window at the top left and the others after it, from the most recently active; **Switch View** lists them in the same order. Tile again after working in another window and that window comes first.

Options ▸ User Interface ▸ Keyboard Shortcuts lists the commands with their shortcuts. Search a command by name, or **Filter** the list (**All**, **All Defined**, **All Not Defined**, **All Reserved**); select a command, type the keys in **Press new keys** and click **Assign**. The same keys may go to several commands: typed in the drawing area, the status bar shows the first, the arrow keys go through the others and `Enter` or `Space` runs the one shown; **Export…** writes every shortcut to an XML file and **Import…** reads such a file back as your shortcuts (a file naming no command here is refused, and your shortcuts stay); **Remove** takes away the keys you added. Your shortcuts are kept for the next sessions and start the command when typed, as the built-in ones do.

A tool never changes the active view by itself. Started in a view where it cannot work (a roof or a room in a 3D view, a level in a plan, an area outside an area plan), it opens **Go to View**: the views that suit it, with the plan of the active level chosen first. **Open View** opens the chosen view and starts the tool there; **Cancel** leaves everything as it was. **Edit Boundary** of a floor, roof, stair or railing, from a section or an elevation, asks the same way for a plan of its level, and a new section or elevation leaves

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the plan you drew it in active. **Move**, **Copy**, a linear **Repeat in Series** and **Mirror on Line** also work in sections and elevations: along the view; the axis is the vertical plane at right angles to the view through its first point, however it is drawn. A radial **Repeat in Series** works in plans and 3D views: in a section or an elevation it stays chosen, but its clicks make nothing and the status bar asks for **Linear**.

Views in separate windows

A view can leave the drawing area: **Separate window** in its menu (right-click it in the **Project Index**, on its thumbnail in the views strip, or on the title bar of a tiled window), or drag the title bar of a tiled or cascaded window out of the drawing area. On the computer the view opens in a system window of its own, which you can move to another monitor too, resize and maximise; on the web it floats in the page, because the browser opens no other windows. A view floating in the program also has **System window**, to move it into a system window.

You can separate several views at once. Each keeps its tools, its view bar and, in 3D, its orientation cube; the selection shows in all of them, and the active view is the last one you clicked. **Back to the main window**, at the top of the view, puts it back in the drawing area; a floating view also goes back when dragged onto the band at the top of the drawing area, which lights up. Closing the system window brings it back the same way. Where each separate view was stays with the project, and when it opens again the separate views come back to their places.

Drawing in every view

A wall is drawn the same way in a plan and in a 3D view, on the level's plane: from the first click the wall being drawn follows the cursor, in the plan as its footprint and in 3D as its see-through solid, at its height and thickness; a dashed line carries its axis on, and others hold it in line with the points of the chain already clicked. The temporary dimension, blue with its witness lines and ticks, gives the length and the angle; type a length and press `Enter` to draw the wall that long. The snaps (endpoints, midpoints, intersections, perpendicular, other walls' faces and axes) work in 3D too. The wall selected in a 3D view shows, as in a plan, its temporary dimensions, the flip arrows and the ends to drag. In sections and elevations a wall is not drawn: **Go to View** offers a plan or the 3D view.

Roofs and floors are drawn in the 3D view too, on the plane of their level: **Pick Walls** takes the wall under the cursor (its top lights up in orange, `Tab` takes the whole chain), the lines are drawn with every sketch tool and with temporary dimensions, and **Edit Footprint** of a roof or a floor opens its sketch in 3D. **Edit Boundary** of a wall works in 3D on the wall's plane, and in a section or elevation that sees it face on.

While you draw or edit a sketch you can switch views (thumbnails, **Project Index**, separate windows): the tool stays active and goes on in the new view if it can work there. A chain of walls begun in a plan goes on in 3D from its last point, and a profile sketch begun in an elevation goes on in 3D. The wall being drawn and the sketch show in every open view at once.

View bar

At the bottom centre of every view, a dark bar of buttons:

Button	What it does
Select and modify	Ends the navigation mode, back to selecting, `Esc`
Pan, **Orbit** (3D), **Zoom**	With one on, the left button drags the view; `Esc` ends it
Zoom to Fit	Fits the whole view in the window, `Home`
Zoom Options	The zoom menu, see below
The graphic style	**Wireframe**, **Lines**, **Shaded**, **Shaded with Edges**, **Flat Colors**
The detail	**Coarse**, **Medium** or **Fine**
Sun and shadows	In 3D views, the sun and its shadows
Crop	**Crop View** crops the view to its crop region, or stops cropping; **Show Crop Region** shows or hides its frame
Isolate Temporarily	Hides or isolates elements or categories in this view until you reset it; framed while active
Reveal Hidden	Shows the hidden elements, to show them again
Full screen	The view alone in the window; **Exit full screen** comes back

The scale of the view is on the right of the status bar, for example **1 : 100**: it opens the list of scales; under **Custom** type another one, as `1 : 75`, and press `Enter`. A sheet has no scale of its own: its views have theirs.

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Zoom

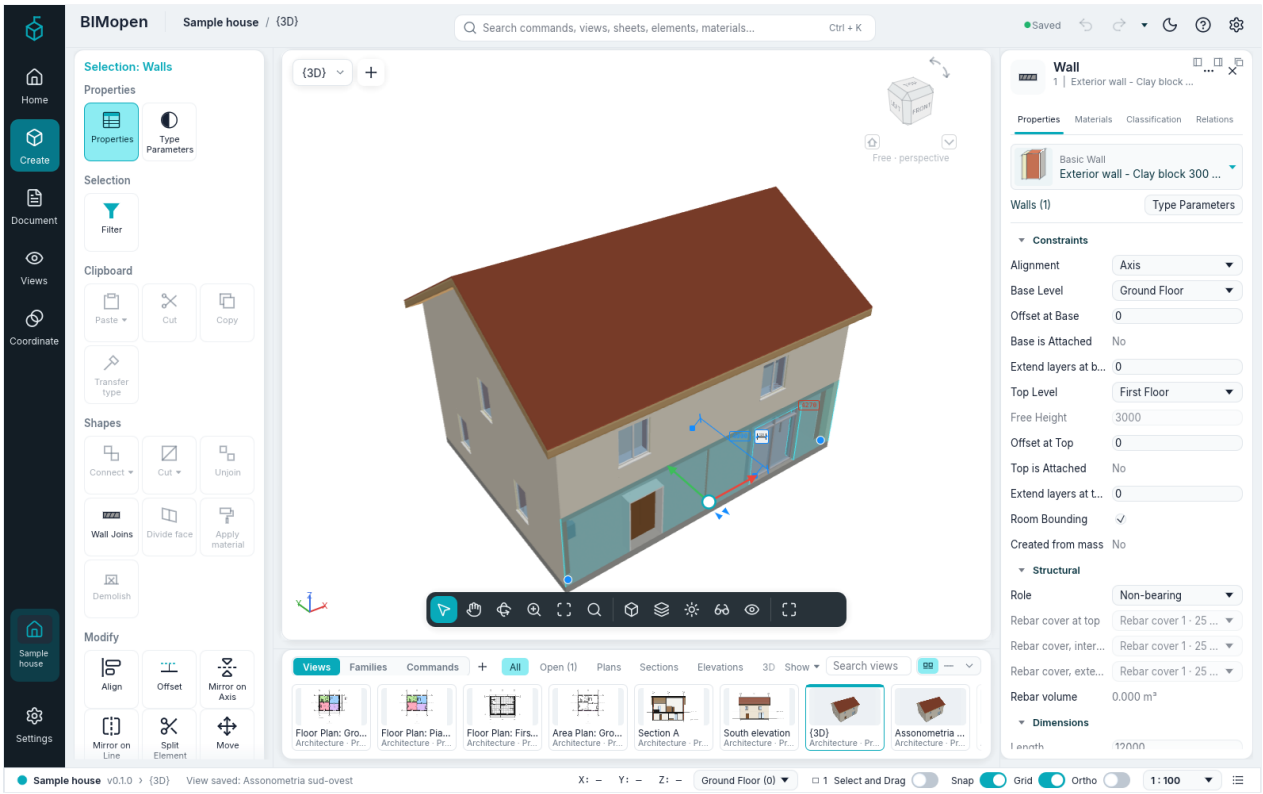
Zoom Options, on the view bar, opens the zoom menu: **Zoom In**, **Zoom Out**, **Zoom in Region** (drag a rectangle on the view; `Esc` gives up), **Zoom Out (2x)**, **Zoom to Fit**, **Zoom All to Fit** (every open window), **Zoom Sheet Size** (the view at the size it prints), and **Previous Pan/Zoom** and **Next Pan/Zoom**, which go back and forth through the views of that window. Typed: `ZR`, `ZO`, `ZF`, `ZA`, `ZS`, `ZP`.

The navigation wheel is not available.

Moving around in the views

In plans, sections, elevations and sheets	
Mouse wheel	Zooms in and out at the cursor
Drag with the middle button	Pans
Drag with the left button	A selection window, or moves the selected elements (Selecting and modifying)
`Home`	Fits the whole view in the window

In 3D views	
Drag with the right button, or `Shift` + middle button	Orbits around the model
Drag with the middle button	Pans
Drag with the left button	A selection window, or moves the selected elements
Mouse wheel	Zooms in and out
`Home`	Fits the whole model in the window



A small tripod in the lower corner of the 3D view shows the directions of the axes: X red, Y green, Z blue. A saved 3D view remembers where you left it. On the selection, the move arrows let you drag it: drag an arrow to move along it, the centre to move freely in plan; the distance shows as you drag, and the release moves as **Move** does.

In 3D, the wheel zooms keeping the model point under the cursor in place, and the orbit turns about the centre of the selection, or about the point the view looks at when nothing is selected.

The **Orientation cube**, in the top corner of the 3D view, turns with the model. Its edges and corners are bevelled and every part is a target of its own: the six faces (**TOP**, **BOTTOM**, **FRONT**, **BACK**, **RIGHT**, **LEFT**, their names written on them in perspective and turning with them) look straight at that side; the twelve edge strips look from between two sides; the eight

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corner triangles from between three (an isometric view). The part under the pointer is tinted with the highlight colour, the one the view looks from now stays highlighted; a click turns the view there in a brief transition. Drag the cube to orbit.

Under the cube the view's orientation is written: **Front · perspective**, **Top – Front · perspective**, **Isometric · perspective**, or **Free · perspective** when the view looks from no part of the cube.

When a face is seen straight on, four triangles round the cube (**Turn up**, **Turn down**, **Turn left**, **Turn right**) turn the view a quarter to the face beyond that side. The two curved arrows at the top right, **Roll left** and **Roll right**, turn the picture a quarter turn counterclockwise or clockwise.

The small house button at the bottom left, **Starting view**, goes back to the starting view; right-click it for **Set as starting view**, which makes the present view the starting one (each 3D view keeps its own, with the project). The small button at the bottom right, **Cube menu**, or a right-click on the cube, opens the menu: **Orthographic** / **Perspective** (for now the view is perspective only: **Orthographic** is greyed, «Not available yet»); **Set as starting view**; **Zoom to Fit**, which fits the model in the window; **Options...**, which opens the **Orientation cube** page of **Options**. Right to left (Arabic) the cube sits at the top left and the two small buttons change sides; the arrows stay where they point.

The context menu of a view

Right-click in a view while no tool is active. If the cursor is on an element that is not selected, the element is selected first. The menu has, with a selection, the modify commands first (**Move**, **Copy**, **Rotate**, **Align**, **Offset** and the others), then:

- **Cancel**: clears the selection;
- **Select All Instances ▸ Visible in View** or **▸ In Entire Project**: the elements of the same type; **Select Previous**: the selection made before;
- **Hide in view ▸ Elements** or **▸ Category**: hides the selected elements, or their whole categories, in this view;
- **Custom Graphics ▸ By category...**: opens **Visibility and Graphics** on the categories of the selection;
- **Visibility and Graphics...**.

While a tool is active, a right-click opens the tool's menu instead, and the step in progress stays: **Cancel** (what `Esc` does, which still cancels the step), **Repeat [command]**, **Snap Overrides ▸** (the snap of the next point, as the two-letter keys do; in every view, see Snap), **Zoom in Region**, **Zoom Out (2x)**, **Zoom to Fit**, **Previous Pan/Zoom**, **Next Pan/Zoom** and **Properties**. It is the same in plans, sections, elevations, 3D views and on sheets. With the menu open, `Esc` or a click in the view only closes it.

Snap

Additional settings ▸ Snaps... (in the BIMopen style the **Snaps...** command) opens the **Snap** window, the same in both styles:

- **Snaps Off** turns everything off (as the status bar's **Snap** switch, which stays as it was when the program opens again).
- **Length dimension snap increments** and **Angular dimension snap increments**, each with its box: values separated by semicolons (by default 1000; 100; 10 mm and 90°; 45°; 15°; 5°; 1°). The zoom chooses the step: zoomed in, finer steps.
- **Object snaps**: a box for **Endpoint**, **Midpoint**, **Nearest**, **Work Plane Grid**, **Quadrant**, **Intersection**, **Center**, **Perpendicular**, **Tangent** and **Remote Objects**, with their two letters; **Check all** and **Check none**. With **Nearest** off, the line under the cursor still offers its ends and its midpoint, even far away. **Remote Objects** (off at first) lines the point up, across or along, with the ends of lines far away, with a dashed guide.
- **Temporary overrides**: the list of the two letters that hold for one point.
- **Restore Defaults** puts everything back as it was at first.

The settings belong to the program, not to the project: they are kept between sessions.

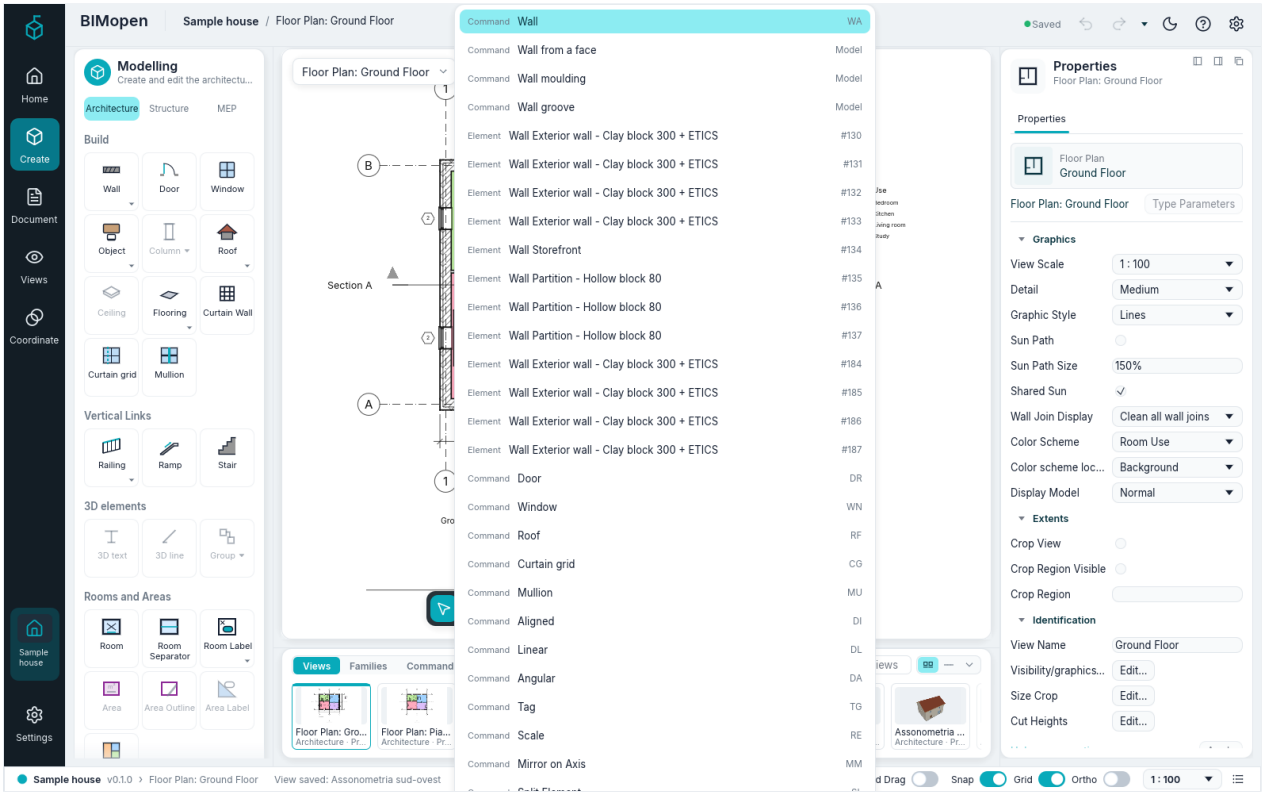
While drawing from a point (walls, lines, grids, reference planes, sketches, and **Rotate**'s second ray) the segment snaps its direction the same way in every view: horizontal or vertical, parallel or perpendicular to a line nearby, or a multiple of the angular increment the zoom gives; only when the cursor is a few pixels from that direction, otherwise the angle stays free. A dashed guide follows the direction, an arc with the value shows the angle, and the status bar says for instance *Snap: Angle 30° · step 15°*. The length is rounded to the length increment. **Angle measured from**, on the options bar, counts angles from the view's **Horizontal axis** or from the **Previous segment**. In the 3D view everything is measured on the work plane (the level's plane). At the command line `snap {"angles":[45,15],"objects":{"quadrant":false}}` changes the settings and `snap` alone shows them.

The `anglesnap 0,0 3,0.02` command (also through MCP, and `snap` with two points) returns the direction a segment snaps to, by the same rules as every view.

At the top of the menu, **Repeat [command]** starts the last tool you used again; `Enter` does the same when no tool is active and the command line is empty. The menu is the same in plans, sections and 3D views.

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Command search and command line



Press `Ctrl+K`, or click the search box in the middle of the top bar, and type a few letters: the list shows the commands, views, sheets, schedules, elements and materials that match. Click a result, or choose it with the arrow keys and press `Enter`: a command starts, a view opens, an element is selected and shown, a material opens in **Materials**. The search finds every command of the tool panel, the contextual card's too (with a view selected on a sheet, «activate» finds **Activate View**), and runs it as its tile does.

The search box is also the command line. You do not need to click it: whatever you type in a view while no field is being edited goes there. `Enter` runs a keyboard shortcut (`WA`, `DR`, `ZF` …) or a command with its arguments (`wall 0,0 5,0`) as typed; `Up` and `Down` bring back the lines typed before. The log of what has been done, the commands run, their results and the errors in red, is in **Command log** at the right end of the status bar and in the **Commands** mode of the strip. The command line chapter describes the commands.

Status bar

On the left, the status bar shows the project, the version and the active view, then:

- while a tool is active, what the tool expects, for example `WALL next point [Esc to finish]:`;
- with no tool, the category and the type of the element under the cursor, as **Walls : Exterior wall - Clay block 300 + ETICS**;
- while you drag, the length of the move, as **Move: 1500**;
- otherwise the last line of the command log.

On the right: the coordinates of the cursor (**X**, **Y**, **Z**) in metres; the active level, whose menu opens the plan of another level; the number of selected elements, which opens **Filter**; the switches **Select and Drag** (drag elements without selecting them first), **Snap**, **Grid** and **Ortho**; the scale of the view; and **Command log**.

Selecting, in short

The selecting and modifying chapter describes selection in full. The essentials:

- click an element to select it; click an empty point to clear the selection;
- `Ctrl` + click adds an element to the selection, `Shift` + click takes it out; `Tab` highlights the next element under the cursor, or a chain of joined walls;
- drag from left to right to select what is entirely inside the rectangle, from right to left to select what it touches;
- drag a selected element to move it;

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- `Ctrl+A` selects the elements of the open plan's level, or of the whole model in a 3D view; `Del` deletes the selection; `Esc` clears it.

On a touch screen the same is done with the fingers and the pen: see touch mode.

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Interface styles

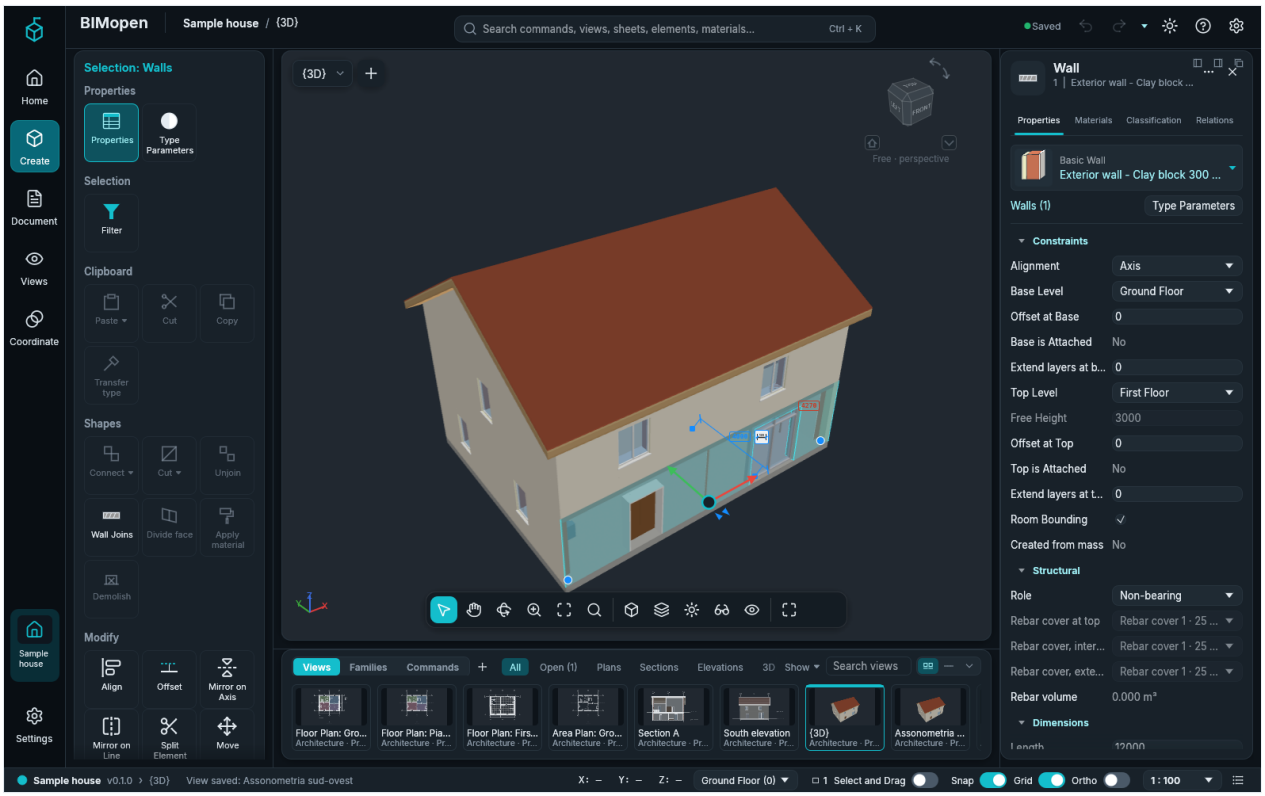
BIMopen has two interface styles and a touch mode. All of them run the same commands, with the same names, keyboard shortcuts and results; only the look and the place of the tools change.

- The **BIMopen** style, described in The interface and pictured throughout this manual: the rail, the tool panel, the command search, the element panel and the strip of views. BIMopen starts in this style.
- The **Classic** style: the ribbon with its tabs, the options bar, the **Properties** palette and the **Project Index** in panels.
- **Touch mode**: the layout for touch screens and pens, on a touch monitor, a tablet or a phone.

Choosing the style

1. Click the gear at the right end of the top bar, or open the application menu and choose **Options**.
2. Under **User Interface**, set **Interface** to **Classic** or **BIMopen**. BIMopen starts in the **BIMopen** style.

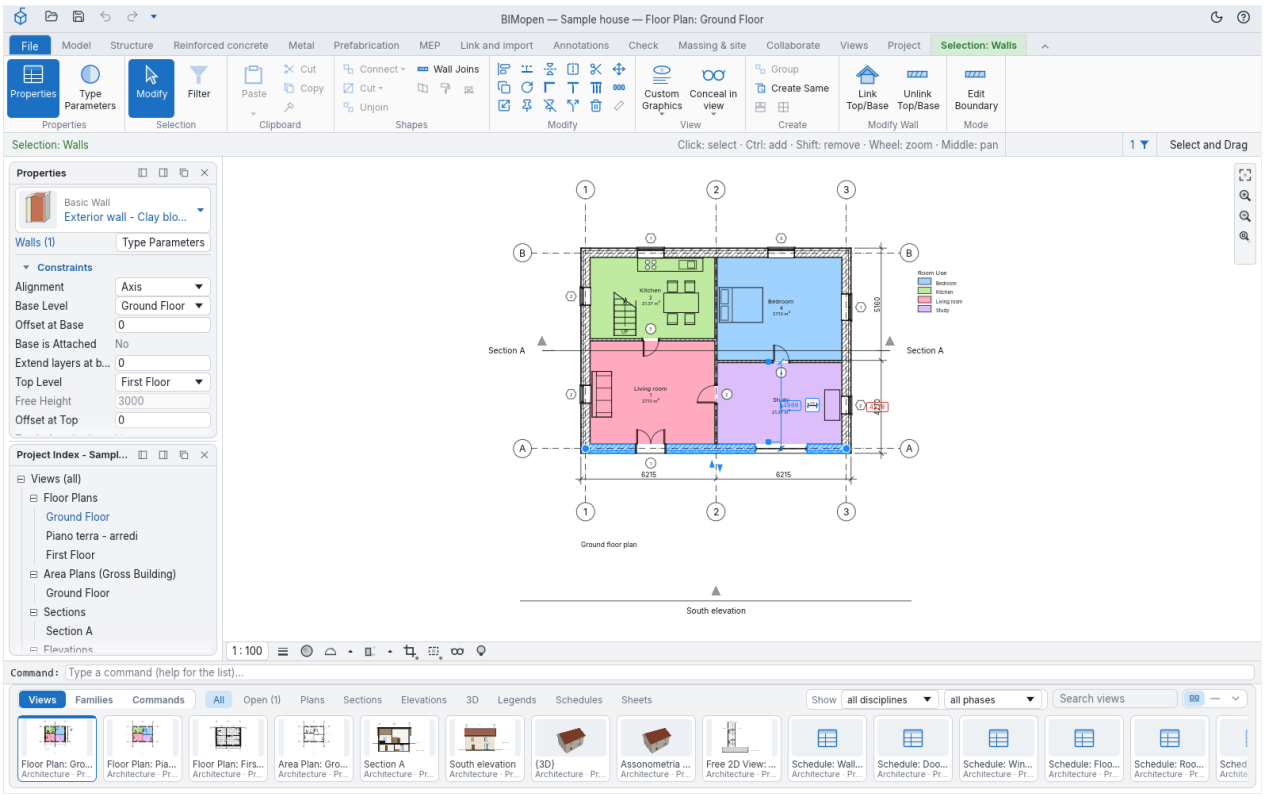
The window changes at once. The choice is remembered, on the desktop and in the browser, and the next session starts in the same style. **Theme** (**Light** or **Dark**) applies to both styles.



In Arabic the whole window is mirrored, in every style: the rail and the tool panel, or the palettes, are on the right, the element panel on the left.

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The Classic style



Part	Where	What it holds
Title bar	Top	The quick access toolbar (**Open...**, **Save**, **Undo**, **Redo** and the tools you add), the project and the active view, the light or dark theme and the **?** menu
Ribbon	Under the title bar	**File**, which opens the application menu, then the tabs (**Model**, **Structure**, **Reinforced concrete**, **Metal**, **Prefabrication**, **MEP**, **Link and import**, **Annotations**, **Check**, **Massing & site**, **Collaborate**, **Views**, **Project**) with their groups of buttons, and the contextual tab while a tool works or something is selected
Options bar	Under the ribbon	The options of the active tool, as the options strip of the **BIMopen** style, and at its right end what the tool expects or the element under the cursor, the selection count and **Select and Drag**; right-click it for **Dock at Bottom**
Properties and **Project Index**	Left	The element panel's **Properties** tab, and the **Project Index** of **Views**, as panels that dock on either side or float
Drawing area	Centre	The views, each with its view control bar at the bottom (scale, detail, crop, isolate) and the navigation bar on its right edge (**Zoom to Fit**, **Zoom In**, **Zoom Out**, **Zoom Options**)
Command line	Under the drawing area	The log and the prompt; what you type in a view goes there
Views and families strip	Bottom	The same strip as in the **BIMopen** style

The rail entries of the **BIMopen** style hold the ribbon's tabs: **Create** the tabs from **Model** to **Massing & site**, **Document** the sheets of **Views** and **Annotations**, **Views** the rest of **Views** and the **Project Index**, **Coordinate Check** and **Collaborate**, **Settings Project**.

Everything of the Classic style

Every function of the **Classic** style is in the **BIMopen** style, once:

- every button of the ribbon is a tile of the tool panel, in the rail entry that holds its tab, and the contextual **Modify** tab is the card on top of the panel while a tool works or something is selected;
- BIMopen's mark at the top of the rail opens the application menu; the top bar has undo, redo, the quick access tools and the **?** menu;

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- **Views** holds the whole **Project Index**, with its menus;
- a thumbnail of the view strip can be dragged onto the open sheet, or onto a sheet's thumbnail, and its right-click menu has the browser's actions and **Open in the active window**;
- the view selector's menu arranges the view windows (**Switch View**, **Tile**, **Cascade**, **Maximize**, **Replicate**, **Close Hidden**); `Ctrl+Tab` goes to the previous window;
- the right button on a tile offers **Add to Quick Access Toolbar**: the tool shows in the top bar, works from any section of the rail, and its right button offers **Remove from Quick Access Toolbar**;
- the key tips work as on the ribbon: tap `Alt` (or press `F10`), then a tab's letter opens the rail section that holds it (its letter shows on the rail) and a tool's letters start it.

Touch mode

Touch mode lays the interface out for touch screens and pens: phones, tablets and touch monitors. It runs the same commands as the two styles, with the same names; only the layout changes, and every control is big enough for a finger.

Turning it on

Options ▸ User Interface ▸ Touch mode has three values:

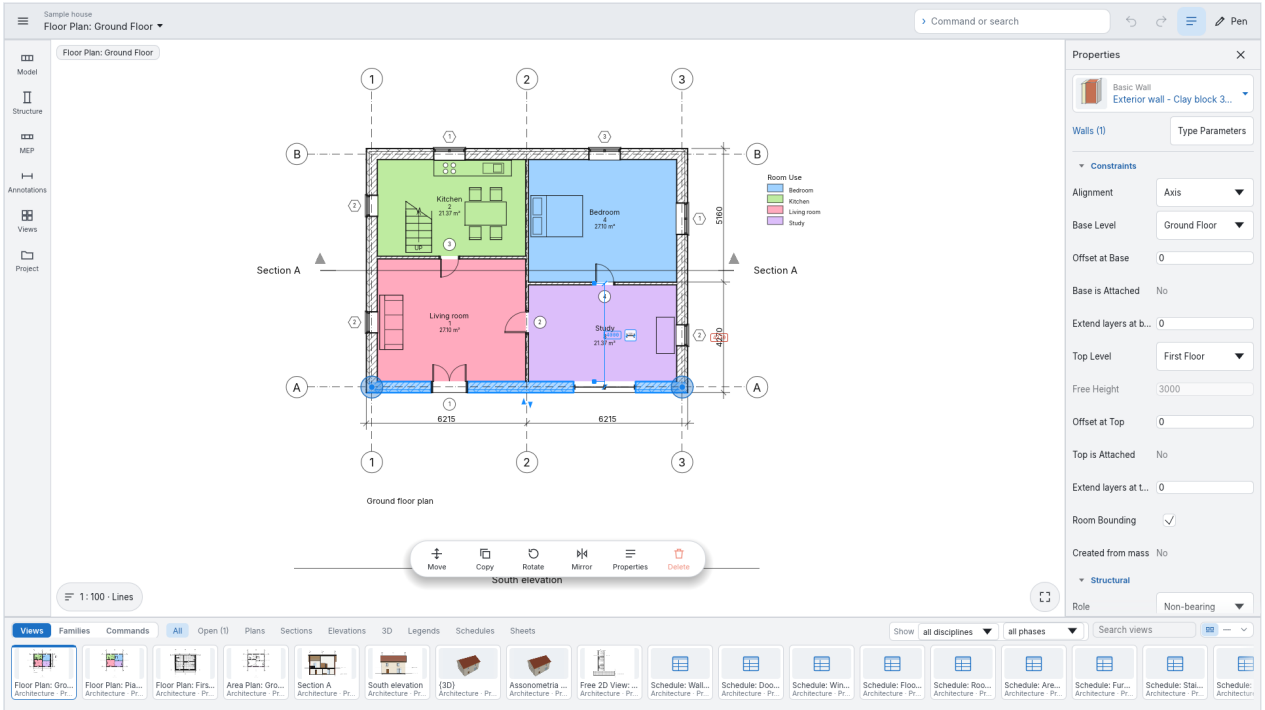
Value	Effect
Automatic	On where the device is touch first: in a browser with a finger as pointer or a window at most 900 pixels wide, and on Android. On the desktop it stays off
Always	Always on, on the desktop too (a touch monitor, a tablet PC)
Never	Never on

BIMopen remembers the choice with the other preferences. **Menu ▸ Leave touch mode** brings back the other interface until the next start.

Phone, landscape and tablet

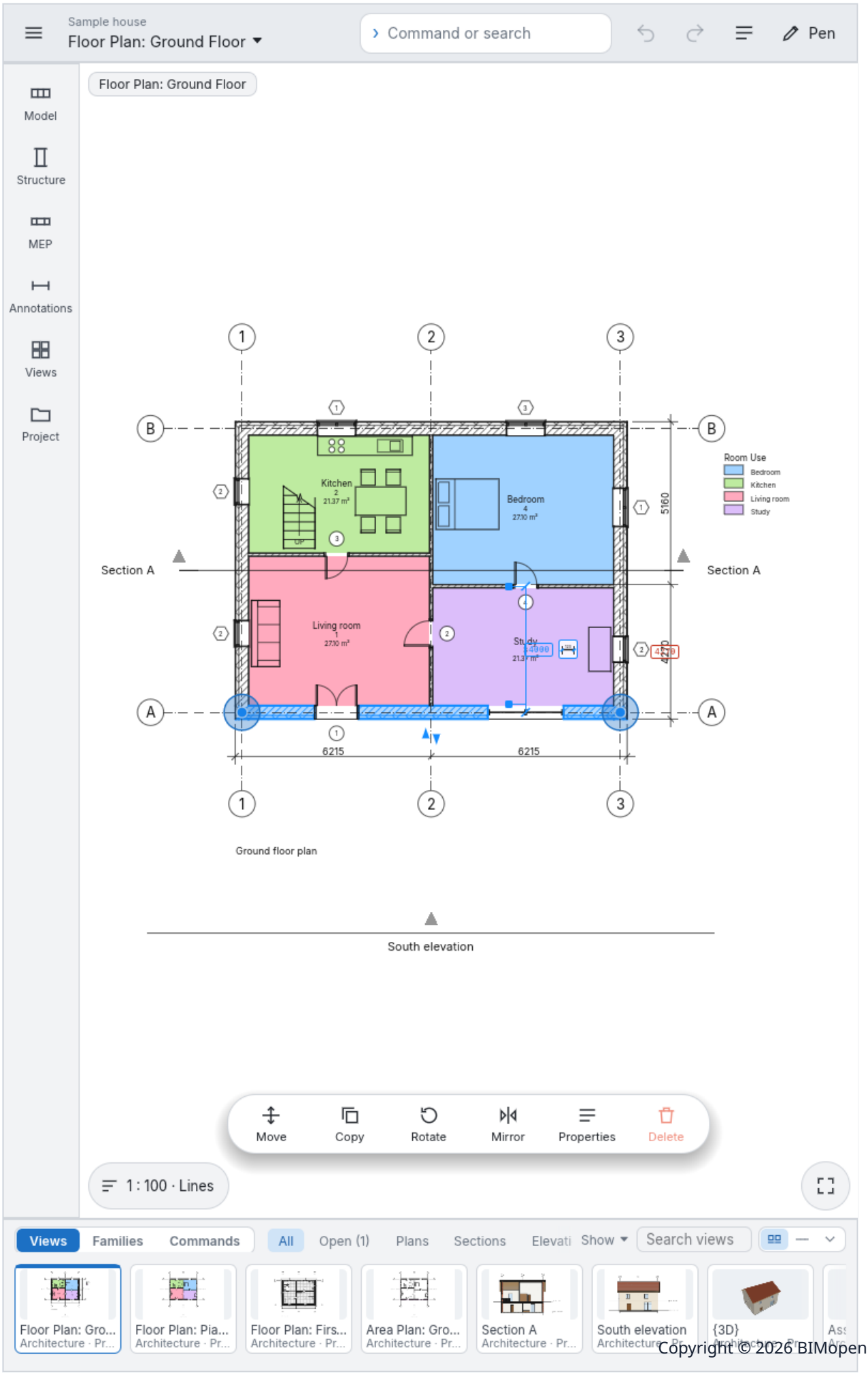
The layout follows the room the window has.

On a touch monitor, or any wide window (here 1920 × 1080): the rail on the left opens a palette of tools for each category (**Model**, **Structure**, **MEP**, **Annotations**, **Views**, **Project**), the properties stay on the right, the views and families strip at the bottom, and the **Command or search** field in the top bar, beside **Undo**, **Redo** and **Pen**.

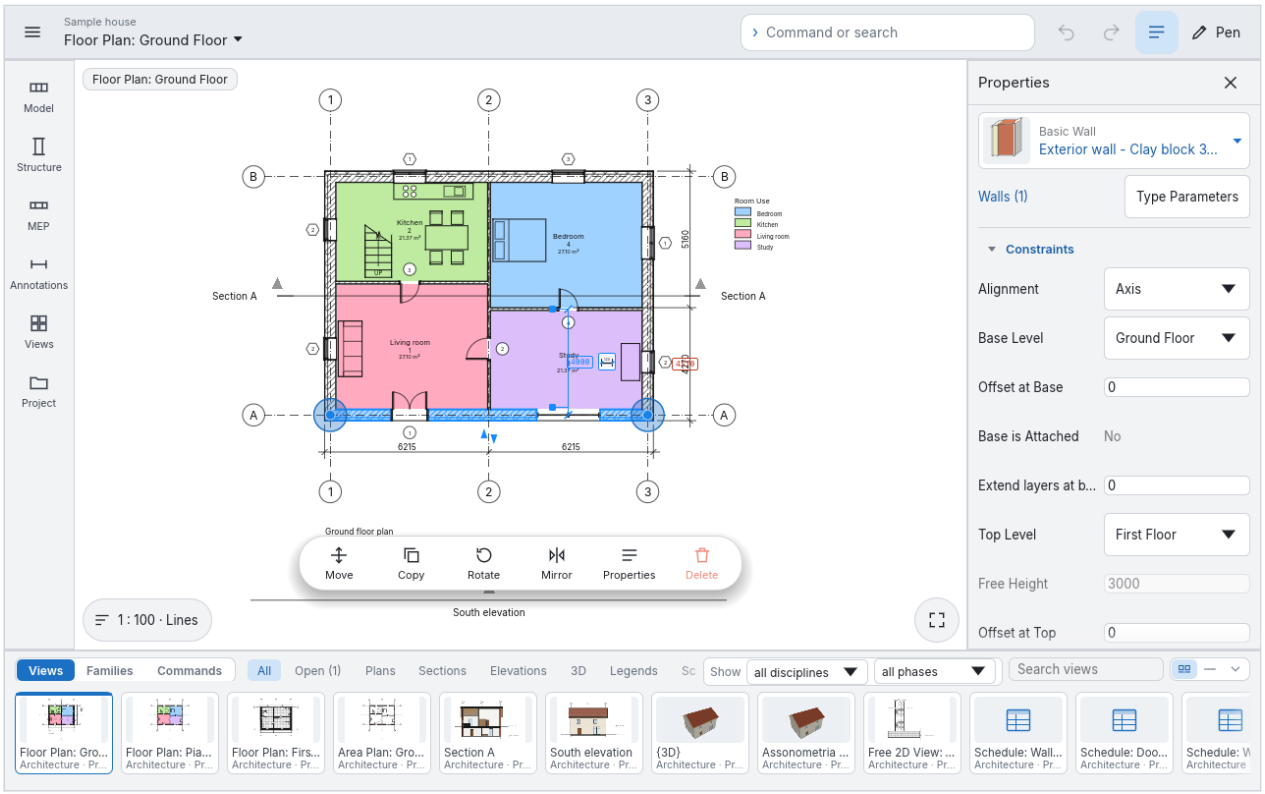


On a tablet, upright (800 × 1280) or sideways (1280 × 800): the same layout as on a monitor. Where the window is narrow the properties are put away, to leave the room to the drawing: **Properties** in the top bar brings them back.

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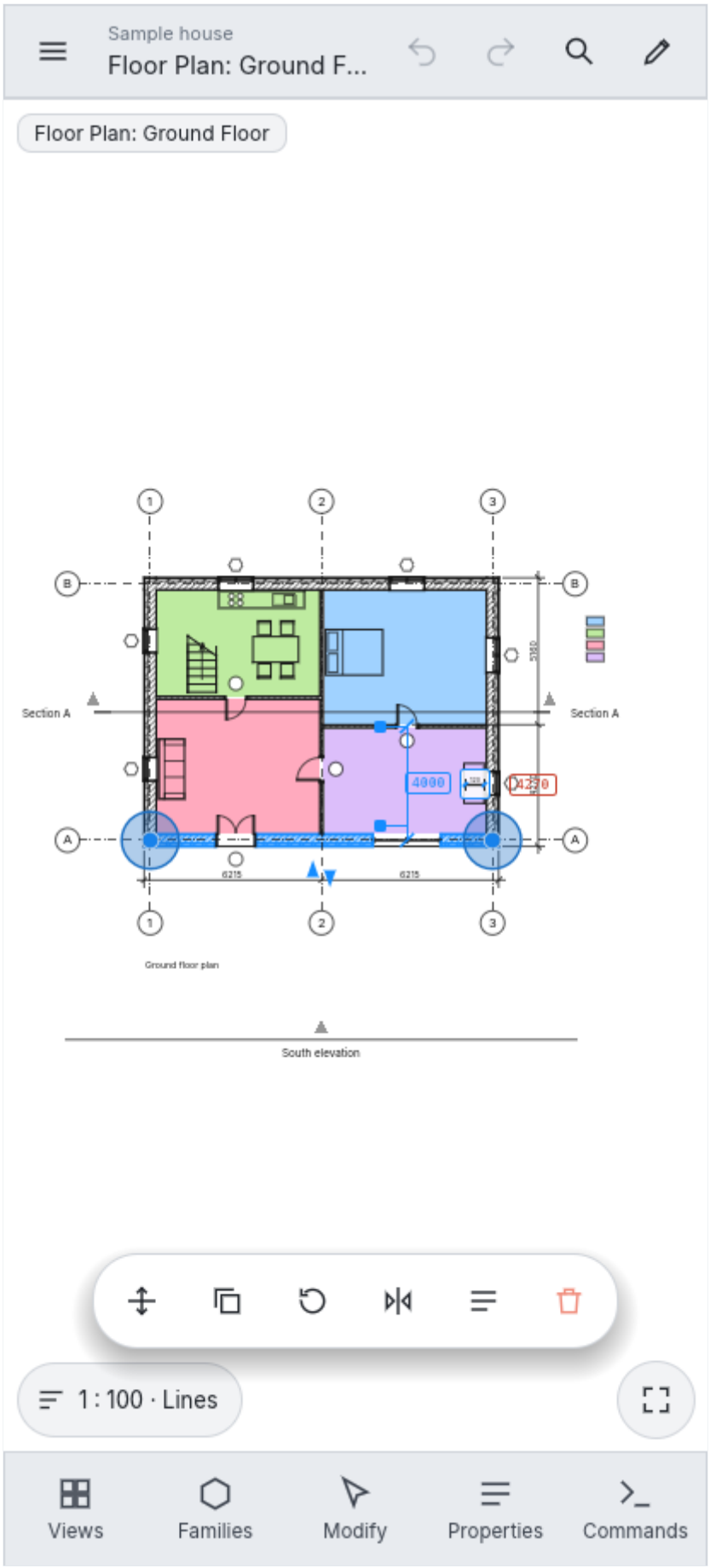


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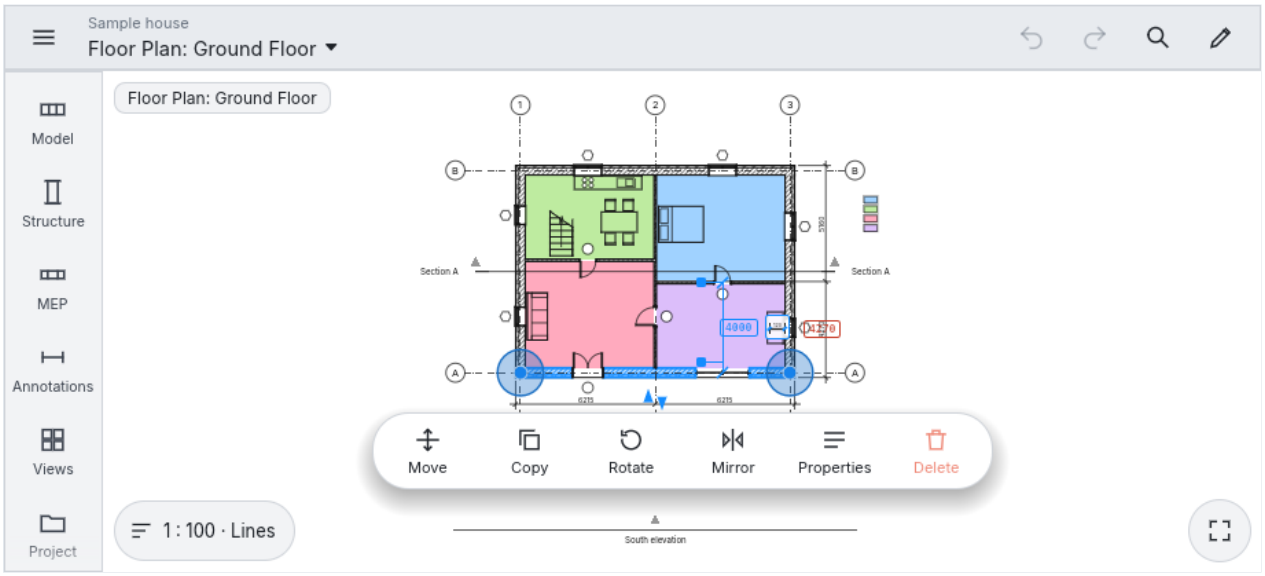
On a phone, upright (narrow and tall): a compact top bar with the menu, the project and the view (a tap opens the views), **Undo**, **Redo**, **Command or search** and **Pen**; the drawing takes the whole width; five big buttons at the bottom, **Views**, **Families**, **Tools** (**Modify** while something is selected), **Properties** and **Commands**, each opening a sheet from the bottom. Drag the sheet's handle up to make it taller, down to make it smaller or close it; a tap outside closes it.

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On a phone sideways (short and wide): a rail on the left with the categories **Model**, **Structure**, **MEP**, **Annotations**, **Views** and **Project**, plus the views and the properties; the sheets open on the right.

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The tools are large tiles with an icon and a name. A tool still to come is greyed, **Not available yet**; a tool that does not work in the 3D view (**Level**, **Grid**, **Reference Plane**, the annotations) is greyed there, **Not available in the 3D view**. A tap on a greyed tile says why.

While a tool works, a green strip under the top bar shows its options and **Done**, which ends it (or finishes a sketch). While you draw a wall, **Length…** opens the command line: type the length of the next wall and, if needed, its angle, as `4.5` or `4.5 90`.

A tap selects, and the selected walls show big grips at their ends. Over the drawing a floating bar offers **Move**, **Copy**, **Rotate**, **Mirror**, **Properties** and **Delete**; a long press opens the context menu. At the bottom left the view button shows the scale and the graphic style and opens **Scale**, **Detail**, **Graphic style**, **Sun and shadows**, **Crop** and **Isolate Temporarily**; at the bottom right **Zoom to Fit**.

Gestures and the pen

Gesture	What it does
tap	selects, or places the point of the active tool
drag with one finger	as a mouse drag: a selection window, a grip, an element
long press	the context menu
two fingers	pan; pinch to zoom; in the 3D view twist to turn the model
double tap	fits the view in the window

Pen in the top bar is for a pen: the pen draws and selects, fingers only move the view (one finger pans, two fingers pan, zoom and turn). A pen palette on the side of the drawing has **Select**, **Wall**, **Door**, **Window**, **Detail line**, **Aligned**, **Text** and **More tools…**. In a browser a pen is told from a finger by the browser; in the desktop app a touch that reports a pressure is taken for a pen.

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Projects and files

This chapter describes how a project is created, opened, saved and exchanged with other programs, its settings and the options of BIMopen itself.

Creating a project

application menu ➤ New ➤ Project... opens **New project**:

- **Template file**: BIMopen's default template, or the one set in **Options ➤ File Locations**; a template file chosen with **Browse...**; or **None**, which starts from two levels and one type of each kind.
- **Create new**: **Project** or **Project Template**.

A project made from a template has everything the template had: types, settings, views, sheets, project information and print settings. Saving it asks for a name, so the template is not overwritten.

Templates

A template is a `.bimtpl` file. **application menu ➤ Save As ➤ Template...** saves the open project as a template; a project created as a **Project Template**, or a template opened, is saved as a template too.

Opening a project

`Ctrl+O` or **application menu ➤ Open...** opens a project. **application menu ➤ Recent Documents** lists the latest projects, the most recent first; a pin keeps a project in the list. Without a file, BIMopen starts on **Recent Files**, with the recent projects, **Open...**, **New** and the **Sample project** (unless **Enable Recent Files page at startup** is unticked in the options). Dropping a project file on the window opens it too.

A project already open in another BIMopen window opens read-only: **Save** asks for a new name. If BIMopen stopped without saving (a crash, a power cut), opening the project again brings the unsaved work back as unsaved changes.

Saving

`Ctrl+S`, or a click on the save state in the top bar, saves the project; until then the file stays as it was last saved, and the top bar says **Unsaved changes**. **application menu ➤ Save As ➤ Project...** writes it to a new file and goes on there.

Save As ➤ Options... opens **File Save Options**: **Maximum backups** keeps that many earlier versions beside the file (`house.0001.bimopen`, `house.0002.bimopen`...), deleting the oldest; **Compact file** rewrites the file without its obsolete parts at the next save.

When a changed project stays unsaved for the interval set in **Options ➤ General** (**Save reminder interval**, 30 minutes to start with), **Save Reminder** offers to save it, not to save it, or to change the interval.

One project is open at a time: the project opened takes the place of the one open, and the views of that one close with it. Opening the project already open leaves it as it is, when it has no unsaved changes; with changes, answering **No** to the question to save them drops them and opens the file again as last saved, as closing it without saving and reopening it.

Before **New**, **Open...**, a recent document, a dropped file, `open` at the command line, the sample project, or closing the window, a changed project asks **Save changes to ...?**: **Yes** saves it, **No** drops the changes, **Cancel** keeps it open.

In the browser, **Open...** uses the browser's file chooser, and **Save** downloads the file; backups, crash recovery and read-only opening need a file on disk and are desktop-only.

Project settings

Project information

Settings ➤ Settings ➤ Project information sets the project's name, number, client, address, issue date and status: the title blocks of every sheet show them.

Project units

Settings ➤ Settings ➤ Project units lists the quantities (length, area, volume, angle, slope and the others) with a sample of each, and the **Decimal symbol/digit grouping**. **Format...** of a quantity sets its unit, its **Rounding** (a number of decimals, or

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Custom with an **Increment factor**, such as 5 mm), its **Unit symbol**, **Suppress trailing zeros**, the plus sign for positive values and digit grouping.

Sun Settings

Settings ➤ Additional settings ➤ Sun Settings opens the sun settings of the active view, the same as the sun button of the view control bar (see Views and graphics).

Options

application menu ➤ Options has four tabs; the choices are kept between sessions.

Tab	What it sets
General	The save reminder interval
User Interface	The theme (**Light** or **Dark**), the language, the interface style and size; **Enable Recent Files page at startup**; in the **Classic** style, the ribbon tab after a selection ends or a tool closes (**Stay on the Modify tab** or **Return to the previous tab**) and **Display the contextual tab on selection**; **Tooltip Assistance** (**None**, **Minimal**, **Normal**, **Extra High**)
Graphics	**Invert background color**, **Selection color**, **Highlight color**, **Use anti-aliasing for 3D views**, **Temporary Dimension Text Appearance** (size and background)
File Locations	**Default template file path** (the template of **New**) and **Default path for user files** (where the file dialogs open)

Copying settings from a project

Settings ➤ Settings ➤ Copy Settings from Project copies settings from another project into this one. In **Select Items To Copy**, choose the project with **Browse...** (**Copy from:**), then tick what to copy: the types of walls, floors, roofs, doors and windows, stairs, railings, curtain panels and mullions, line styles, filled region types, materials, fill and line patterns, line weights, styles by category, view templates, filters, viewport types, title blocks, the project units, the print settings and the project information (**Check all**, **None**).

What is copied brings along what it uses: a wall type its materials, a material its patterns, a view template its filters. When names are already in the project, **Duplicate Types** asks whether to **Overwrite** them or to transfer the **New Only**. The whole transfer is one step of undo.

Exchanging files

Exporting to IFC

application menu ➤ Export ➤ IFC... writes the model in IFC4: walls with their axis and body, openings, doors, windows, floors, roofs, stairs, railings, curtain walls, furniture and rooms as spaces, on their storeys.

Export ➤ Options ➤ IFC Options... opens **IFC Export Classes**: the IFC class of each category; **Not Exported** leaves a category out, and **Default** goes back to the standard classes. The classes are kept with the project.

Opening IFC files

application menu ➤ Open ➤ IFC... makes a new project from an IFC2x3 or IFC4 file, in metres or millimetres: storeys become levels, walls are rebuilt from their axis (or from the long side of their body), slabs become floors, doors and windows go in the walls their openings cut, and spaces become rooms. The messages count what was not read.

Open ➤ IFC Options... opens **Import IFC Options**: what each IFC class becomes (walls, floors, doors, windows, rooms or **Not Imported**), and **Default**. The choices are kept between sessions.

.rvt files

Opening a `.rvt` file is experimental: today it reads only the project name and the level names, so the new project has its levels and no walls, doors, windows or floors. The messages say so after the import. To exchange models with other BIM software, use IFC4.

Other exports

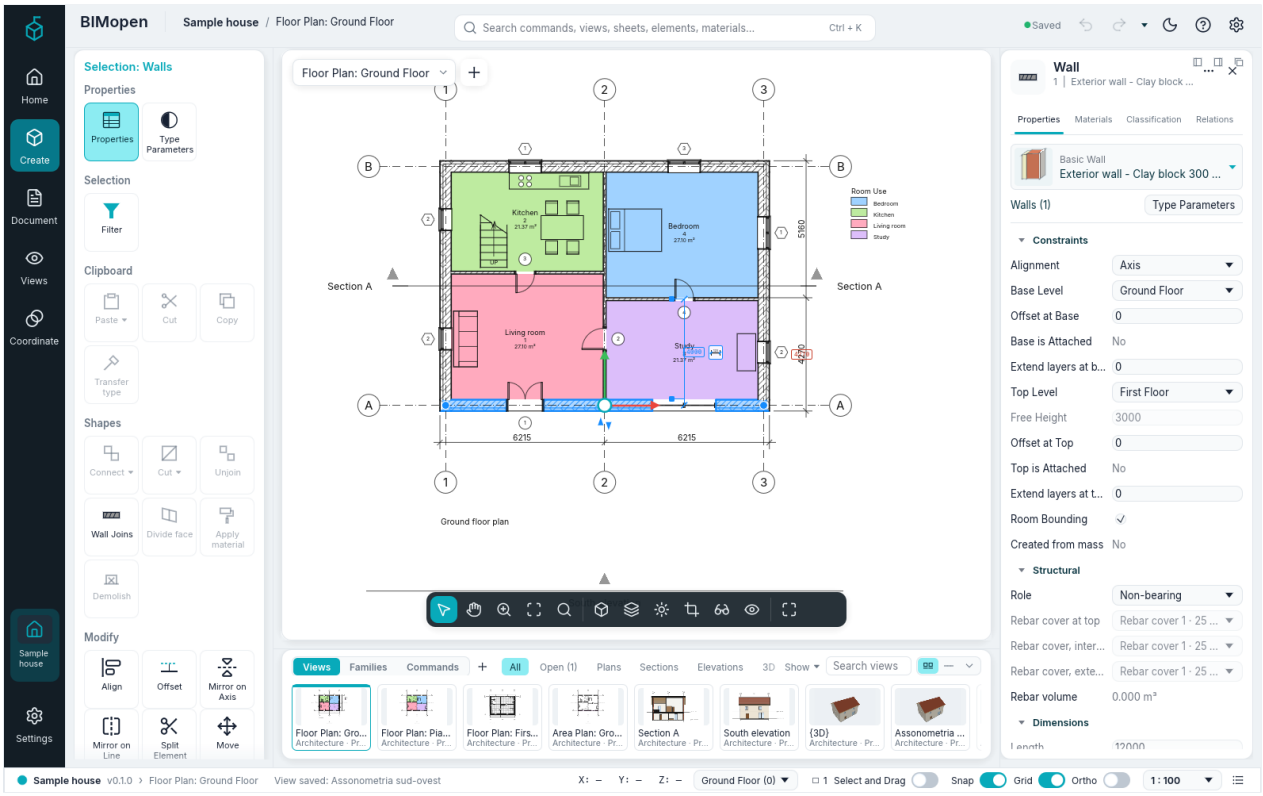
Printing to PDF and exporting images are described in Sheets, schedules and printing; schedules export as delimited text from the same chapter. `bimopen-cli convert` exports from the command line (see Command line and MCP).

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Selecting and modifying

Before most changes you select what to change. This chapter describes how elements are highlighted and selected, how a selection is refined, how selected elements are dragged, and the tools of the **Modify** group. Everything here works the same in the desktop app and in the browser.

Highlighting and selecting



Move the cursor over an element: its lines are drawn wider, the status bar at the bottom of the window names it (category and type, as **Walls : Exterior wall...**), and after a moment the same words appear under the cursor. Click to select the highlighted element. A selected element is drawn in the selection colour in every view where it shows, and the count at the right end of the status bar says how many elements are selected.

A right click on an element selects it and opens the context menu. `Esc`, or a click on an empty spot, clears the selection.

To...	Do this
add an element to the selection	`Ctrl` + click
take an element out of the selection	`Shift` + click
select what is wholly inside a window	drag from left to right
select what a window touches	drag from right to left (the window is dashed)
add, or take away, what a window catches	drag with `Ctrl`, or with `Shift`
select again what was selected before	`Ctrl+Left`, or right click ▸ **Select Previous**

Windows work in plans, sections, sheets and 3D views. In a 3D view the left button selects; the right button orbits and the middle button pans (see The interface).

Elements close together: TAB

When elements lie close together or one over the other, the one highlighted may not be the one you want. Keep the cursor still and press `Tab`: the next element under the cursor is highlighted and the status bar names it. `Shift+Tab` goes back. Click to select the highlighted element. In a 3D view `Tab` goes from the nearest element to the ones behind it; on a sheet, through views placed one over the other.

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Chains of walls and lines

Walls and lines joined end to end form a chain. With the cursor on one of them, `Tab` highlights the whole chain (the status bar says **Chain of n joined elements**); a click selects it. A chain can mix walls with room separators and area outlines; detail lines chain with the detail lines of their view. Where three or more meet, the chain goes on along the one that makes the smallest angle with the last; a wall that stops against another one between its ends takes the chain on along that wall.

Part of a chain. Select the first element of the part, move the cursor over the last one and press `Tab`: the elements between them are highlighted. `Tab` again highlights the whole chain, a third time the last element alone. Click to select what is highlighted. On a closed chain the part goes round by the side where the cursor is: near the end of the last element that one way round reaches first.

Elements joined at a point. Select an element and keep the cursor on it, near an end where others are joined. `Tab` highlights it with the nearest one joined there; `Tab` again, every element joined there. Click to select them.

A selected chain. Where two selected walls or lines meet, a round control appears. Drag it to move the joint: the selected ends there move together and stay joined; elements not selected are left where they are. Drag the chain with `Shift` to move it whole, horizontally or vertically; drag it with `Ctrl` to place a copy.

Filtering a selection

Click the count in the status bar, or **Filter** on the contextual card, to open **Filter**. It lists the categories in the selection, how many elements each has, and the total. Untick a category to leave its elements out: the selection and the count in the status bar follow at once. **Check all** and **Check none** tick or untick every category. **OK** keeps the selection, **Cancel** brings back the one you started with.

Select All Instances

Right-click an element ▸ **Select All Instances**, then **Visible in View** (the elements of the same type shown in the active view) or **In Entire Project**. Typing `SA` and `Enter` does the same as **Visible in View**. The element must have a type: rooms, for example, have none.

Dragging elements

Drag a selected element to move it; the whole selection moves together, and one drag is undone by one `Ctrl+Z`. A temporary dimension shows the move, and the status bar its length (**Move: 1500**). With **Select and Drag** turned on in the status bar, an element can be dragged without selecting it first; with it off, a drag that starts on an element not selected draws a window.

- A wall joined at its ends to walls that are not selected moves square to itself only, and the joined walls stretch to stay joined. Walls that stop against it follow too. Hold `Shift` to move it freely; the joined walls still follow.
- Any other element moves freely; hold `Shift` to keep the move horizontal or vertical.
- Doors and windows move with their wall and keep their place along it. Dragged on their own, they slide along the wall; dropped on another wall, they move into it.
- Hold `Ctrl` when you start dragging to place a copy instead.

Elements are dragged in plans, sections and elevations (along the view), on sheets (views), and in 3D views, where they move on the horizontal plane where you grabbed them. The arrow keys move the selection a small step, rounded to a tidy value at the zoom of the view; with `Shift`, ten steps.

The Modify tools

The **Modify** group of the contextual card has three rows: **Align**, **Offset**, **Mirror on Axis**, **Mirror on Line**, **Split Element**, **Split with Joint**; **Move**, **Copy**, **Rotate**, **Shorten/Lengthen**, **Shorten/Lengthen one element**, **Shorten/Lengthen several elements**; **Repeat in Series**, **Scale**, **Fix**, **Unpin**, **Delete**. The status bar says what each tool expects next, and the options strip shows its options. `Esc` cancels the step in progress; pressed again, it ends the tool.

Move, **Copy**, **Rotate**, the two **Mirror** tools, **Repeat in Series** and **Scale** work on the selection. Started with nothing selected, they let you select first (clicks, `Ctrl`, windows, `Tab`), then go on when you press `Enter` or **End** on the options strip.

Move, **Copy** and **Rotate** also work in 3D views: their points are taken on the plane of the level clicked.

Before each click the tool shows what the click would do: the selection drawn where it would go (moved, copied, turned, mirrored about the axis under the cursor, offset to the cursor's side, aligned to the reference), following the cursor, with the distance or the angle beside it.

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Where a tool asks for a second point, you can type a distance (or an angle) instead and press `Enter`: it shows next to the cursor while you type, and is measured towards the cursor.

Move

Click a base point, then the destination. The tool ends and the elements stay selected. Options:

- **Constrain**: the move is along or square to the selected wall (horizontal or vertical for other elements);
- **Disjoin**: walls joined to the moved ones do not follow, and a door or window can be moved onto another wall;
- **Copy**: copies are placed instead; with **Multiple**, one at each click until `Esc`.

Copy

As **Move**, but a copy is placed. With **Multiple**, a copy at each click until `Esc`; **Constrain** as for Move.

Rotate

The centre of rotation shows at the middle of the selection; drag it to put it elsewhere. Click the first ray, then the second: the selection turns by the angle between them, shown by a temporary angular dimension. Instead of the second ray, type the angle and press `Enter`, or type it in **Angle** on the options strip and press `Enter`. **Copy** turns a copy; without **Disjoin**, walls joined to the turned ones follow.

Mirror

Mirror on Axis: click a wall, a grid or a line as the axis. **Mirror on Line**: click two points. **Copy** is ticked: a mirrored copy is placed; untick it to mirror the elements themselves.

Repeat in Series

Linear: click two points. **Number** is how many elements in all. With **Move To: 2°** the two points give the spacing; with **Last**, the last element goes to the second point and the others are spread evenly. **Constrain** keeps the direction along or square to the selection.

Radial: drag the centre where the copies turn about, click the first ray, then the second (or type the angle and press `Enter`, also in **Angle**). With **2°** the angle is between neighbours, with **Last** it is the whole span.

Offset

Moves a wall or a line square to itself, or places a copy (**Copy** is ticked).

- **Numerical**: set **Offset**, then click the element on the side it should go to. Press `Tab` first to offset the whole chain (of the same category) as one: its corners are kept.
- **Graphical**: click the element (`Tab` for the chain), then two points: the distance between them, square to the element, is the offset. You can type it instead. `Ctrl` copies.

Align

Click the reference, then the element to align: the face of the element nearest to your click goes onto the face of the reference nearest to its click. **Prefer** chooses **Wall faces** or **Wall centerlines**. After aligning, a padlock shows on the aligned face: click it to keep the two aligned, so that moving the reference moves the other one too. Without **Multiple Alignment** the next click picks a new reference; with it (or holding `Ctrl`) you can align several elements to the same one. `Esc` once starts a new alignment, twice ends the tool.

Shorten/Lengthen

Walls and lines are trimmed or extended; click each one on the part to keep.

- **Shorten/Lengthen**: two elements, which then meet at a corner;
- **Shorten/Lengthen one element**: the boundary, then the element;
- **Shorten/Lengthen several elements**: the boundary, then each element in turn.

Split Element and Split with Joint

Split Element: click a wall or a line where it should become two. With **Delete Inner Segment**, click two points: the part between them is removed, with the doors and windows in it. **Split with Joint**: set **Joint Gap**, then click: the two parts are that far apart.

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Scale

Walls and lines (and the position of dimensions) can be scaled; with anything else selected, **Scale** is not available. **Graphical**: click the origin, the end of a first vector, the end of a second one: the elements are scaled by the ratio of the two lengths (you can type the second one). **Numerical**: set **Scale Factor**, then click the origin. Doors and windows keep their distance from the middle of their wall.

Fix and Unpin

Fix keeps the selected elements where they are: they cannot be dragged, moved by a tool or deleted, and a message says so. A pinned element shows a pushpin when selected; click the pushpin, or use **Unpin**, to free it.

Delete, undo and redo

Delete (or the `Del` key) removes the selection and what it hosts; pinned elements stay. `Ctrl+Z` undoes the last change and `Ctrl+Y` redoes it: each tool, drag or nudge is one step.

Typed shortcuts

Type the letters and press `Enter`: `MV` **Move**, `CO` **Copy**, `RO` **Rotate**, `MM` **Mirror on Axis**, `DM` **Mirror on Line**, `AR` **Repeat in Series**, `OF` **Offset**, `AL` **Align**, `TR` **Shorten/Lengthen**, `SL` **Split Element**, `SG` **Split with Joint**, `RE` **Scale**. `PN` pins and `UP` unpins the selection, `SA` selects all its instances visible in the view. Keyboard shortcuts lists them all.

Not yet available

- `Tab` to reach a part of an element: a curtain panel or mullion, a part of a stacked wall;
- **Select All Instances** from a type in the Project Index;
- **Group and Associate** of **Repeat in Series**;
- splitting walls along a horizontal line in elevations and 3D views;
- cut, copy and paste through the clipboard, **Paste Aligned**, **Create Same**.

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Walls, doors and windows

This chapter describes how walls are drawn, how they join and attach to roofs and floors, how their types and layers are set up, and how doors and windows are placed in them, flipped, moved and drawn at each level of detail.

Drawing walls

Create ▸ Build ▸ Wall (or type `WA`) starts the wall tool, in a plan or in the 3D view. Pick the wall type in the type selector at the top of the element panel; **Type Parameters** changes the type, and with it every wall of that type. The contextual card **Draw Wall** shows the **Draw** group, and the options strip shows:

Option	What it does
Height or **Depth**	The wall rises from the level (**Height**) or goes down from it (**Depth**)
A level, or **Unconnected** with a value	The wall reaches that level, or is that high (or deep)
Level	In a 3D view only: the level the walls stand on; takes the level of the surface of the first click
Alignment	The plane of the wall that runs along the clicks: **Axis** (the default), the core centre line, or one of the finish and core faces, exterior or interior
Continue	On, each click ends a wall and starts the next one; off, each two clicks make a wall on its own
Offset	The alignment line runs this far from the clicks, towards the exterior
Radius	Not available: it needs curved walls, and is shown disabled
Join at the ends	Not available yet, shown greyed: the walls drawn always join; an end's join is prevented afterwards from its control
Automatic joining	Not available yet, shown greyed and ticked: the walls join those they meet

In the 3D view the wall being drawn follows the cursor as a solid, with its thickness and its height up to the top level (or the height typed), on the plane of the level, with its length and angle as a temporary dimension; **Shift** keeps it square and **Esc** takes it away. **Rectangle**, the polygons and **Pick Lines** work there too: the walls of the whole shape follow the cursor, and the line of the level under the cursor is picked as in plan.

While drawing:

- the wall's exterior is on the left of the drawing direction: drawn from left to right, the exterior face is at the top. The preview shows the exterior face thicker;
- the **space bar** puts the exterior of the next wall on the other side;
- after the first point, type a length and press Enter: the value shows in a box at the cursor, and the wall ends that far, in the direction of the cursor (grids, room separators and area outlines, sections and elevations take a typed length the same way);
- **Esc** ends the chain; **Esc** again, or **Modify**, ends the tool.

The **Draw** group picks the shape:

Shape	How
Line	Click the start and the end of each wall
Rectangle	Click two opposite corners: four joined walls, with the exterior outside
Inscribed Polygon	Set **Sides** in the options strip, click the centre and a corner: the corners are on the circle
Circumscribed Polygon	The same, but the sides touch the circle
Pick Lines	Click a line of the plan (a line or an edge of an element): a wall along it. Press **TAB** over a line to take the whole chain of lines joined to it, then click

The alignment and flipping

Select a wall: blue dots mark the ends of its alignment line, and two arrows stand on its exterior side. Clicking the arrows turns the wall over about its alignment line: the first and the last layers swap, and the doors and windows keep their orientation.

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Changing **Alignment** of a placed wall does not move it; the next flip turns it about the new plane. Changing the type of a wall, or its layers, keeps its alignment line where it was.

Attaching walls to roofs and floors

A wall can follow a roof or a floor instead of a fixed height.

1. Select the walls.
2. **Selection: Walls ▸ Modify Wall ▸ Link Top/Base**.
3. In the options strip, **Attach Wall: Top** or **Base**.
4. Click the roof or floor, in a plan, a section or the 3D view.

With **Top**, the walls rise or drop to the underside of the roof (following its slopes: the gable walls go up to the ridge) or of the floor above. With **Base**, they stand on the top face of the floor below. From then on they follow it: a steeper roof, a roof raised by its offset, a moved floor carry the walls with them.

To detach them: select the walls, **Selection: Walls ▸ Modify Wall ▸ Unlink Top/Base**, then click the element to detach them from, or press **Detach All** in the options strip. The walls keep the height they have at that moment, and can be attached again at any time. When a roof or floor with walls attached to it is deleted, the walls keep their height too.

The properties **Top is Attached** and **Base is Attached** show the state (read-only); while the top is attached, **Free Height** is read-only.

Editing a wall's boundary

A wall need not be a rectangle: its outline can be sketched in an elevation or a section parallel to it, with holes, as for a gable or a stepped wall.

1. Select the wall and click **Mode ▸ Edit Boundary** in its contextual card. From a plan or a view that is not parallel to the wall, **Go to View** lists the sections and elevations parallel to it: choose one and click **Open View**.
2. Sketch mode opens with the wall's outline in magenta. Change it with **Line** and **Rectangle** (see sketch mode): pick a line and press `Del`, draw new ones, or a **Circle**. Every closed loop inside the outline is a hole through all the layers; a loop inside a hole is solid again.
3. Click **Finish Edit Mode** (or **Cancel Edit Mode** to leave the wall as it was).

The wall takes that outline in 3D, in sections and in plan, where it is cut only where the outline reaches the cut plane; its doors and windows still cut it, also over a hole, which stays a hole. A top attached to a roof or a floor still clips it. Plan, sections, elevations and the 3D view all show one solid, cut from the same outline and holes. The layers that wrap line the holes as they line doors and windows (see Wall types and layers). A curtain wall keeps the panels and mullions inside its new outline. **Mode ▸ Reset Boundary** makes it a rectangle again; `Ctrl+Z` undoes either.

The boundary can also be edited in the 3D view: the sketch lies on the wall's plane and the view stays 3D. Only from a plan does **Go to View** appear, and nothing changes until a view is opened with **Open View**. While sketching, the wall's original outline is drawn as dashed reference lines; the sketch snaps to their ends and onto them.

Wall joins

Walls whose ends meet join by themselves, and so do a wall ending on the side of another and two walls crossing each other. At **Medium** and **Fine** detail each layer is resolved against the layers of the other wall:

- at a corner (an L), each layer meets the same layer of the other wall: the finishes wrap round the corner and the cores meet;
- at a T, the core of the wall that ends runs through the finishes of the other wall up to its core; its own finishes stop at the face, which is open there;
- where two walls cross (an X), a core runs through the finishes of the other wall;
- pieces of the same material merge with no line between them; a thin line separates two different materials; the free outline is drawn thick.

At **Coarse** detail joined walls are one fill, with no line between them. Sections and the 3D view use the same pieces, so nothing overlaps.

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The Wall Joins tool

Shapes ▸ Wall Joins in the contextual card, then click the grey square that appears on a corner. The options strip offers:

- **Configuration: Butt** (one wall runs through, the other stops against it), **Miter** (both end on the bisector) or **Square off** (the wall that stops ends square); **Previous** and **Next** change which wall runs through;
- **Display: Clean Join**, **Don't Clean Join** (the walls butt, with a thick line between them) or **Use View Setting**.

Each change can be undone. **Esc** ends the tool.

The plan's property **Wall Join Display** (in **Graphics**) is the view setting: **Clean all wall joins**, or **Clean same wall type** to keep a thick line where walls of two types meet, unless their corner is set to **Clean Join**.

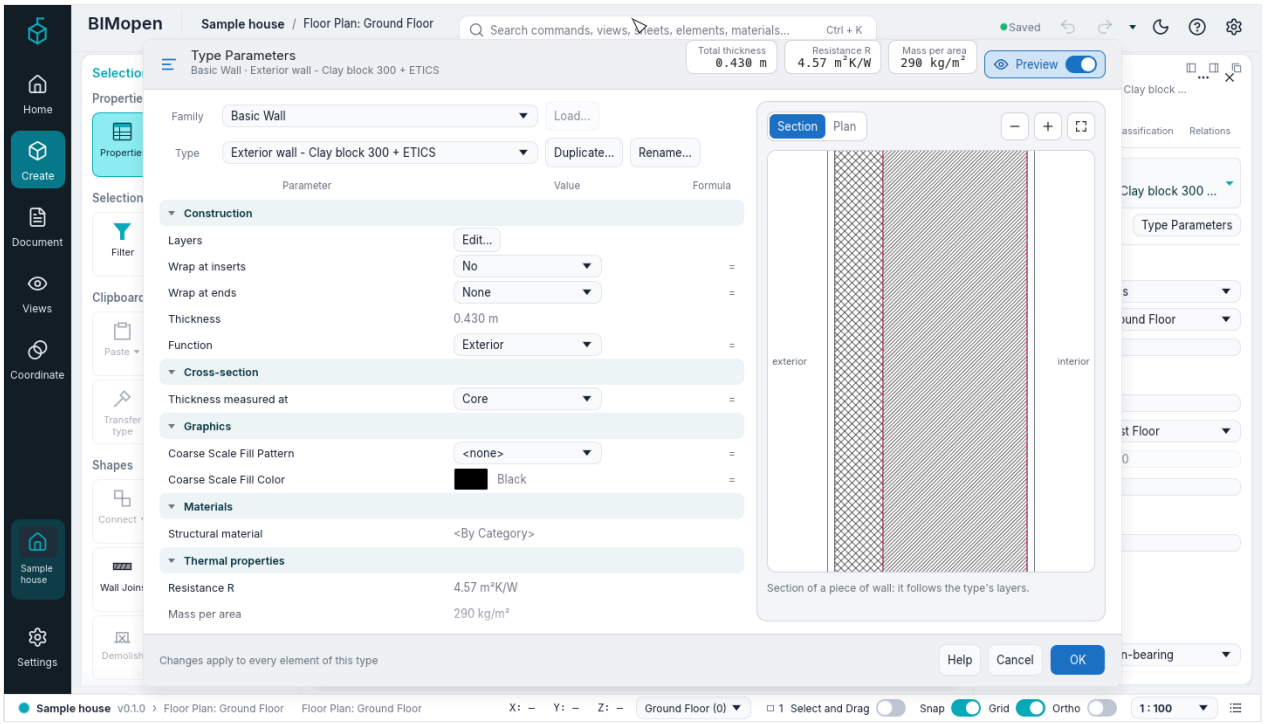
Disallowing a join

To keep an end of a wall from joining (an expansion gap, say), select the wall, right-click the blue dot of that end and choose **Disallow Join**: both walls then end square, apart. A small mark beside the end shows it; click it, or right-click the dot and choose **Allow Join**, to join it again.

Wall types and layers

Type Parameters (the tile of the tool panel while a wall is selected, the button under the type selector of **Properties**, or a type's menu in the Project Browser) shows the parameters of a wall type; a change applies to every wall of that type.

Type parameters



At the top are the family and the type and, for a type with layers, the three figures of **Type layers**: **Total thickness**, **Resistance R** and **Mass per area**. Below, **Family** with **Load...** (a family from the library; greyed for walls and the other system families, which are duplicated rather than loaded) and **Type** with the other types of the family, **Duplicate...** (a copy, which becomes the type of the selected walls; its name does not follow its layers) and **Rename...**. The parameters follow in a table **Parameter · Value · Formula**, in groups that open and close with a click on their title:

- **Construction**: **Layers** with **Edit...**, which opens **Type layers**; **Wrap at inserts** (**No**, **Exterior**, **Interior**, **Both**); **Wrap at ends** (**None**, **Exterior**, **Interior**); **Thickness** (the sum of the layers in metres, read-only); **Function** (**Exterior**, **Interior**, **Foundation**, **Retaining**, **Cavity**, **Core-shaft**).
- **Cross-section**: **Thickness measured at** (**Core**, **Exterior face**, **Interior face**).
- **Graphics**: **Coarse Scale Fill Pattern** and **Coarse Scale Fill Color** (a swatch that opens the colour picker), how cut walls of the type are filled at **Coarse** detail.
- **Materials**: **Structural material**, read-only: the material of the layer marked structural in **Type layers**, «» when none.

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- **Thermal properties**: **Resistance R** and **Mass per area**, read-only and computed from the layers (they show «—» while a material lacks its thermal values); **Solar absorptance** (0 to 1, 0.7 at first) and **Roughness** (from **1 · Very smooth** to **5 · Very rough**).
- **Identification**: **Keynote** (typed, or **…** opens the project's **Keynotes**: codes and descriptions to add, change or delete; a row picked and **OK** give its code to the type), **Assembly Code**, **Assembly Description** (read-only), **Model**, **Manufacturer**, **Type Comments**, **URL**, **Description**, **Type Mark** (what a wall tag shows), **Fire resistance** and **Cost**.

The **Formula** column marks with «=» the values that can be changed. Every change applies at once and Ctrl+Z undoes it, one step each; **Cancel** undoes all the changes made since the dialog opened, **OK** keeps them, **Help** opens this page.

On the right the preview draws the type as **Type layers** does: **Section** (the wall cut through, its layers side by side) or **Plan** (the end of the wall), with the buttons to zoom out, zoom in and fit; floors and roofs show their layers stacked; doors, windows, components and curtain walls a picture of one of them. The **Preview** switch in the header, beside the figures (an eye and an on/off switch), hides it and the table widens into the freed space, while the dialog keeps its size and place; switched on again it shows the preview, and the choice is kept. Floors, roofs and the other families use the same dialog, with their own parameters.

Type layers

Layers ▸ Edit… opens **Type layers**, the same dialog for walls, floors, roofs and ceilings. At the top are the family and the type, and three figures updated at every change: **Total thickness** (in metres), **Resistance R** (m^2K/W: the sum of thickness / thermal conductivity of the layers, plus 0.17 for the two surfaces) and **Mass per area** (kg/m^2: the sum of thickness × density). R and mass show when every material has its **Thermal conductivity** and **Density** (the template's materials have them, others are completed in **Materials**); otherwise they show «—».

On the left is the list of the layers, from the **EXTERIOR SIDE** at the top to the **INTERIOR SIDE** at the bottom. Each layer is a card with its number, a sample of its material (colour and cut pattern), its **Function** (**Structure**, **Substrate**, **Thermal insulation**, **Air space**, **Finish 1**, **Finish 2**, **Membrane**), its **Material** (a click opens **Materials** to choose it), its **Thickness** in millimetres and three toggles:

- **Wraps**: the layer may wrap at openings and ends (never a layer of a wall's core);
- **Structural material**: the layer's material is the type's structural material (only one, in the core);
- **Variable**: the layer's thickness may vary (only one, in floors and roofs).

Two dashed lines, **Core boundary · above** and **Core boundary · below**, delimit the core. A click on a card selects it; **Insert** adds a layer under the selected one; **Delete**, **Up** and **Down** act on the selected layer and are greyed without a selection. In a wall a core boundary is also selected with a click and moved with **Up** and **Down**. A membrane layer has no thickness, and every other layer is at least 4 mm thick: other values are refused, the dialog stays open and the status line at the bottom says why.

For walls, on the row of the buttons:

- **Wrap at inserts** (**No**, **Exterior**, **Interior**, **Both**): the wrapping layers outside the core on that side turn into the jambs of doors and windows; the layers behind them are pulled back;
- **at ends** (**None**, **Exterior**, **Interior**): the same at the free ends of the walls.

On the right the preview follows every change. **Section** shows the wall cut through, its layers side by side and as tall as **Sample height** (6000 mm at first), or the floor, roof or ceiling with its layers stacked; **Plan** (walls only) shows the end of the wall with the layers that wrap round it. The selected layer is outlined and the core lies between two dashed lines. The buttons above the drawing zoom it out, zoom it in and fit it; the wheel, a pinch and a drag zoom and move it too, and a click on a layer selects it in the list. Below, for walls, **Vertical structure (in section)** gathers **Modify**, **Merge Regions**, **Profiles**, **Assign Layers**, **Split Region** and **Reveals**, not available yet (greyed).

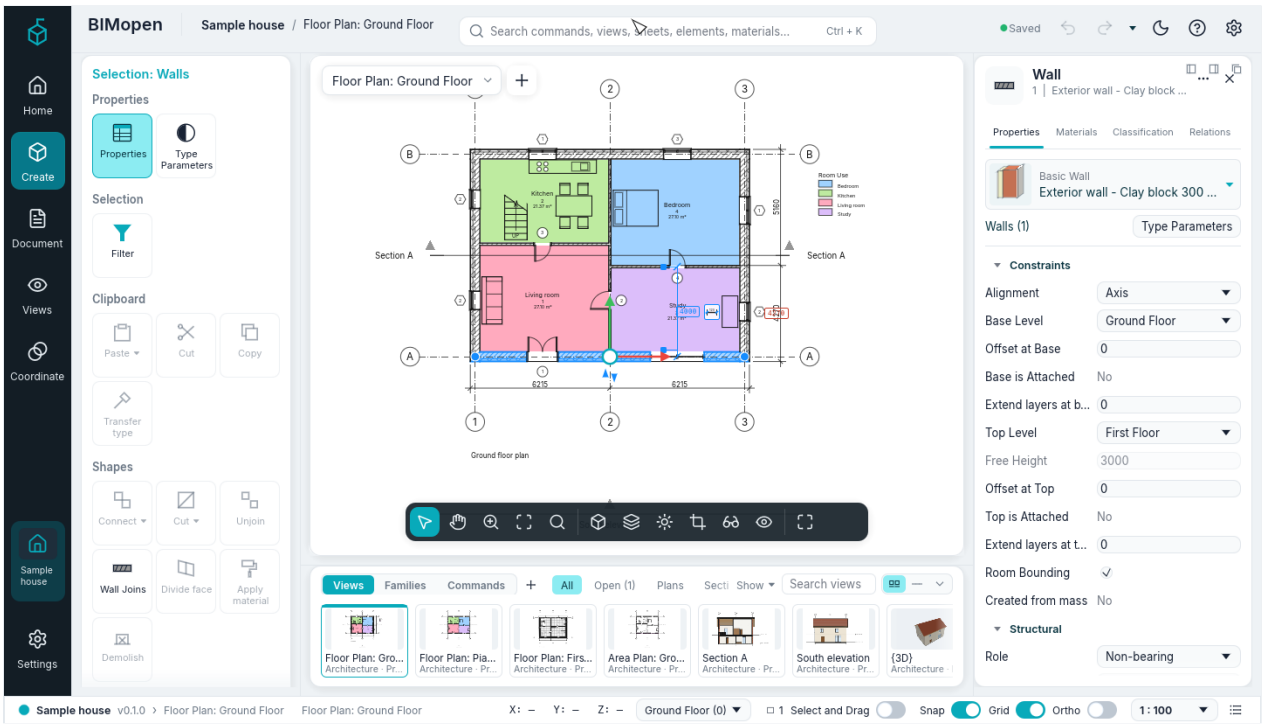
The preview draws each layer exactly as plans and sections draw it at **Medium** detail: the material's cut pattern in its colour, on the view's background in **Lines** or over the material's shading colour in the shaded styles, following the active view's graphic style and the theme. A material changed in **Materials** changes the preview, the plans, the sections and the 3D view alike; the overrides of a view (**Visibility/Graphics**, filters, an element's overrides) change that view only, never the preview.

The **Preview** switch in the header, beside the figures, hides the preview and the list takes the whole width, while the dialog keeps its size and place; switched on again it shows the preview, and the choice is kept. At the bottom, **OK** applies all the changes in one step, which Ctrl+Z undoes; **Cancel** closes without changing anything; **Help** opens this page. In a narrow window the preview stands above the list.

Wrapping shows in every view: in plan, in the 3D view, in sections and in elevations the ticked layers line the jambs, the head and the sill of doors and windows, the sides of the holes of an edited boundary (see Editing a wall's boundary) and the free ends. A door or window type overrides the wall's wrapping with its **Wall Closure**.

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Wall properties



With a wall selected, **Properties** shows at the top the type selector: the picture of the type, its family and its name. A click opens the list of the types grouped by family (**Basic Wall**, **Curtain Wall**), each with a small picture — a wall's layers side by side in their materials' colours, a curtain wall's grid — and the current type highlighted; typing in **Search a type...** keeps the types whose name contains the text, the arrows and Enter pick one. Under it **Walls (1)** says what is shown and **Type Parameters** opens the dialog described above.

The parameters come in groups that open and close with a click on their title:

- **Constraints**: **Alignment** (**Axis**, **Core Centerline**, **Finish Face: Exterior**, **Finish Face: Interior**, **Core Face: Exterior**, **Core Face: Interior**), **Base Level**, **Offset at Base**, **Base is Attached** (read-only), **Extend layers at base**, **Top Level** (a level, or **Unconnected**), **Free Height** (typed only with **Unconnected**, greyed otherwise; at the command line it is refused, saying why, while the wall has a top level), **Offset at Top** (only with a level, greyed otherwise), **Top is Attached** (read-only), **Extend layers at top**, **Room Bounding** (untick it and the rooms around the wall ignore it) and **Created from mass** (read-only).
- **Structural**: **Role** (**Non-bearing**, **Bearing**, **Shear**, **Combined**). A wall other than **Non-bearing** is structural, and its **Rebar cover at top**, **Rebar cover, interior side** and **Rebar cover, exterior side** can be chosen (25, 35 or 40 mm); they are greyed otherwise. **Rebar volume** is read-only.
- **Dimensions**: **Length**, **Thickness**, **Area**, **Volume** and the ends of the wall's line.
- **Identification**: **Comments**, **Mark** and **Classify as** (**Exterior wall**, **Interior wall**, **Partition**, **Foundation wall**; until one is chosen, after the type's function).
- **Phasing**: **Phase created** (**Existing**, **New construction**) and **Phase demolished** (**None**, **Existing**, **New construction**). They are kept with the wall; views do not draw phases yet.

Extend layers at base and **at top** carry the layers outside the core below the base and above the top by that distance (a negative one shortens them), in the 3D view, in sections and in elevations; the core stays between the base and the top. At the top they extend where the top is flat, not where the wall is attached to a roof or a sloped plane.

With several walls selected the filter says **Walls (n)**, the type selector shows their type (or «varies») and each row shows the value the walls share, or «varies» where they differ; a change goes to all of them in one step. **Help on properties** at the bottom opens this page; **Apply** confirms the value being typed. Every change is one step that Ctrl+Z undoes.

Placing doors and windows

Create ▸ Build ▸ Door (`DR`) or **Window** (`WN`), in a plan, a section, an elevation or the 3D view. Pick the type, then move over a wall: the opening follows the cursor along it, and temporary dimensions measure from its centre line to the centre lines of the nearest walls across the host on each side. A door swings to the side of the wall the cursor is on; the **space bar** puts its hinge on the other jamb. Click to place it: the wall is cut in every view. A window takes its type's **Default Sill Height**.

Doors and windows get a **Mark** that counts up by category (1, 2, 3...), from the highest.

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Tags

Tag on Placement, on the options strip of the door and window tools, tags each opening as it is placed. The options strip then offers **Horizontal** or **Vertical**, and **Leader** with its length. A door's tag shows its **Mark**; a window's tag shows its type's **Type Mark**. **Document ▸ Tag All** tags the openings not tagged yet.

Flipping a door or window

Select a door or window in a plan: arrows across the wall, on the side it opens to, turn its swing over; arrows beside it, along the wall, put its hinge on the other jamb. The context menu does the same: **Flip Hand** (the hinge) and **Flip Facing** (the swing).

Moving a door or window to another wall

Select it, **Selection: Doors ▸ Host ▸ Pick New Host** (or **Selection: Windows**), then click another wall: the opening moves there, where you clicked, with its type and sill. It stays selected.

Dragging a door or window keeps it in a wall: the pointer is projected onto the wall's alignment line, so the opening slides along the wall, between its ends, with the temporary dimensions to the nearest walls, in a plan, a section, an elevation or the 3D view; there a click on a window's glass picks the window, not what is seen through it. Dropped over another wall, it moves into that wall; released over empty space, it stays at the last place on its own. The arrow keys move it along its wall the same way.

Door and window properties

Instance: **Level**, **Sill Height**, **Width** and **Height** (of the type), **Head Height**, **Comments**, **Mark**. Changing **Sill Height** or **Head Height** moves the opening up or down; its size stays.

Type: **Wall Closure** (**By host**, or a wrapping of its own), **Function** (**Interior** or **Exterior**), **Door Material** (**Glass Material** for windows) and **Frame Material**, **Trim Width**, **Thickness** (of a door's leaf, or the depth of a window's frame), **Default Sill Height** and **Window Inset** (windows: how far the frame sits from the exterior face), **Trim Projection Ext** and **Trim Projection Int** (doors), **Leaves** and **Type Mark**. A door cannot become a window by changing its type.

How walls, doors and windows are drawn

The **Detail** of the plan decides:

	Coarse	Medium	Fine
Walls	One fill (or the type's coarse pattern), joined walls merged	Each layer with its material's cut pattern	As **Medium**
Windows	The frame as one outline, the glass along it, the wall below seen as lines	The frame cut at each jamb and at the mullion of a two-leaf window, at its inset; one line of glass; the exterior sill and the interior sill board	As **Medium**, with double glass
Doors	The leaf as a line and its swing	The frame at each jamb, the leaf with its thickness, its swing	As **Medium**, with the trims out of the faces

In every case the layers end at the jambs of the opening, closed by the cut line, or wrap into them as the types ask. In sections and in 3D, doors and windows show their frame, leaves, glass and sills, in the type's materials.

Curtain walls

A curtain wall is a wall of a curtain wall type, drawn with the **Wall** tool: pick **Curtain Wall** (one glass panel the length of the wall), **Exterior Glazing** (grid lines by the type) or **Storefront** (grid lines and mullions) in the type selector, then draw as for any wall, in plan or in 3D, with **Continue** and the **Draw** shapes. A basic wall changes into a curtain wall by picking a curtain wall type in the type selector; the doors and windows it hosts are deleted, because a curtain wall does not host them, and a message says so (undo brings them back).

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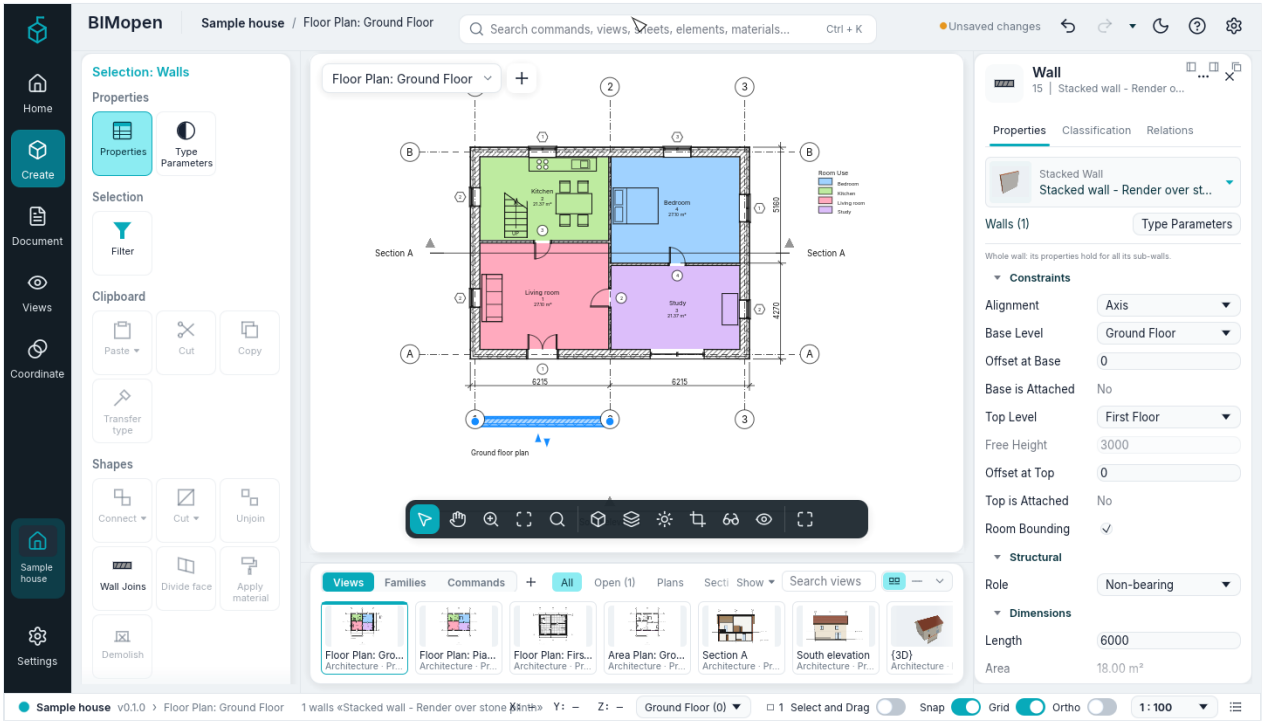
In plan each glass panel is drawn with its thickness and the mullions are cut. A type with **Automatically Embed** ticked (the **Storefront**) drawn along a wall cuts its own hole in it, in plan, sections, 3D and IFC. The wall stays one wall: beside and above the hole it goes on with its layers, its net area leaves the hole out, and moving or resizing the curtain wall takes the hole with it.

- The type's **Vertical Layout** and horizontal layout (**Fixed distance**, **Fixed number**, **Maximum spacing**) with their spacing, and the instance's **Number of Vertical Grids** and **Vertical Justification**, place the grid lines by themselves.
- **Create ▸ Build ▸ Curtain Grid**: click near the top or bottom edge for a vertical line, near a side for a horizontal one, in an elevation, a section, a plan or the 3D view (on the face under the cursor); it snaps to the middle and the thirds of the panel.
- **Create ▸ Build ▸ Mullion**: **Grid Line**, **Grid Line Segment** or **All Grid Lines**, with the mullion type picked in the type selector; click the grid line in an elevation, a section or the 3D view.
- Select the curtain wall, then **Panel** in its contextual card: pick a panel type (a **Curtain Wall Door**, a glazed or solid panel, or none) and click the panel to change, in an elevation, a section or the 3D view.

Panels and mullions have their own schedules, and the IFC export writes the curtain wall with its plates and members.

- **Curtain Grid ▸ Placement**: **All Segments** (the whole line), **One Segment** (only across the panel clicked) or **All Except Picked** (click to place the line, click the segments to leave out, then click away from the line).
- With a curtain wall selected, **Add/Remove Segments**: click a segment of a grid line to take it off (the two panels on either side become one, and its mullion goes) or to put it back. Panels stay rectangular: a segment that would make an L-shaped panel is refused.
- Hover over a curtain wall and press TAB: the panel under the cursor is outlined; click to select it alone. The element panel shows its **Width** and **Height**, and the type selector changes its type.

Stacked walls



A **stacked wall** is made of sub-walls one above the other: a rendered wall over a stone plinth, for instance. Each sub-wall is a basic wall type, with its own layers and materials. The template has one, **Stacked wall - Render over stone plinth** (the exterior wall over a 0.9 m stone plinth): pick it in the type selector of **Wall** and draw it as any wall.

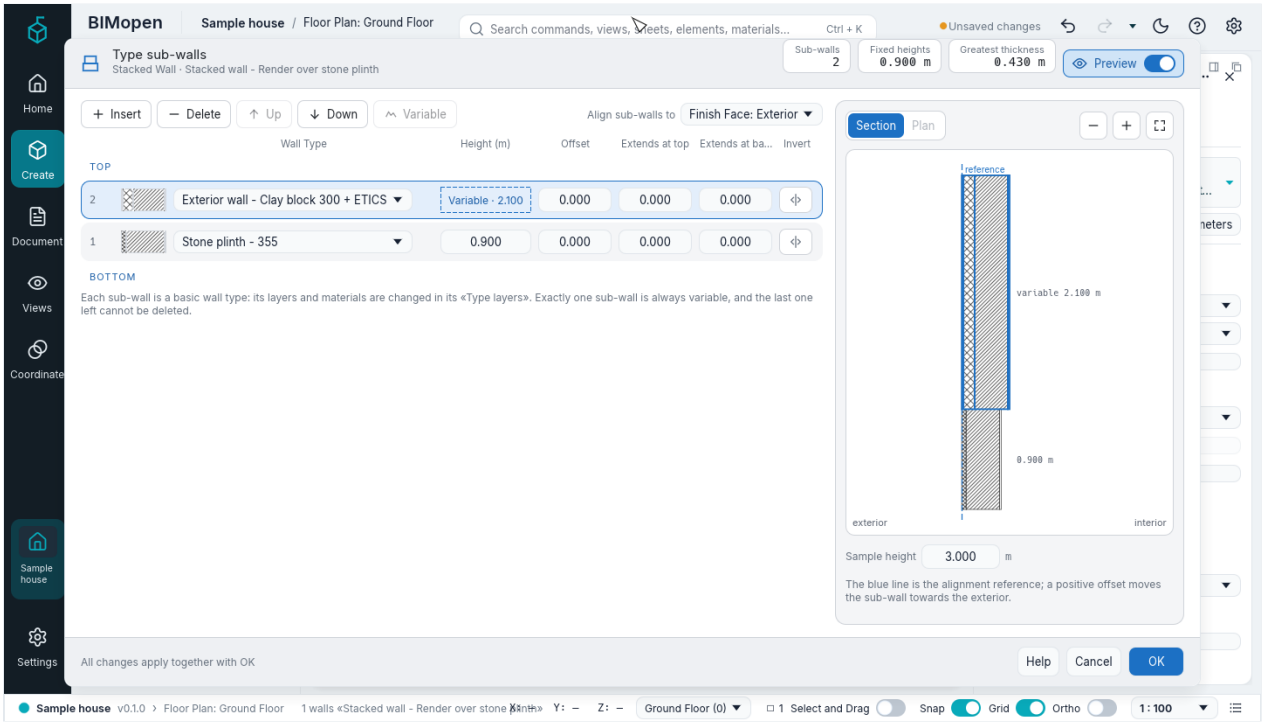
The stacked wall holds the line, the levels and the height; its sub-walls follow them. In plans, sections, elevations and the 3D view each sub-wall is drawn with its own type's layers, in the colours of their materials. The sub-walls join the walls they meet, each at its own height: at a corner of two stacked walls plinth meets plinth and the wall above meets the wall above, a basic wall meets each sub-wall in turn, and two stacks of different heights or thicknesses meet over each height they share; every layer closes round the corner, in plan and in 3D. The **Wall Schedule** lists the sub-walls, one row each. A door or window placed on a stacked wall cuts every sub-wall it passes through, its sill measured from the stacked wall's base.

A click selects the whole wall; **Tab** goes on to each sub-wall, from the bottom up (the status bar says «Tab · next element»). The Properties of the whole wall hold for all its sub-walls; its **Alignment** is the plane it is drawn on, as for any wall: the faces of the band its sub-walls fill, or the core of the variable sub-wall. Giving a basic wall a stacked type, or a stacked wall a basic one,

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keeps it on its alignment. Of a sub-wall only **Room Bounding** and **Role** change: the rest comes from the stacked wall and its type, and its type is changed in the stacked type. **Break up into single walls**, in the context menu and with the wall selected, makes each sub-wall an independent basic wall where it stands; they do not go back together (Ctrl+Z undoes it). The boundary of a stacked wall or of a sub-wall cannot be edited yet.

The sub-walls of the type



Type Parameters of a stacked wall type shows **Sub-walls** with **Edit...**, which opens **Type sub-walls**. At the top are the type and three figures: how many sub-walls, the sum of the fixed heights and the greatest thickness. The table lists the sub-walls from **TOP** to **BOTTOM**, numbered from the bottom: **Wall Type** (a basic wall type), **Height** in metres, **Offset** from the reference line (positive towards the exterior), **Extends at top** and **Extends at base** (how far the layers outside the core go beyond the sub-wall; negative, they stop short) and **Flip**. Exactly one sub-wall is **Variable**: its height is what remains of the wall's height.

Insert adds a sub-wall below the selected one, **Delete** removes it (never the last one; deleting the variable one makes the one above variable), **Up** and **Down** move it, **Variable** makes it the variable one. A positive **Extends at base** hooks the top of the sub-wall below («Hooked»): its layers stop where those of the one above begin. **Align sub-walls to** chooses the plane of each sub-wall that lies on the reference line, among those of **Alignment** (**Axis**, **Finish Face: Exterior**...).

On the right is the section of the type, as tall as **Sample height**, the reference line in blue, the exterior on the left; a click on a sub-wall selects its row. The slim **Preview** strip hides and shows it, and the dialog keeps its size and place. **OK** applies the changes in one step (Ctrl+Z undoes it); **Cancel** leaves the type as it was.

Levels

Create ▸ References ▸ Level draws a level in a section or an elevation with two clicks, at the height of the first one; while drawing, a temporary dimension gives the distance to the nearest level. The options strip shows:

- **Make Plan View**: on, the new level gets its floor plan; off, it is a reference level with no plan, which walls and other elements can still take as their base or top;
- **Plan View Types...**: the kinds of plan the level gets (**Floor Plan**; ceiling plans are not available yet);
- **Offset**: how far above the clicks the level goes.

Work planes

Every view draws and places on a work plane. A plan's work plane is its level, a 3D view's the lowest level; a section or an elevation has none until one is set. The plane is kept with the view.

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Setting the work plane

Create ▸ Work Plane ▸ Define opens the **Work Plane** dialog. **Current Work Plane** shows the plane the view uses now. Under **Specify a new Work Plane**:

- **Name**: pick a level, a grid or a named reference plane in the list (picking one selects **Name**), then **OK**;
- **Pick a plane**: after **OK**, click a wall (its face on the side of the cursor in a plan, the face turned to you in a section or a 3D view), a grid, a reference plane or, in a section, a level;
- **Pick a line and use the work plane it was sketched in**: after **OK**, click a wall or a room separator: its level becomes the work plane.

A plane that does not lie in the view, such as a grid picked in a plan or a level in a section, is not set there: **Go to View** lists the views it lies in (the sections and elevations facing a vertical plane, the plans of a horizontal one) and the 3D views, and **Open View** opens the chosen one with that work plane. **Undo** takes a new work plane back.

Showing the work plane

Create ▸ Work Plane ▸ Display draws the work plane's grid in the view, over the model, when the plane lies in the view (in a 3D view always); **Show** again hides it. In a section with no work plane it opens the **Work Plane** dialog first. Click the border of the grid to select it: the options strip shows **Spacing**, the distance between its lines (1 m to start with). While the grid is shown, the points of walls and other lines drawn in the plan snap to the crossings of its lines.

In a 3D view, a click where there is no element under the cursor lands on the view's work plane when one has been set for it: a component placed east of the house with the work plane set to the first floor stands on the first floor.

Reference planes

Create ▸ Work Plane ▸ Reference Plane (or type `RP`) draws a reference plane with two clicks, like a grid. Drawn in a plan it is the vertical plane through the line, shown as a green dashed line in every plan and in the sections and elevations that cross it. Drawn in a section or an elevation it is the plane through the line across the view's depth (a horizontal line makes a horizontal plane), shown in the views facing the same way. Select one to give it a **Name** in the element panel: a named reference plane is listed in the **Work Plane** dialog and its name is written at the start of its line. Reference planes are snapped to, moved, copied, rotated, mirrored and deleted like grids.

At the command line

The same operations, typed:

```
attach                      attach the selected walls to the roof over them
attach 412                  attach their top to element 412 (a roof or a floor)
attach base 398             attach their base to the floor 398
detach all                  detach them from everything
walljoin 0,0 {"kind":"Giunto d'angolo","cleanup":"Non calcolare unioni"}
walljoin 57 1 disallow      wall 57 does not join at its end
rehost 120 12,4             move door or window 120 into the wall at 12,4
door {"point":[3,0],"tag":{"vertical":true,"leader":true,"length":1}}
refplane 3,-2 3,11 Entrance axis   a reference plane in the open plan, named
workplane Entrance axis     the open view's work plane, by name
showworkplane on 0.5        show the work plane's grid, 0.5 m apart
```

Not available yet

Curved walls (**Arc**, **Circle**, **Radius**), **Pick Faces**, structural walls, stacked walls, wall sweeps and reveals, vertically compound layers (regions), **Join Geometry**, phases, and skylights in roofs.

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Floors, roofs, stairs and components

Floors, roofs, stairs and railings are drawn in **sketch mode**: their lines are sketched first, and the element is made only when the sketch is finished. Components (furniture and plumbing fixtures) are placed with a click. Curtain walls are described in Walls, doors and windows.

Sketch mode

Create ▸ Build ▸ Flooring, **Roof**, **Stairs** and **Railing** open sketch mode. The contextual card is named after what is sketched (**Edit | Create Floor Boundary**, **Create Roof Footprint**, **Create Stairs Sketch**, **Create Railing Path**) and has:

- the **Mode** group: **Finish Edit Mode** makes the element; **Cancel Edit Mode** drops the sketch and leaves the model as it was;
- the **Draw** group, with the drawing tools of the element;
- for stairs and railings, the **Tools** group.

Nothing is made while sketching: the lines are magenta, and the model changes only with **Finish Edit Mode**. The whole element is then one step of the undo history: one `Ctrl+Z` takes it away.

Drawing lines

Tool	How it draws
Line	Click point after point: each click ends a line and starts the next. The length and angle of the line under the cursor are shown beside it; `Shift` keeps it horizontal or vertical. With **Continue** off on the options strip, each line is drawn on its own. Clicking the start of the chain again closes the loop Type a length and press `Enter`: the line ends that far along the cursor's direction (a stair's run too).
Rectangle	Two opposite corners: four lines
Pick Walls	Click a wall: its line is added to the sketch (see below). Press `Tab` over a wall, then click, to pick the whole chain of walls joined end to end
Pick Lines	Move over a wall, a grid, a line or the edge of a floor or roof: the line under the cursor is highlighted and named (a wall's faces, each layer boundary, its two **Core Boundary** lines and its axis); `Tab` goes to the next one there. Click to add a sketch line on it (set off by **Offset** towards the cursor)
Slope Arrow	Click the tail, then the head of the arrow (floors and roofs)

Offset on the options strip moves the drawn lines (and the lines of picked walls) by that distance.

The arcs of the **Draw** group (and a circle, as two halves) are one sketch line each: picked, measured and dragged as arcs, kept by the floor, roof, stair, wall boundary or filled region, and given back as arcs by **Edit Boundary** or **Edit Sketch**. In every view a sketch point snaps to the ends and middles of the sketch's lines and arcs, to arcs' centres, where lines cross and onto a line or an arc; from the point being drawn from, to the foot of a perpendicular and to the point where a line touches an arc. The marker and its name show where the point snaps; `S C` and `S P` ask for the centre or the perpendicular for the next point.

Floors and roofs need closed loops. A loop drawn inside the outer one is an **opening**: the floor or the roof has a hole there. When **Finish Edit Mode** finds lines that do not close or that cross, a message says so and the sketch stays open to be fixed (**Continue**).

Picking and changing sketch lines

`Esc` ends the chain being drawn; a second `Esc` puts the drawing tool down (so does **Modify**). While the sketch is open its lines are elements of the view, and **Modify** works on them as on other lines; nothing else can be picked until the sketch is finished:

- the line under the cursor is highlighted; a click selects it, `Ctrl` adds lines, `Shift` takes them away, a window selects several, `Tab` the chain of joined lines;
- the end of a line selected on its own drags alone: the line next to it stays where it is and the join is broken; with a chain selected (`Tab`), the ends where its lines meet drag together; a dragged line takes the lines joined to it along;
- **Move**, **Copy**, **Rotate**, **Mirror on Line**, **Repeat in Series**, **Align**, **Offset**, **Shorten/Lengthen** and **Split Element** work on them, and the sketch stays open;
- one selected line shows temporary dimensions to the nearest parallel line on each side (another sketch line, the wall's original outline, a wall, a grid, a level): click the value and type a distance to move the line; the symbol beside it makes a permanent dimension, kept while the sketch is open;

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- its properties are in the element panel: **Defines Slope** and **Slope**, for a roof line picked from a wall its **Overhang**, and its **Length**.

Align takes as its reference the line under the cursor, and `Tab` goes to the next one there, so a sketch line can be aligned to a wall's layer or core boundary. While drawing or dragging an end, the points snap to the corners of the walls' faces, layers and core boundaries, then onto those lines. `Del` deletes the selected lines. `Ctrl+Z` while sketching takes back the last change to the sketch, not to the model. All this works in plans, in sections and elevations (a wall's boundary) and, for selecting, dragging ends and deleting, in the 3D view.

Editing the sketch of an element

Select a floor, a roof, a stair or a railing: the **Mode** group of its contextual card has **Edit Boundary** (floors), **Edit Footprint** (roofs), **Edit Sketch** (stairs) or **Edit Path** (railings). Sketch mode opens with the element's lines, openings, slopes and arrow; **Finish Edit Mode** changes the same element, **Cancel Edit Mode** leaves it unchanged.

The sketch opens with **Modify** active: its lines can be selected and changed at once.

In a plan the sketch is drawn on the level; in the 3D view it lies on the element's own plane (the floor's top, the roof's base) and the 3D view stays open.

A new floor can be sketched in the 3D view too: **Flooring** started there puts the sketch on the plane of the level, the line being drawn follows the cursor with its length, and **Finish Edit Mode** makes the floor without leaving the 3D view. In the 3D view the options strip's **Level** chooses the work plane: the level whose plane the sketch lies on and the floor goes on. **Pick Walls** takes the wall whose face is under the cursor. Roofs by footprint, stairs and railings are sketched in a plan.

Floors

1. Click **Create ▸ Build ▸ Flooring**. Sketch mode starts with **Pick Walls**.
2. Click the walls around the floor, or press `Tab` over one of them and click to pick them all. Each line goes on the **interior face** of its wall; the lines of walls that meet are trimmed or extended to meet at the corner.
3. Or draw the boundary with **Line** or **Rectangle**; draw a loop inside it for an opening.
4. Click **Finish Edit Mode**.

On the options strip, **Extend into wall (to core)** puts picked lines into the wall up to the face of its core, and **Offset** moves them (a negative offset moves them away from the walls). The type of the new floor is the one in the type selector at the top of **Properties**.

Sloped floors

A floor slopes in one direction, set in one of two ways while sketching:

- a **Slope Arrow**: its properties are **Specify** (**Height at Tail**, from the heights of its two ends, or **Slope**), **Height Offset at Tail**, **Height Offset at Head** and **Slope**. The heights are measured from the top of the floor;
- one boundary line with **Defines Slope** ticked and its **Slope** in degrees: the floor rises from that line, inwards.

The floor is sloped in 3D and in sections. Its **Slope Angle** in **Properties** changes the slope afterwards.

Floors over walls

When a floor crosses walls (an upper floor resting on the walls below, for instance), its slab is cut out of them: in 3D and in sections the wall stops under the floor and goes on above it, and walls go on through the floor's openings.

Floor properties

Property	What it is
Level, **Height Offset From Level**	The level and the height of the top face from it
Slope Angle	The slope of a sloped floor, in degrees
Thickness	From the type's layers (read-only)
Perimeter, **Area**, **Volume**	Without the openings (read-only)
Comments, **Mark**	Free text

Type Parameters has the floor's **Structure** (its layers) and its default thickness.

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Roofs

1. Click **Create ▸ Build ▸ Roof**. Sketch mode starts with **Pick Walls**; the options strip has **Defines Slope**, **Overhang** and **Extend into wall (to core)**.
2. Pick the walls (`Tab` for the whole chain). Each line goes on the **exterior face** of its wall, moved out by the overhang (from the face of the core with **Extend into wall (to core)**).
3. Or draw the footprint with **Line** or **Rectangle**; a loop inside is an opening.
4. Click **Finish Edit Mode**.

Every line with **Defines Slope** rises inwards at its **Slope** (30° by default); a small triangle with the slope is drawn beside it. The shape comes from those lines:

- all the lines define the slope: a **hip** roof;
- two opposite lines do not: a **gable** roof, the walls under the gables rise to the ridge when attached;
- no line defines the slope: a **flat** roof, or a roof sloping along a **Slope Arrow** (then no line may define a slope).

To change one line, press `Esc` twice, click the line and change **Defines Roof Slope**, **Slope** or (lines picked from walls) **Overhang** in **Properties**.

Select walls, click **Modify Wall ▸ Link Top/Base**, then the roof: they rise to its underside, gables included (Walls, doors and windows tells more).

Roof properties

Property	What it is
Base Level, **Base Offset From Level**	Where the eaves are: by default on top of the walls of the level
Slope	The slope of every line that defines one (or of the arrow)
Thickness, **Maximum Ridge Height**, **Area**, **Volume**	Read-only
Comments, **Mark**	Free text

In plan a roof shows its outline, ridges and hips; in 3D its slopes, overhangs and layers.

Stairs

1. Click **Create ▸ Vertical Links ▸ Stairs**. Sketch mode starts with **Run**.
2. Click where the first riser is, then where the run ends: the run snaps to whole treads, and the status bar counts the risers made and left (for example `9 risers created, 8 remaining`).
3. To turn, click again away from the end of the first run and draw the next one: a landing joins them. At any angle the landing is bounded by the two runs' risers, its outer corner where their outer sides meet.
4. When every riser is placed, click **Finish Edit Mode**.

Dashed guides show where the next run can start: the last run's extension beyond its end and the lines of its first and last risers. A click near them snaps onto them. In a 3D view the stair is drawn the same way, on the plane of the active level.

A curved run (a spiral stair): with **Run** active, click **Center-ends arc**, then the centre, the start of the run line (its distance from the centre is the radius of the walking line) and the end, going round past half a turn if needed, but under one turn (further, the run stops short and the status bar warns that the runs would overlap). The risers radiate from the centre a tread apart on the run line; riser lines drawn radiating from a centre make a curved run too. A second curved run round the same centre and on the same run line (a click near them takes them) is joined to the first by a landing along the arc.

A stair can also be drawn by its lines. In its sketch, click **Draw ▸ Boundary** and draw the left and right edges (green), then **Riser** and draw the riser lines (black) between them, with **Line** or **Pick Lines**. Parallel, evenly spaced risers make a run, as wide as the risers; a landing joins two runs out to the boundary lines. Each riser is as high as the storey divided by the desired number of risers; with fewer riser lines the stair stops short of the top (**Properties** shows the actual number). **Edit Sketch** gives the same lines back. Each run drawn with **Run** is at once boundary and riser lines like these: they can be moved or deleted like the others. A landing follows its boundary lines: split its far edge with **Split Element** and drag a piece to notch it, or draw it wider than the runs.

The **Width** on the options strip is that of the new stair (by default the type's). **Tools ▸ Railing Type** chooses the railings of the new stair: a railing type, put on both sides, or **None**.

The number of risers comes from the height between **Base Level** and **Top Level** (with their offsets) and the type's **Maximum Riser Height**; changing a level or an offset in **Properties** changes the risers. **Edit Sketch** reopens its boundary and riser lines:

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change or delete them (click, `Del`) and draw runs again; the railings of the stair follow. **Multistory Top Level** repeats the stair, with its railings, storey by storey up to that level; a storey whose height is not the stair's gets no copy, and a warning names it. With the stair selected, the blue flip arrows beside its first riser make it go up from the other end; **Up Label**, **Up Arrow**, **Down Label** and **Down Arrow** in **Properties** show or hide the plan symbol's parts. The square grip under the **UP** text drags it elsewhere.

In plan the stair shows its treads up to the cut line, the break line, the arrow and the **UP**/**DN** text; in 3D its treads and risers, or a monolithic slab, its landings and railings.

How a stair is built

Type Parameters of the stair decide its construction:

- **Monolithic Stairs** ticked: each run is a concrete slab cut to the steps; its underside runs parallel to the slope, **Structural Depth** below the inner corners of the steps, and the landings are as thick: the sloping underside of a run meets the landing's underside along a line, without a step; each run ends with its last tread: the landing, flush with it, or the floor above makes the last riser, so steps and slab show as one solid with no line across the risers.
- otherwise treads (**Tread Thickness**) with risers (**Riser Type Straight**, **Riser Thickness**) or open (**None**), on stringers: **Right Stringer** and **Left Stringer Closed** (a board the treads are housed in), **Open** (a board cut to the steps, the treads on top) or **None**, and **Middle Stringers** under the treads, with their **Stringer Thickness** and **Stringer Height**.
- **Begin with Riser** and **End with Riser** cleared: the board of the stair's first or last riser is not built.
- **Riser Type Slanted** leans each riser board, its top under the front of the tread above; **Riser to Tread Connection** runs each tread under the riser behind it (**Extend Tread Under Riser**) or the riser down behind the tread (**Extend Riser Behind Tread**).
- **Nosing Length** sets how far a tread runs past its riser; **Extend Below Base** moves the foot of the stringers (negative: below the base level); **Tread Material**, **Riser Material**, **Stringer Material** and **Monolithic Material** colour the parts; **Break Symbol in Plan** and **Text Size** control the plan symbol.

The landing stops behind the last riser of the run that reaches it, following its back face; in a monolithic stair it starts at the riser's face. In a monolithic stair whose top floor has its edge on the last riser, the run's underside goes on under that floor until it meets the floor's underside, as under a landing.

A section across the stair shows it cut with its slab. **Volume** in **Properties** and in the schedule of stairs is the volume of its solids.

The type's **Text Font**, **Nosing Profile** (**Default** or **Rounded**) and **Apply Nosing Profile** (front only, front and left, or front, left and right: past a side without a closed stringer), **Open Stringer Offset**, **Trim Stringers at Top** (**Match Level** stops the side stringers at the top level) and its identification (**Code**, **Model**, **Manufacturer**, **Type Comments**, **URL**, **Description**, **Assembly Code**) are in **Type Parameters**, as is a monolithic stair's **Underside of Stringer** (**Smooth**, a sloping slab, or **Stepped**: the slab follows the steps). Under **Calculation Rules**, **Use Stair Calculator** with a **Target**, **Maximum** and **Minimum Result** checks 2 risers + 1 tread: a stair outside the range is made, and a warning says so.

Ramps

1. Click **Create ▸ Vertical Links ▸ Ramp**. Sketch mode starts with **Run**.
2. Click where a run starts, then where it ends: it slopes at most the type's **Ramp Max Slope (1/x)** and rises at most its **Maximum Incline Length** (while drawing, a longer run stops there); the length made and left show beside the cursor.
3. Draw the next run away from the end of the first: a landing joins them. **Boundary** and **Riser** draw a ramp by its lines, with a riser line at the start and the end of each run.
4. Click **Finish Edit Mode**.

The type's **Shape** is **Thick** (its **Thickness**) or **Solid** (down to the base), and **Ramp Material** colours it. Its railings, **Edit Sketch** and **Properties** work as for stairs.

With **Arc from Center and End** in the **Draw** group, click the centre, the start of the run line and then its end round the centre: the run curves round it, two lines radiating from the centre at its ends and two boundary arcs at its sides. The type's **Show Up Arrow in All Views** draws the whole up arrow and **UP** in the plan of the top level too; its **Text Font** and identification are as for stairs.

Railings

1. Click **Create ▸ Vertical Links ▸ Railing**.

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2. Draw the path with **Line**: one chain of connected lines (a path in pieces, a closed path or one that goes back over itself is refused at **Finish Edit Mode**).
3. To put the railing on a stair, click **Tools ▸ Pick New Host** and then the stair: the path drawn on it climbs with the stair.
4. Click **Finish Edit Mode**.

Edit Path reopens a railing's path. **Create ▸ Vertical Links ▸ Railing ▸ Place on a host** puts a railing on each side of the stair or ramp you click: on the options bar **Position** chooses **On treads** (on the stair's edge) or **On stringer** (over the stringer, where the stair has one), **Offset from Path** moves it towards the inside of the stair and **Invert** switches between the two positions. The same two values are in the **Properties** of a stair's railing.

The railing's **Type Parameters** hold for every railing of that type: **Railing Height** (that of the top rail), **Baluster Spacing**, **Baluster Offset** (balusters and posts moved sideways) and the identity data; the preview shows the railing on a sample stair, seen from the side. **Rail Structure (Non-Continuous) ▸ Edit…** opens the list of the rails below the top one (**Name**, **Height** above the walking surface, **Offset** sideways, a square or round **Profile**) with **Insert**, **Delete**, **Up** and **Down**. **Baluster Placement ▸ Edit…** opens the **Main pattern**: the baluster or **None**, where it starts and ends (**Host**, **Top rail** or one of the rails) with their offsets, the distance between balusters, **Break pattern at** (at the end of each segment, at angles greater than a value, or never), **Justify**, **Excess length fill** and **Use baluster per tread** with how many on each tread; and the start, corner and end **Posts** (the post or **None**, its offsets, **Space** along the path, **Offset** sideways) with **Corner posts at**. The changes in these two windows apply with **OK** and are undone with a single `Ctrl+Z`. The **Top rail**, **Handrail 1** and **Handrail 2** rows are greyed: not available yet.

The railings of a stair run on over its landings as one rail, at their height above the line of the nosings and above the landings. Where a run leaves a landing one riser higher, the type's **Tangent Joins** (the rail goes straight on) and **Angled Joins** (it turns) decide the join: **Extend Rails to Meet** (tangent joins only) carries the sloping rail on down to the landing, **Add Vertical/Horizontal Segments** rises upright at the first riser, **No Connector** leaves a gap.

The rails are mitred at every joint, where the path turns and where the slope changes: with **Rail Connections Weld** each rail is one piece, with **Trim** its pieces are cut on the joint's vertical plane. A post stands at the start and the end of each railing and at every change of direction in plan and wherever the slope changes, as where a run meets a landing. **Use Landing Height Adjustment** with **Landing Height Adjustment** raises (or lowers) the rail over the landings by that much.

A stair railing's path follows the stair's outline. Every post stands whole on the tread, landing or floor under it: where it would fall on a riser's line it moves by half its section, so no post sinks into the steps or hangs in front of the first one. Between two posts the balusters are evenly spaced, never wider apart than the spacing.

Place on a host works in plans and in the 3D view: the stair under the cursor shows where the two railings go. **On stringer** can be chosen only for a stair that has a stringer: on a monolithic stair it is greyed, on the options bar and in **Properties**. A new railing type has both **Angled Joins** and **Tangent Joins** set to **Add Vertical/Horizontal Segments**, **Weld**, no landing height adjustment, a bottom rail at 0.100 m (30×30), 20×20 balusters from the bottom rail to the top rail every 0.120 m, broken at the end of each segment and fitted to its length, and 50×50 posts standing on the host: where a run meets a landing the post stands on the landing at the riser's line, with the upright piece of the rail, and nothing runs over the treads. Where the rail rises upright at the start of a run, a low rail stays on the landing without entering the first tread. **Edit Joins** (one join at a time) is greyed: not available yet.

Components

1. Click **Create ▸ Build ▸ Component**. The type selector lists the components loaded in the project; a preview follows the cursor, in plan and in the 3D view.
2. Press `Space` to turn the preview by 90°.
3. Click to place the component on the level of the plan.

Mode ▸ Load Family opens the library, by folder (**Furniture**, **Plumbing Fixtures**): choose a family and click **Open** to add its types to the project.

Select a component and click **Host ▸ Pick New Host**, then a floor: the component moves onto the floor and stays on its top (click an empty point to put it back on the level). Its properties are **Level**, **Host**, **Offset from Level**, **Rotation**, **Mark** and **Comments**; **Type Parameters** has its width, depth, height and material.

Schedules and IFC

Floors, roofs, stairs, railings and furniture have their schedules (**Views ▸ Create ▸ Schedules ▸ Schedule/Quantities**), one row per element. **Export ▸ Export IFC4…** writes floors as slabs (with their openings and slopes), roofs, stairs with their flights and landings, railings, furniture and plumbing fixtures. A multistory stair is written once in each storey it is repeated in, with its railings.

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Not yet in BIMopen

Ceilings, roofs by extrusion, shafts and vertical openings, slab edges, fascias, gutters and soffits come after this chapter's realignment.

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Views and graphics

This chapter describes the views of the model, how each one is drawn, and how to change what a view shows and how: graphic styles, scale and detail, the cut heights of a plan, crop regions, hiding and isolating, **Visibility and Graphics**, filters and view templates. Line weights, line patterns and styles by category are described in the same terms by the program's dialogs of **Settings** on the rail.

3D views

Views ▸ Create ▸ 3D View ▸ Default 3D View opens the default 3D view, **{3D}**, framing the whole model; opened again, it shows the model as it was left. Drag to orbit (or **Shift** and the middle button), drag with the middle button to pan, and turn the wheel to zoom at the cursor; see The interface.

To keep a framing, **Views ▸ Create ▸ 3D View ▸ Save View**: type a name and press Enter. The new view is listed under **3D Views** in the **Project Index** and opens at once; it keeps the framing it is left with. Renaming **{3D}** in the **Project Index** (right-click ▸ **Rename…**) does the same: **{3D}** becomes a named view, and the next default 3D view is a new **{3D}**.

A selected element is washed in blue and half see-through, its edges in blue: what is behind it stays visible.

Floors cut the walls they pass through: where a slab crosses a wall the wall stops under it and goes on above it, so the slab's edge and the wall's face are one surface.

Graphic styles

The **Graphic Style** button of the view control bar, at the bottom of every view window, offers:

Style	What is drawn
Wireframe	Every edge, also those behind other faces; no faces
Lines	Faces as white paper and the edges in front of them in dark ink, on the light and on the dark background alike
Shaded	Faces lit, in the colour of their material
Shaded with Edges	As **Shaded**, with the edges in front
Flat Colors	Faces in the colour of their material, the same whatever the light

The style belongs to the view and can be undone. The keyboard shortcuts `WF`, `HL` and `SD` choose **Wireframe**, **Lines** and **Shaded**.

The edges are the real ones: where two faces meet at an angle, and the outline of each face and each opening. A wall with a window shows its own outline and the window's, and no line where the pieces of the wall meet.

Plans, sections and elevations have a graphic style too. A plan is drawn in **Lines** until another style is chosen. In **Lines** the surfaces, cut or seen, are not coloured: what is cut shows its material's cut pattern; at **Coarse** detail the type's coarse fill pattern, or else only its outline. In **Shaded**, **Shaded with Edges** and **Flat Colors** every element seen or cut is filled with the shading colour of its material (a part with no material takes its category's, from **Styles by Category**, else the category's default colour; the 3D view follows the same rule): walls (each layer where it is cut), floors, roofs, doors, windows, furniture and components, stairs and railings. Plans are flat; in sections and elevations **Shaded** darkens a little the faces turned away from the viewer, such as the slopes of a roof, while **Flat Colors** keeps them all alike. At **Medium** and **Fine** detail the cut patterns stay on top of the colour. Sheets, prints and exported drawings show the views as the screen does.

The rooms' colour scheme shows in every style. With **Color scheme location** at **Background** it lies under the model's lines: in a plan over the floors and under the furniture, in a section over the surfaces seen through each room it cuts. At **Foreground** it covers the model, under the annotations.

Sun and shadows

Plans, sections, elevations and 3D views show the sun (drafting views, legends, sheets and schedules do not). The view control bar has two buttons, each with a small arrow for its menu (the right button opens it too):

- **Sun Path**: its menu has **Show Sun Path** / **Hide Sun Path**, **Sun Settings…** and **Fit to Model** (frames the whole sun path);
- **Shadows**: its menu has **Shadows On** / **Shadows Off**, **Graphic Display Options…** and **Preview Sun Study**.

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While either is on, the bar also shows the sun's hour and date, each with **−** and **+** (a quarter of an hour, a day). The same buttons and menus are in the **Sun and shadows** menu of the BIMopen style's view bar, in the touch mode's view sheet and in the view's context menu. **Sun Path** is also a check box of the view's **Properties** (**Graphics**), with **Sun Path Size** (100–500 %, 150 % by default) and **Shared Sun**.

Shadows are computed from the model and the sun, the same in every view: in plans they fall on what the plan shows and, where nothing stands, on the plane of a level (**Ground Plane at Level** in the sun settings, otherwise on the ground, the lowest level); in sections and elevations they fall on the façades and on what is behind them, from eaves, balconies, recesses and other buildings; in 3D views the graphics card draws them (the processor when there is none). In **Lines** (hidden line) a shadow is a grey of the paper, in the shaded styles it darkens the materials' colours. Faces cut by the view stay in the light.

The sun path is a ring round the model with the points of the compass at true north, the day's arc with its hours, the analemma (where the sun stands at that hour through the year: one smooth figure of eight, every day at the clock of the chosen date), the part a study goes through, and the sun with its hour and date; in a 3D view it is a dome over the ring. In plans, sections, elevations and 3D views the sun can be dragged: along the day it snaps to quarters of an hour, along the year it follows the analemma day by day (across a change of summer time the clock moves by the change, so the sun does not jump); `Shift` while dragging takes the other, and **Along the Day** / **Along the Year** in the sun path's menu choose which comes first. The shadows follow the drag: in plans, sections and elevations, while the sun moves, they are a coarser draft, each view showing its last shadows until the new ones are ready, and they are drawn in full when the sun is released. With the view under the cursor (and nothing selected, or after a click on the sun), `Left` and `Right` move the hour, `Up` and `Down` the date. With **Lighting** the sun stands still and has no path.

The sun path's menu (in the view bar, in the view's menu and in touch mode) also has **Visual Aids** that help read it: **Vertical Hour Lines** (from each even hour a dashed line down to the ground and out to the ring, with the hour there; in plans a dot on the ring), **Arc Projected on the Ground**, **Solar Noon** (the arc's highest point with its clock time, and the north–south line), **Band of the Year** (the sky the sun crosses in the year, in a light yellow, the arcs of the solstices and of the equinox, the analemmas of the even hours), **Sunrise and Sunset** (marks on the ring with their times; the view bar shows them with the length of the day) and **Analemma**; all but the projection are on at first. In a 3D view what stands behind the model is hidden by it. **Sun Settings** also show the year's sunrises and sunsets as a chart: a click or a drag on it picks the day.

Shared Sun: the views share the project's sun unless they have one of their own (the **Shared Sun** property). Moving the shared sun in one view moves it in every view that shares it; a view with its own sun is not touched. **Shared Sun in All Views** in the sun settings puts every view back on it.

Each change is one step of **Undo**; the `sun` command does the same at the command line and over MCP (see Command line and MCP).

Sun settings

Sun Settings... opens a dialog with the sun's azimuth, altitude, time and date in its header:

- on the left the **presets**: one unsaved row for each mode, the built-in ones (**Summer Solstice**, **Winter Solstice**, **Spring Equinox**, **Autumn Equinox**, **Summer, from 9 to 17**, **Light from top right**) and the project's own; **Save...**, **Duplicate...**, **Rename...** and **Delete** (only the project's presets are renamed or deleted);
- in the middle the mode and its fields:

Mode	What it takes
Still	the **Date** and the **Time**; **Shared Sun in All Views**
One Day	the **Date**, **From Sunrise to Sunset** or a **Start Time** and an **End Time**, the **Frame Interval** (15, 30 or 45 minutes, 1 hour); the number of **Frames** follows
Multi-Day	**From** and **To** (day/month), the **Time**, the **Interval** (1 hour, 1 day, 1 week, 1 month)
Lighting	**Azimuth (°)** and **Altitude (°)**; **Relative to View**: the light follows the view's direction

Then the project's **Location** (a city, or **Custom** with its latitude and longitude: one for the whole project; a custom place takes the time zone of the nearest city; a new project starts in **Rome**, the sample house stands in **Cagliari**) with **Daylight Saving Time**, which is automatic: the place's rule (Europe, North America, Chile, South-eastern Australia, New Zealand, Egypt, or none) and the exact Sunday of the sun's **Year**, so the times on the sun path and in the studies are those of the clock; the rule can be changed. Then **Ground Plane at Level** with its **Ground Level**; - on the right the preview: the sky seen from above, with the day's path, the analemma and the sun. **Preview** in the dialog's header, right of the figures, is a switch that hides and shows it; the dialog keeps its size, and without the preview the settings take its room.

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Apply changes the view's sun and keeps the dialog open, **OK** changes it and closes it, **Cancel** changes nothing. Projects made before the project had a location keep each view's place and date: the first view whose sun was changed gives the project its location, and the views whose sun was changed keep theirs as their own.

Sun study

Preview Sun Study (in the **Shadows** menu) shows the study bar over the view: ⏮ **−10**, ◂, ▸ **Play** / ⏸ **Pause**, ▸, **+10** ⏭ and **frame n of N**. A sun of one moment is studied through its day. **Save as Image** saves the frame shown as a PNG named after the view, the date and the hour; **Export Study…** saves every frame as a PNG or JPEG image sequence (a film needs a video encoder BIMopen does not have: the choice is greyed, with the reason). **×** closes the bar; another view closes it too.

Graphic display options

Graphic Display Options… sets for the active view: the **Graphic Style**, **Silhouette Edges** (**None**, **Medium**, **Thick**: a thicker line round the model's outline), **Cast Shadows**, **Ambient Shadows** (corners and recesses darkened), the sun (its button opens the **Sun Settings**), **Sun**, **Ambient Light** and **Shadow Intensity** (0–100) and, for 3D views, the **Background** (**None**, **Sky**, **Gradient**). The preview draws the view with the options chosen; the **Preview** switch in the header hides and shows it, as in the sun settings. View templates carry these options with the graphic style, and the sun and the shadows as **Sun and shadows**.

Plans and their cut heights

Views ▸ Create ▸ Plan ▸ Floor Plan opens **New Plan**: pick the levels and the scale, then **OK**. With **Do not duplicate existing views** on, the levels that already have a plan are left out; turn it off to make a second plan of a level.

A plan is a horizontal cut of the model at the **Cut Plane** of its cut heights. Walls crossing the cut plane are drawn cut, with the cut patterns of their layers' materials at **Medium** and **Fine** detail; lower walls and the floors of the level are seen from above.

To change the cut heights, select nothing (so that **Properties** shows the view), then **Cut Heights ▸ Edit…**. Each plane is measured from a level, with an offset:

Plane	Level
Top	The plan's own level (**Associated Level**), the **Level Above**, a level by name, or **Unlimited**
Cut Plane	Always the plan's own level
Bottom	The plan's own level, the **Level Below** or a level by name
View Depth	As **Bottom**, or **Unlimited**

OK sets the four planes together, as one change. They must stay in order, from the top down (top, cut plane, bottom, view depth): otherwise the window explains what is wrong and stays open. Floors below the bottom, down to the view depth, are drawn *beyond* in thin lines; so are floors a little under the bottom, such as a sunken slab, even with the view depth at the bottom.

Light and dark background

Options ▸ Theme ▸ Dark (or **Invert background color**) turns the drawing area dark. As on an inverted background, black lines, text and patterns are drawn light, colours keep their hue, and colour fills such as the rooms' are darkened, so the text over them stays readable. **Options ▸ Interface size** makes the whole interface bigger, from 100 % to 250 %, on top of the screen's own scale: useful on screens with many pixels. It is kept for the next start.

Opening a view from its symbol

Double-click a view's symbol to open its view: a section line or its head, an elevation symbol, a callout's boundary (a reference callout opens the view it points at), a level's line in a section or elevation (its floor plan opens), or a view placed on a sheet. The view's window becomes the active one.

Views ▸ Create ▸ Elevation ▾ ▸ Framing Elevation makes an elevation of a grid: click beside the grid, on the side to look from (before the click, a line and an arrow show the view). The elevation looks at the nearest grid, as wide as the bay between the grids that cross it and as deep as the next grid parallel to it.

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Scale, detail and crop region

The view control bar shows the view's **View Scale** and **Detail** (in 3D views too).

- At a smaller scale the model is drawn smaller on paper while text, tags and symbols keep their size on paper: at 1:200 a text covers twice as much of the model as at 1:100.
- At **Coarse** a cut wall is one fill; at **Medium** and **Fine** its layers show, each with its material.

Sections cut the model the same way. Walls, floors and roofs that meet in the cut are drawn as one: a thick outline round them, a thin line only where two materials meet, none between two parts of the same material; walls attached to a roof stop under it. Each cut layer has its material's cut pattern (at **Coarse**, the type's **Coarse Scale Fill Pattern** in its **Coarse Scale Fill Color**, set on wall, floor and roof types, else only its outline). Furniture and plumbing fixtures are never cut: they are seen whole. A window is cut with its frame in the thick cut line and its glass in thin lines; what lies beyond the cut is drawn in thinner projection lines. Beyond the cut, and in elevations, a face in front hides what is behind it, and a window's glass is opaque: each window shows its frame and glass, not the rooms behind it.

Crop View clips the view to its crop region, **Show Crop Region** draws the region; drag its edges to resize it. To type its size: **Properties ▸ Size Crop ▸ Edit...**, then **Width** and **Height**; the region keeps its centre.

Hiding and isolating

Isolate Temporarily (the glasses on the view control bar) acts on the selected elements, in the active view only, and does not change the project:

Command	Shortcut	What stays visible
Isolate Element	`HI`	Only the selected elements
Hide Element	`HH`	Everything but the selected elements
Isolate Category	`IC`	Only the categories of the selected elements
Hide Category	`HC`	Everything but those categories

The view gets a coloured border. Each use narrows what is shown: hiding one element and then another hides both. **Reset Isolate Temporarily** (`HR`) shows everything again; **Apply Hide/Isolate to View** hides for good what was hidden for now.

To hide for good in the view, select the elements, then right-click ▸ **Hide in view ▸ Elements** or **Category**, or the contextual card's **View ▸ Hide in View**, or type `EH` (elements) or `VH` (category).

Reveal Hidden (the light bulb on the view control bar, or `RH`) frames the view in magenta and shows what is hidden in magenta, the rest in halftone. A hidden element can then be selected and shown again: right-click ▸ **Unhide in View ▸ Elements** (or **Category**). Click the light bulb again, or type `RH`, to leave.

Visibility and Graphics

Views ▸ Graphics ▸ Visibility and Graphics (or `VG`, or **Properties ▸ Visibility/graphics overrides ▸ Edit...**) shows, for the active view, one row per category:

- the check box shows or hides the category; **Show model categories in this view** shows or hides them all;
- **Projection lines** and **Cut lines**: **Override...** sets the weight, colour and pattern;
- **Surface Patterns** and **Cut Patterns**: the material's patterns, hidden, in another colour or replaced by another fill pattern;
- **Halftone** blends the lines with the background;
- **Transparent** makes the surfaces see-through in the 3D view;
- **Detail** draws the category at another level than the view.

Clear overrides takes the selected rows back to the styles by category. Changes show at once: **Apply** keeps them and the window, **Cancel** (or the window's ×) takes back what was not applied, **OK** keeps them and closes.

One element can have its own graphics in a view: select it, then right-click ▸ **Custom Graphics ▸ By Element...** (or the contextual card's **View** group). **View-Specific Element Graphics** has **Visible**, **Halftone**, **Transparent**, the surface and cut patterns and the projection and cut lines. They apply only in this view, over the category's and the filters'.

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Filters

Views ▸ Graphics ▸ Filters lists the project's filters: **New**, **Duplicate**, **Rename**, **Delete**. A filter has categories and up to four rules on their parameters, all of which an element must meet.

In **Visibility and Graphics ▸ Filters**, **Add** applies a filter to the view; its row then sets visibility, lines, **Halftone** and **Transparent** for the elements that pass it. Where filters disagree, the one higher in the list wins: **Up** and **Down** move the selected filter.

View templates

Views ▸ Graphics ▸ View templates ▸ Create template from current view keeps the view's scale, detail, graphic style, cut heights and graphics as a template, by name. **View templates ▸ Apply template to current view** gives them to another view of the same kind: a plan (a second plan of a level too), a section or elevation, or a 3D view. A view does not follow later changes of its template until it is applied again, and the graphics of single elements are never part of a template.

Managing views

Right-click a view in the **Project Index**:

- **Duplicate View ▸ Duplicate** copies the view with its properties; **Duplicate with Detailing** copies its dimensions, text, tags and 2D details too. The copy opens;
- **Rename...** asks for the new name (**Rename View**); names stay unique among views.

Views ▸ Windows ▸ Tile shows all the open views side by side, filling the drawing area: with three windows, the active one on the left at full height and two stacked on the right.

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Annotation and 2D details

This chapter describes dimensions and constraints, text notes, tags, rooms and areas with their colour schemes, legend views, and the 2D drawing tools: detail lines, regions, detail components, repeating details, insulation and free 2D views. Annotations belong to the view they are made in: a dimension placed in a plan does not appear in another plan.

Before each click, every tool shows what it would make where the cursor is: the dimension with its value, the note's box, the tag on the element under the cursor, the region a room or area would fill, the line, region, component or insulation being drawn.

Dimensions

Annotations ▸ Dimension holds the dimension tools: **Aligned**, **Linear**, **Angular**, **Radial**, **Arc length**, and the spot dimensions **Spot Elevation**, **Spot Coordinate** and **Spot Slope**. They work in plans, sections, elevations, free 2D views and legends. Pick the dimension type in the type selector while the tool is on; **Type Parameters** changes the type, and with it every dimension of that type.

Aligned and linear dimensions

With **Aligned**, the options bar offers which part of a wall is picked (**Wall centerlines**, **Wall faces**, **Center of Core**, **Faces of Core**) and what a click picks:

- **References**: click references one after the other (wall faces or centre lines, grids, door and window edges, detail lines, the edges of legend symbols), then click an empty spot to place the dimension line there. While you choose, the string follows the cursor. Press **TAB** over a wall to step through its references.
- **Entire Walls**: one click dimensions a whole wall. **Options** opens the automatic dimensioning options: **Openings** with their **Centers** or both **Widths**, **Intersecting Walls** and **Intersecting Grids**. A grid that falls on a wall end adds no witness line.

Linear measures along the horizontal or the vertical of the view, between points such as wall ends: where you place the line decides which, and the **space bar** switches between the two.

Angular measures the angle between two walls, grids or lines that meet; more lines through the same point make a string of angles, each with its value; the last click picks the sector. **Radial** and **Arc length** dimension an arc: click it, then where the dimension goes.

An ordinate type has **Ordinate Dimension Settings** in its type parameters: **Edit...** opens a dialog with the **Text Orientation** (**Aligned** with the witness lines or **Horizontal**), the **Text Location** (**In line with the witness line** or **Above the witness line**), the **Origin Visibility**, the **Origin Marker** (**Zero value**, **Circle** or the type's **Tick Mark**) and the **Dimension Line Style** (**Continuous** or **None**, which leaves only the witness lines and ticks).

Changing a dimension

- Select a dimension and click its value: **Dimension Text** opens. **Use Actual Value** keeps the measure and adds text above, below, before or after it; **Replace with text** shows words instead (a number is refused).
- Drag the blue control under the text to move it away from the line; dragged back near the line, it snaps into place. Text moved far away gets an arc leader to its segment; **Leader** on the options bar removes it.
- **Selection: Dimensions ▸ Witness Lines ▸ Edit Witness Lines**: click elements to add witness lines, click an existing witness line to remove it, then click an empty spot to finish. Each witness line of a selected dimension has a control in its middle: right-click it ▸ **Move Witness Line**, then click the new reference; or ▸ **Delete Witness Lines**. The control at the element's end of a witness line drags its gap from the element; near another witness line's gap it takes the same.
- Select an element the dimension measures and click a value next to it: type a new distance and press Enter, and the element moves: a wall, a grid, a door or window (along its wall) or a detail line.

The type sets the tick (diagonal, filled arrow or dot), the extensions of the dimension and witness lines (**Witness line control**: a gap to the element, or **Fixed to Dimension Line** with its **Witness Line Length**), the **Dimension Line Snap Distance** (while placing, the line snaps at multiples of it from what it measures; 0 turns it off), the **Line Weight** and **Tick Mark Line Weight**, the **Centerline Symbol**, **Centerline Pattern** and **Centerline Tick Mark** (**Default** is the type's tick) of witness lines to centre lines, the colour, the text size, offset, background, bold, italic and underline, the units (**Units Format** opens **Format**: **Use project settings**, or a unit with its rounding, which may be **Custom** with a **Increment factor**, its symbol and options), and for linear dimensions the **Dimension String Type**: **Continuous**, **Baseline** (stacked values from the first reference) or **Ordinate** (values from the first reference along one line). **Show Opening Height** writes the height of a door or window under the width measured jamb to jamb. Radial types can drop the **R** prefix and their centre mark (**Center Marks**, with its size). With arrows, a segment too short for them has them outside, and the line runs on past them by **Flipped Dimension Line Extension**.

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Annotations ▸ Dimension ▸ Dimension Types opens the type parameters of each family's current type, to set before dimensioning.

A selected baseline string shows a flip control at its origin: click it, or press the **space bar**, to measure from the other end. **Baseline Offset** in its properties sets the distance between its lines on paper.

Temporary dimensions

Select an element and temporary dimensions show its distances to the nearest references parallel to it: walls, grids, levels, reference planes, floor edges, room separation lines and detail lines. They show for every category in every view the element is drawn in: plans, sections, elevations and 3D views, where they lie on the element's level. A selected level shows its distances to the levels above and below it, in a section or an elevation.

Click a value, type a new one and press Enter to move the element: a length takes a unit of its own (`2.5 m`), and a formula starts with `=` (`=8000/2`). Click the dimension symbol beside it to make it permanent.

Each witness line has a blue control. Drag the one on the reference onto any other reference to measure from there; or right-click it ▸ **Move Witness Line**, then click the new reference. A click on a control steps through the references of the same object: a wall's centerline, its faces, its core faces and its core centerline. The element remembers the references chosen, for each view, and they follow when the reference moves; right-click ▸ **Reset References** goes back to the nearest ones.

With several elements selected none are shown, and the options bar offers **Activate Dimensions** to show them for all the selected elements.

While a component, a door or a window is placed, temporary dimensions show its distances to the nearest references; **Tab** steps to the next ones. Rest the pointer for a moment: the element waits while the pointer goes to one of its values, and a click on the value types the distance it goes at. While drawing, the length and the angle are shown; **Rotate** shows the angle as an angular dimension, and an angle typed turns the selection.

Project ▸ Additional settings ▸ Temporary Dimensions sets where walls measure from (**Centerline**, **Faces**, **Center of Core**, **Core Faces**) and doors and windows (**Centerline** or **Openings**), for the whole project. The same dialog sets the text of every temporary dimension, for the application: from 8 to 20 pt (8 at first), on a transparent or an opaque background. Beside the settings a preview shows three walls measured as chosen; the **Preview** switch in the dialog's header hides it, the dialog keeping its size, and the choice is kept.

Constraints

Select a dimension: a padlock is shown on each segment. Click it to lock the segment: the distance then holds when either element moves, and a move that would break it is refused. Click it again, or right-click it ▸ **Toggle Lock**, to unlock; right-click ▸ **Show Related** selects the two elements the lock ties. A string of two segments or more shows **EQ**: click it and the references become equally spaced, and stay so when one of them moves. **Equality display** in the properties shows the values instead of **EQ**.

Selecting an element a constraint holds shows the constraint as blue dashed lines, unless the category **Constraints** is hidden in **Visibility and Graphics**. Deleting a dimension that holds a constraint asks what to do: **OK** deletes the dimension and keeps the constraint; **Delete Elements** deletes both. **Modify ▸ Align** with its padlock closed is a constraint too.

Spot dimensions

- **Spot Elevation**: in a plan, click a floor, a roof or the level; in a section, any point. The value is in metres with its sign.
- **Spot Coordinate**: the north and east coordinates of a point in a plan.
- **Spot Slope**: the slope of a roof pitch in a plan, or of a sloped line in a section, with an arrow pointing down.

The options bar offers **Leader** and, with a leader, **Shoulder**: with no leader one click places the spot; with a leader, click the point and then the text; with a shoulder, the point, the bend and the text. While you hover, the value shows at the cursor. For spot elevations it also offers **Relative Base:** (for a type whose **Elevation Origin** is **Relative**) and, in plans, **Display Elevations:**: the actual elevation, the top, the bottom, or both. For spot slopes, **Slope Representation:** draws an **Arrow** or, in sections, a **Triangle**.

A selected spot has grips at its point, its shoulder and its text: drag the point and the spot measures what is there. A spot on a floor or a roof is deleted with it. While placing a spot, **Esc** lets go of the one half placed; a second **Esc** ends the tool.

The spot type sets **Elevation Indicator**, **Top Indicator** and **Bottom Indicator** (the top or bottom indicator, then the elevation's, then the value; each one's **… as Prefix/Suffix** puts it after the value instead), the **Symbol**, **Include Elevation** and the **North/South Indicator** and **East/West Indicator** of coordinates (with **Indicator as Prefix/Suffix**), and the **Leader Line Length** and **Slope Direction** (**Down** or **Up**) of slopes.

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Text

Annotations ▸ Text ▸ Text (`TX`): click where the note goes and type; Enter starts a new line. A click gives a note that does not wrap; dragging gives it a width, and the words wrap in it. Clicking elsewhere finishes the note, and **Esc** ends the tool. `Tab` while typing moves the text to the next tab stop. The text size is on paper: the note keeps its printed size at any view scale.

The **Format** panel sets, before placing or on a selected note:

- the leader: **No Leader**, **One Segment** (click the tip, then the note), **Two Segments** (tip, elbow, note) or **Curved**;
- the alignment, left, centre or right, and **Bold**, **Italic** and **Underline** (Ctrl+B, Ctrl+I, Ctrl+U while typing or on the selected notes), which apply to the whole note;
- on a selected note, **Add Left Straight Leader** and the other three leader buttons, and **Remove Last Leader**;
- **Left Attachment** and **Right Attachment** (**Top**, **Middle**, **Bottom**): where the leaders meet the note; chosen while placing, they stay for the next notes.
- **Paragraph Format**: the paragraphs as a list, with **Bullets**, **Numbers**, **Lowercase letters** or **Uppercase letters**; the text of each item starts at the first tab stop, and wrapped lines are indented by one.

Near the line of another note's corner, a new note snaps to it, and a dashed line shows it.

On a selected note, drag the cross to move the box (the leader tips stay), the round side controls to change its width, and the rotation control to turn it. A grip in the middle of each leader makes an elbow when dragged, or bends a curved leader. Double-click inside it to edit the words. Its properties include **Arc Leaders**, **Left Attachment** and **Right Attachment**, and **Keep Readable**, which never shows it upside down. The text type sets the colour (of text, border and leaders), the background (**Opaque** masks what is behind), **Show Border** with its offset, the leader arrowhead, the **Line Weight** of the border and leaders, the size, the **Tab Size** (12.7 mm), bold, italic and underline. The small **Text Types** button of the **Text** panel opens **Type Parameters**, where **Duplicate...** makes a new text type.

Annotations ▸ Text ▸ Find/Replace finds words in the notes of the current selection, the current view or the whole project (**Scope:**), with options for case and whole words. **Find All** lists the matches with the view they are in; **Find Next**, or a click on a row, selects that note and shows its whole text under **Context**. **Replace** changes the match highlighted (the first when none is), **Replace All** every one, and **Close** closes the dialog. On selected notes it is also under **Tools** of their tab.

Symbols

Annotations ▸ Symbol ▸ Symbol (`SY`) places a 2D annotation symbol in a plan, a section, a free 2D view or a legend: a **North Arrow**, a **Centerline Symbol**, a **Flow Arrow**, a **Target** or a **Revision Delta**, the last two with a text of their own. The symbol follows the cursor before the click; the **space bar** turns it a quarter, or **Rotation** on the options bar sets its angle. With **Number of leaders:** above zero, the click places the symbol and the next clicks put the ends of its leaders. The type sets the symbol's size on paper and its leaders' arrowhead; **Duplicate...** in **Type Parameters** makes a new one.

Tags

Annotations ▸ Tag ▸ Tag by Category (`TG`): hover an element (a door, window, wall, room or area) and click to tag it. The options bar sets **Horizontal** or **Vertical**, the **Leader** and its length; the **space bar** switches the orientation, while placing or on a selected tag. Door and window tools have **Tag on Placement**, so each new door or window comes with its tag.

Tag All opens **Tag All Not Tagged**: tick the categories, the tag type of each, the leader and the orientation, for all the elements of the view or only the selected ones; only elements with no tag get one.

With a leader, **Attached End** ends it on the element; **Free End** ends it where you clicked, and a selected tag shows a grip to drag that end anywhere (**Leader End** in the properties switches between the two). A tag lines up with the other tags of its kind while you place it or drag it: near the line of another tag's centre it snaps to it, and a dashed line shows it. Tags hidden in the view (**Visibility and Graphics**) are turned on when you tag, and the message says so.

Double-click a tag to edit what it shows: a door's mark or a wall's type mark in place; a room's or area's number and name together in **Edit Label Values**. A tag takes another type of its own category from the type selector, and **Pick New Host** (**Host** panel) moves it to another element of the same category. Room label types choose whether they show the number, the area and the volume. A room or area label set to **Pattern** keeps an angle: it turns with **Rotate**, and the **space bar** lines it up with each wall near it in turn (**Rotation** in its properties). Deleting a room's label leaves the room, and the message says so.

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Rooms

Home ▸ Rooms and Areas ▸ Room (`RM`), then click inside a region enclosed by walls or room separators: the room fills it, around free-standing walls too. In the options bar:

- **Upper Limit** and **Offset**: the room reaches that level plus the offset (4000 mm by default); a higher level gives a room over several storeys, such as an atrium;
- **Tag on Placement**, on by default, with the tag's orientation and leader;
- **Room**: **New**, or a room that is not placed, to put it back;
- **Highlight Boundaries**: walls and separators that bound rooms turn orange, and the warning lists them.

A room placed in open space is kept with a warning, and takes its shape once the space is closed; a second room in a region already taken is marked **Redundant Room** and not counted twice. A number another room already has, a room lower than the computation height (1200 mm) and a base above it are warned about too, and so is a room separator drawn over another.

Walls bound the rooms of a level when they stand at its computation height: 1200 mm while **Automatic Room Computation Height** is ticked in the level's properties, otherwise its **Computation Height**.

A section shows the rooms it cuts when they are selected or under the cursor (**TAB** reaches a room behind other elements). A room selected there has an arrow at its top and one at its base: drag them to change its **Limit Offset** and **Offset at Base**.

Room Separator draws lines that split a space into rooms without walls. A wall bounds rooms while its **Room Bounding** property is on. Floors and roofs above cap the room's volume.

Deleting a room removes it from the plan but keeps it in the project, **Not Placed**, with its data; its tags go with it. Deleting a room label leaves the room. A room label dragged out of its room asks whether to **Move to Room**.

Rooms show through their tags. In **Visibility and Graphics**, **Rooms** has **Interior Fill** and **Datum** (the cross lines). **Settings ▸ Area and Volume Computations** opens on **Computations**: the room boundary at the wall finish, centre, core layer or core centre, and **Areas and Volumes** to compute volumes too (by default only areas; the volume is always measured to the wall finish). Its second tab is **Area Schemes**.

Areas

Areas belong to an area scheme: **Gross Building** (which cannot be deleted) and **Rentable** come with every project, and **Area Schemes ▸ New** adds more.

1. **Rooms and Areas ▸ Area ▸ Area Plan**: pick the scheme and the levels. Answer **Yes** to draw area outlines along the exterior walls, **No** to draw them yourself.
2. **Area ▸ Area Outline**: draw the outlines, or pick walls; with the area rules on, an exterior wall of a gross plan gives its outer face.
3. **Area ▸ Area**: click inside the outlines. The area is tagged when placed; **Tag ▸ Area Label** tags it later. Placed where no outlines close a region, the area is kept with a warning and takes its shape once they do; a second area in a region is **Redundant Area**.

Deleting an area leaves it in the project, not placed, without its tags; **Area:** in the options bar of the **Area** tool lists the areas not placed, to put one back with a click.

Colour schemes

A colour scheme fills the rooms or areas of a plan (and the rooms a section cuts) by a value (name, number, department, area type) or by ranges of area. Set it in the plan's **Color Scheme** property; **(none)** turns it off. **Settings ▸ Color Schemes** (**Edit Color Scheme**) creates and duplicates schemes and edits the title, the parameter, the values with their colour and visibility, their order and the ranges. **Apply** keeps the changes and sets the scheme of the view with the dialog open. **Color scheme location**, a view property, draws the fills in the **Background** (under the model) or in the **Foreground** (over the model, under the annotations).

Rooms and Areas ▸ Legend places the legend of the plan's scheme: a swatch and a label for each value. Its **Values displayed** are **By view** (those in the plan) or **All**, and **Columns** flows the swatches into several columns. A plan can have several legends; deleting one leaves the scheme.

Legend views

Views ▸ Legends ▸ Legend opens **New Legend View**: a name and a scale. The legend opens and is listed under **Legends** in the Project Index. **Annotations ▸ Component ▸ Legend Component** places the symbol of a type: in the options bar, the type,

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the **View** it is seen from (a wall in plan or section; a floor or roof in section or plan; a door, window or component in plan, front or back) and the **Host Length**, the piece of wall a door or window sits in. Legend components draw at the legend's detail, can be moved and dimensioned (the edges of doors, windows and components; walls, floors and roofs are not), and are not counted by schedules. Text, detail lines and regions work as in a free 2D view. A legend can be placed on several sheets.

A type dragged from **Families** in the Project Index into a legend becomes its legend component where it is dropped; a type of **Families ▸ Symbols** places that annotation symbol. **Create Same**, in the context menu of a selected element, starts the tool that places it with its type: for a legend component, with its view and host length too.

Detail lines and regions

Annotations ▸ Detail ▸ Detail line (`DL`) draws lines in the open view only, in the line style of the **Line style** panel. The **Draw** panel offers **Line**, **Rectangle**, **Inscribed Polygon**, **Circumscribed Polygon**, **Circle**, **Start-End-Radius Arc**, **Center-ends Arc**, **Tangent End Arc**, **Spline**, **Ellipse**, **Partial Ellipse** and **Fillet Arc**. The options bar shows what the shape takes:

Option	What it does
Continue	Each line (or arc, or spline) starts where the last ended; off, each one stands alone
Offset	What is drawn moves that far sideways; chained lines join at their corners
Radius	A circle, polygon or centre-ends arc of that radius (a circle takes one click); chained lines and rectangles get round corners. **Offset** is then ignored
Sides	The number of sides of a polygon

After the first click, type a length (a line, a chord) or a radius (circles, polygons, arcs, ellipse axes) and press `Enter`; the value shows at the cursor. A centre-ends arc never exceeds 180°: past it, it turns the other way, and a typed angle sets its end. A tangent arc starts at the end of a line or arc you click. `Shift` keeps lines and polygon radii horizontal or vertical and turns chords, arc radii and ellipse axes in steps of 45°, and keeps the sweep of an arc at 90°, 180° or 270°. `Esc` ends a chain or a spline. The same shapes and options are in every sketch (floors, roofs, filled regions, revision clouds, wall boundaries).

Pick Lines makes a detail line on a wall's face, axis or layer, a grid or another line under the cursor (`Tab` for the next one), moved by **Offset**; with **Lock** the line stays on it and follows when it moves. **Fillet Arc** picks two lines and rounds their corner, with the arc under the cursor or of the **Radius** set. A selected detail line has grips at its ends and a temporary dimension of its length: type a value to change it. The end of a line selected on its own drags alone and leaves the next line where it is; with a chain selected (`Tab`), the ends where its lines meet drag together, in one undo step. The cursor snaps to the ends, middles, centres and crossings of the view's detail lines.

Filled region ▸ Filled region opens sketch mode: the boundary is sketched with every shape of the **Draw** panel and with **Pick Lines**, and loops inside another are holes. Curves are kept as short straight segments. `Esc` ends a chain and keeps the sketch; **Modify** and the Modify tools work on its lines. **Finish Edit Mode** makes the region in the current region type, or says why it cannot (an open or crossing boundary). A selected region's **Edit Boundary** opens its sketch again. The type sets its fill pattern, colour and background (**Opaque** hides what is behind). **Filled region ▸ Masking Region** makes a region in the paper colour that hides what is beneath it. A region shows its **Area** and has **Comments**. A selected region has a grip at each vertex: drag one to change its shape.

Detail components, repeating details and insulation

- **Component ▸ Detail component**: a 2D symbol (brick, block, timber, steel section, insulation board, break line) placed in a drafting or detail view; the **space bar** lines it up with each wall or detail line near the cursor in turn (a quarter turn when there is none).
- **Component ▸ Repeating detail**: click a start and an end; the type sets the repeated component, the **Layout** (fill the available space, fixed distance, fixed number, maximum spacing), the **Spacing**, **Inside** and the rotation within the pattern. The instance has the **Number** when the layout is a fixed number.
- **Insulation**: click two points; **Width** and **Offset** (the insulation runs that far to the left of the clicks) in the options bar; with **Pick Lines** ticked, a click on a detail line or a wall draws it along that line. The insulation width and the bulge ratio in its properties. Selected insulation, and a selected repeating detail, have a grip at each end.

Draw order

Select detail items only: **Arrange** offers **Bring to Front**, **Bring Forward**, **Send to Back** and **Send Backward**. Detail items always draw over the model, and a new one goes on top. **Undo** restores the previous order.

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Free 2D views

Views ▸ Create ▸ Free 2D View asks for a name and a scale (or a custom one). The view opens and is listed under **Free 2D Views**; everything drawn in it is 2D, with no model behind it. It can go on one sheet. **Duplicate View ▸ Duplicate with Detailing** copies a view with its detail items, text, dimensions and legend components.

Display Model in the properties of a plan, a section or a detail view draws the model **Normal**, **Halftone** (greyed under the details) or not at all (**Do not display**), leaving the details and annotations.

Views ▸ Create ▸ Detail with **Reference other view** ticked on the options bar draws a callout that points at an existing detail or free 2D view instead of making a new one: its head names that view and a double-click opens it; **Referenced view** in its properties changes it. In a free 2D view the callout is always of this kind.

Hide at scales coarser than in the properties of a section or a callout hides its symbol in the views whose scale is coarser, for example a 1 : 50 detail left out of the 1 : 100 plan.

At the command line

Every tool has its command: `dim`, `lineardim`, `angulardim`, `radialdim`, `arcdim`, `spotelevation`, `text`, `findreplace`, `tag`, `room`, `roomsep`, `areaplan`, `areaboundary`, `area`, `colorscheme`, `legend`, `legendview`, `legendcomponent`, `detailline`, `region`, `mask`, `draworder`, `lock`, `eq`. `help` followed by a command shows its usage; Keyboard shortcuts lists the typed shortcuts.

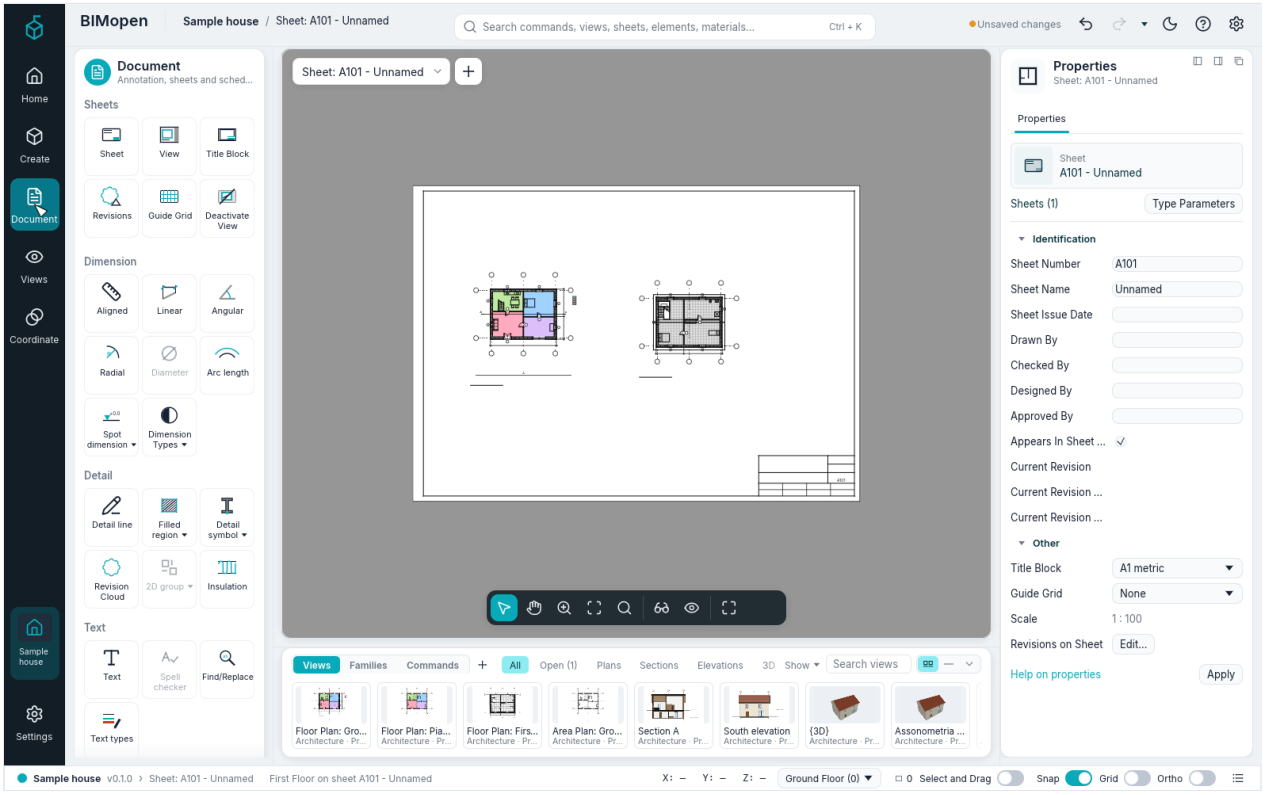
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Sheets, schedules and printing

This chapter describes how the project is documented: sheets and their title blocks, the views placed on them, revisions, schedules, printed drawings and exported images.

Sheets

A sheet is a drawing to print: a title block and the views placed on it. Sheets are listed under **Sheets (all)** in the Project Index, by number.



Creating a sheet

Document ➤ Sheets ➤ Sheet opens **New Sheet**: choose a title block (its paper size is shown), **None**, or one of the placeholder sheets of the sheet list. **OK** creates the sheet with the next free number (**A101**, **A102**...) and opens it.

Right-click a sheet in the Project Index ➤ **Rename...** to change its number and name in **Sheet Title**. The other parameters of the sheet are in the element panel: **Sheet Number**, **Sheet Name**, **Sheet Issue Date**, **Drawn By**, **Checked By**, **Designed By**, **Approved By**, **Appears In Sheet List**, its guide grid and its revisions. Several sheets picked together in the Project Index (with `Ctrl` or `Shift`) are changed together: the panel shows the values they share.

Settings ➤ Settings ➤ Project information sets what every sheet shows: **Project Issue Date**, **Project Status**, **Client Name**, **Project Address**, **Project Name** and **Project Number**.

Views ➤ Create ➤ Schedules ➤ Note Block makes a schedule of the symbols placed with **Symbol**, of one family only (north arrows, targets, revision deltas...): in **New Note Block** choose the **Family** and type the **Note Block Name**, then **OK**. **Note Block Properties** opens first, with the same tabs as any schedule; by default each row shows the symbol's type, its text and the count.

Title blocks

The title block shows the project's information, the sheet's number and name, the people and the dates, the current revision and the scale (**As indicated** when the views have different scales); above it, a table lists the sheet's revisions with their number, description and date. A double click on one of its fields types the value in place.

- Click a line of the title block to pick it: the type selector changes it to another paper size, `Del` removes it (the sheet and its views stay).

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- **Document ➤ Sheets ➤ Title Block** inserts one on the open sheet: choose it in the type selector, its frame follows the cursor, click to place it.
- **Type Parameters** shows its **Sheet width** and **Sheet height**; **Duplicate...** makes a new paper size.

Views on sheets

Drag a view from the Project Index onto the sheet, or use **Document ➤ Sheets ➤ View** (**Views**, then **Add View to Sheet**): the view follows the cursor at its scale, and a click places it as a viewport. A view goes on one sheet only; a schedule on several.

A picked viewport:

- moves with the mouse or the arrow keys, unless it is pinned (**Fix**); near another view, its title lines up with the other's and a dashed line shows it;
- has its **View Scale**, its **Title on Sheet** (also typed by double-clicking the title) and its **Rotation on Sheet** (90° clockwise or counter-clockwise; the title stays horizontal);
- shows the end of its title line, which can be dragged;
- takes a viewport type from the type selector: **No Title**, **Title w/ Line** or **Title w/o Line**, or a type of your own (**Show Title**, **Show Extension Line**, line weight, colour and pattern of the line);
- leaves the sheet with `Del`; the view stays in the project.

Activate View (on the **Selection: Viewports** card or the right button) edits the view where it is on the sheet, the rest of the sheet in halftone; **Deactivate View** goes back to the sheet. A view turned on its sheet cannot be activated: take its rotation off first. A double click on a view opens it in its own window.

Guide grids

Document ➤ Sheets ➤ Guide Grid asks for a name and covers the sheet with a grid. Pick it to drag its corners or to set its **Guide spacing**; the sheet's **Guide Grid** parameter shows the same grid on other sheets, or none. Views dragged near the grid's lines snap onto them. The grid is never printed.

Sheet list

Views ➤ Create ➤ Schedules ➤ Sheet List makes a schedule of the sheets: number, name and any other field. Sheets with **Appears In Sheet List** unticked are left out. **Rows ➤ New** adds a placeholder sheet: a row that is not a sheet yet and is not in the Project Index. **Filter placeholder sheets** shows every row, hides the placeholders or shows only them. A placeholder chosen in **New Sheet** becomes a real sheet.

Revisions

Document ➤ Sheets ➤ Sheet Issues/Revisions lists the revisions of the project, one per row: **Sequence**, **Numbering** (**Numeric**, **Alphabetic** or none), **Date**, **Description**, **Issued**, **Issued to**, **Issued by** and **Show**. **Add** adds one; a row picked by its sequence number is merged into the one above or below (its clouds go with it) or moved up and down. **Numbering** is **Per Project** or **Per Sheet** (the place of the revision among the sheet's revisions); **Alphabetical order ➤ Options...** sets the letters used. An issued revision cannot be changed, and neither can its clouds, until **Issued** is unticked.

Document ➤ Detail ➤ Revision Cloud draws a cloud in a plan, a section, a free 2D view or on a sheet: sketch closed loops with the shapes of the **Draw** group, then **Finish Edit Mode**; a selected cloud's **Edit Sketch** opens them again, unless its revision is issued. The cloud takes the latest revision not issued; picked, it shows its **Revision** (to change), its number, date and people, its **Mark** and **Comments**. **Show** of a revision draws its clouds and tags, keeps them but does not draw them, or hides them.

A sheet lists the revisions of the clouds of its views and of its own clouds. Its **Revisions on Sheet** (in the element panel, **Edit...**) adds other revisions to the list.

Schedules

A schedule is a table of the elements of the model, and a view like the others: it opens in a window of the drawing area, is tiled, cascaded and switched with plans, sections and 3D views, and is listed under **Schedules/Quantities** in the Project Index. It follows every change of the model at once, and every change made in it changes the model.

Creating a schedule

Views ➤ Create ➤ Schedules ➤ Schedule/Quantities opens **New Schedule**: choose a category, a name (the default follows the category until one is typed) and **Schedule building components** or **Schedule keys**. **OK** opens **Schedule Properties**; its second **OK** creates the schedule and opens its view. **Material Takeoff** in the same menu lists the materials of the layers of walls, floors and roofs.

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Schedule Properties

The dialog opens again from the element panel of the schedule (each tab has its **Edit...** button) or from the right button on the schedule in the Project Index. Its tabs:

Tab	What it sets
Fields	The columns, chosen from **Available fields:** and put in order; the fields of the rooms the elements are in; **Add Parameter...** for a project or shared parameter; **Calculated Value...** for a formula (with `+ - * / ^`, `sqrt`, `round`, `IF`, `AND`, `OR`, `NOT` and units) or a percentage
Filter	Up to four criteria, all of which an element must meet
Sorting/ Grouping	Up to four sort levels, each with **Header**, footer and **Blank line**; the grand totals; **Itemize every instance** (off: one row per type)
Formatting	Per field: heading, orientation, alignment, hidden field, **Format...** (units, rounding, symbol), totals and **Conditional Format...** (a background colour where a value meets a test)
Appearance	Grid lines and outline with their line styles, the title and the column headers, a blank row, **Header text** and **Body text** (size, **Bold**, **Italic**); shown when the schedule is on a sheet

Working in a schedule view

- A cell edits the element, or its type for a type parameter; every edit is one step of undo.
- A row selects its elements in every view. **Highlight in model** on the **Edit | Schedule/Quantities** card opens **Show Elements in View**, which brings a plan forward framed on them; **Show** goes on to the next view.
- **Rows ➤ Delete** deletes the elements of the selected rows, after asking.
- Headings picked together are joined under a common heading with **Group**, and parted with **Ungroup**; **Hide column** and **Unhide all columns** work on the column picked.

Schedules on sheets

Drag a schedule from the Project Index onto a sheet. Selected on the sheet, it shows its controls: the Z on the right edge of a segment splits it in two, the arrow in the middle of a segment moves it (dropped on the segment beside it, the two join), the dot at the bottom of a segment moves its last row, and the triangles at the top of the columns set their widths. A double click, or **Edit Schedule** on the right button, opens the schedule view. The same schedule can go on several sheets.

Key schedules and material takeoffs

A key schedule (**Schedule keys** in **New Schedule**) defines named sets of values, one row per key, added and deleted with **Rows**. Its **Key Name** becomes a parameter of the category: choosing a key for an element gives it all the values of that row.

A material takeoff lists, for each element, the materials of its layers with their area and volume; its fields can be sorted, grouped and totalled like any other.

Project and shared parameters

Settings ➤ Settings ➤ Project Parameters adds parameters to the elements of some categories, or to their types, with a name, a type of value and the group they show in the element panel.

Settings ➤ Settings ➤ Shared Parameters opens **Edit Shared Parameters**: the shared parameter file in use (**Browse...**, **Create...**), its groups and the parameters of each group. A shared parameter keeps its name and type, and has an identifier of its own, so that several projects add the same parameter. In **Parameter Properties**, choose **Shared parameter** and **Select...** to add one to the project; **Export...** writes a shared parameter of the project back to the file.

Exporting and reusing schedules

- **Application menu ➤ Export ➤ Reports ➤ Schedule...** writes the schedule as delimited text: **Export Schedule** sets the field delimiter, the text qualifier and which headings and group rows go in the file.
- **Create view template from view...** on the right button of a schedule makes a view template with its fields, filter, sorting, formatting and appearance; **Apply view template...** gives them to another schedule.
- **Save to New File...** on the right button writes the schedule's setup, without the elements, to a file of its own. **Create ➤ Load from library ➤ Insert from File ➤ Insert Views from File...** brings the schedules of another project into this one, with their project parameters.

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Printing

Printing writes PDF files: on the desktop they are saved where you choose, in the browser they are downloaded.

The Print dialog

`Ctrl+P`, or **application menu ➤ Print ➤ Print...**, opens **Print**:

- **File**: the name of the file and, when several views or sheets are printed, **Combine multiple selected views/sheets into a single file** (one page each) or **Create separate files** (one file each, named after the sheet or view).
- **Print Range**: **Current window**, **Visible portion of current window** (only what the window shows), or **Selected views/sheets** with **Select...**.
- **Reverse print order** prints the last page first.
- **Settings**: the print settings in use, changed with **Settings...**; **Preview** shows the pages.

Plans, sections, elevations, free 2D views and sheets print as drawings; the 3D window prints as an image of the view; a schedule prints its table on a page. The page is always white, elements hidden temporarily in a window are printed, and each printed sheet carries the date and time of the print.

View/sheet sets

Select... opens **View/Sheet Set**: tick the sheets and views to print (**Show** sheets, views or both; **Check all**, **Check none**). A set is saved by name with **Save As...** and kept in the project; **Save**, **Reset**, **Rename...** and **Delete** manage the saved sets, and the name list picks one again.

Print Setup

application menu ➤ Print ➤ Print Setup..., or **Settings...** in **Print**, sets:

Setting	What it does
Size and Orientation	The paper (A0 to A4, or the sheet's own size) and its orientation
Paper Placement	**Center**, or an offset from the lower-left corner
Hidden line views	**Vector Processing** (sections and elevations print only the lines that are seen: the lines behind nearer surfaces are removed, not just covered, and lines lying on one another print once, so the PDF opens clean in other programs) or **Raster Processing** (the pages as images, at the raster quality chosen)
Zoom	**Fit to page**, or a percentage of the real size
Colors	**Black Lines**, **Grayscale** or **Colour**
Options	**Show view links in blue**, **Hide unreferenced view tags**, **Hide crop boundaries**, **Replace halftone with thin lines**, **Keep coincident lines** (off: two lines lying on one another in printed and exported sections and elevations become one; on: both stay, each with its own element)

Settings are saved by name with **Save As...**, and kept in the project. Closing the dialog after changing saved settings asks whether to save the changes; **No** keeps them for this session only.

Print Preview

application menu ➤ Print ➤ Print Preview, or **Preview** in **Print**, shows the pages as they will print, in their colours: **Previous Page** and **Next Page** go through them, the wheel zooms and dragging pans; **Print...** prints, **Close** goes back.

Exporting images

application menu ➤ Export ➤ Images and animations ➤ Image... opens **Export Image**:

- the file name and the **Export Range**: the current window, its visible portion, or the views and sheets chosen (one file each);
- the **Image Size**: **Fit view(s)/sheet(s) to** a number of pixels across or down, or a zoom percentage of the real size at a resolution;
- the options, as for printing: view links in blue, crop boundaries and unreferenced view tags hidden;
- the **Format** of views without shading and of shaded views: PNG, JPEG, BMP or TIFF.

The 3D window is exported as it is shown, with its graphic style.

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Command line and MCP

Everything you do in BIMopen is a command. A click on **Wall** and two points in the plan run the same `wall` command you can type yourself. This chapter describes the command line of the window, the `bimopen-cli` program, which runs the same commands without a window, and the MCP server, through which an AI agent can model, check and render a project.

The command line in the window

The command line sits under the drawing area. Its upper part is the log; its last row is where you type.

- You do not need to click it first: whatever you type while no other field is being edited goes there.
- `Enter` runs the line. The command is echoed in the log after a `>`, followed by its result; an error is shown in red and changes nothing.
- `Up` and `Down` bring back the lines typed before.
- On the left of the row, the prompt says what is expected: `Command:` when no tool is active, or the request of the active tool, such as `WALL next point [Esc to finish]:`.
- `help` lists every command with a line of description; `help wall` shows the usage of one. **?** ▸ **Commands** in the top bar does the same as `help`.

Typed shortcuts start the tools

A command typed **without arguments** starts the interactive tool, as if you clicked its button: `WA` (or `wall`) starts the wall tool, `DR` the door tool, `VG` opens Visibility and Graphics. The same command **with arguments** does the work at once, without the tool: `wall 0,0 5,0` draws a wall from the origin to 5 m along x.

Command names, their short forms and their arguments are the same in every language of the interface; most also accept an Italian name, such as `muro` for `wall`. Parameter names, type names and view names follow the language of the interface: in English you type `set Free Height 2.7m`, in Italian `set Altezza libera 2.7m`. The keyboard shortcuts chapter lists every short form.

Points and lengths

You type	It means
`3,4`	The point x = 3 m, y = 4 m
`@2,0`	2 m along x and 0 along y from the previous point
`@4<90`	4 m from the previous point, at 90° (angles counter-clockwise from the x axis)
`2.7m`, `270cm`, `2700mm`	A length with its unit
`2.7` or `2,7`	A length in metres: at the command line a number without unit is in metres

For example, `wall 0,0 @6,0 @4<90 0,4 c` draws a 6 × 4 m rectangle; the final `c` closes it.

> Warning
>
> At the command line `set Free Height 2700` means 2700 **metres**. Write the unit: `2700mm`. In the element panel, instead, a number without unit is in the project's units, millimetres in a new project.

Arguments in JSON

Every command also takes its arguments as a JSON object, which gives access to all its options. The usage line of `help` lists them:

```
wall 0,0 10,0 {"height":3.2,"location":"Linea d'asse del muro"}
roof walls {"slope":30,"overhang":0.5}
sheet A102 Plans {"title_block":"A3"}
export doors.csv {"schedule":"doors"}
```

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Commands you will use often

Command	What it does
`new`, `sample`	A new project, the sample project
`open house.bimopen`, `save house.bimopen`	Opens and saves a project
`undo`, `redo`	Undoes and redoes the last change
`view 3d`, `view Ground Floor`	Opens the default 3D view, or the plan of a level
`wall`, `door`, `window`, `floor`, `roof`	Model elements: see the usage with `help`
`select walls`, `select all`, `deselect`	Selects by category, everything, nothing
`set <parameter> <value>`	Sets a parameter of the selection, as in the element panel
`type <type name>`	Changes the type of the selection, or of the next element
`move 1,0`, `copy 0,3`, `rotate 90`	Modifies the selection
`delete`	Deletes the selection
`list walls`, `params 102`, `info`	Lists elements, the parameters of one, a summary of the project
`schedule doors`	Shows a schedule at the command line
`export house.ifc`	Exports IFC, SVG, PNG or CSV, by the file's extension

Every element has a number, its id, shown in brackets after it is created: `Door «Single swing door 80x210» in wall 102 (106)`. Commands accept it where they need an element: `params 106`, `delete 106`.

bimopen-cli

`bimopen-cli` runs BIMopen without a window: to convert files, to build a project from a script, to check a model, or to serve agents over MCP. Build it with `cargo build --release -p bimopen-cli`.

```
bimopen-cli info <file>
bimopen-cli convert <in> <out> [--level NAME] [--view 3d] [--schedule walls|doors|windows|floors|rooms|materials]
bimopen-cli run [<file> | --sample] [--cmd LINE]… [--script FILE] [--save FILE] [--export FILE]…
bimopen-cli commands
bimopen-cli mcp [<file> | --sample]
```

Subcommand	What it does
`info <file>`	A summary of the project and, in JSON, its levels, element counts and active view
`convert <in> <out>`	Opens `<in>` (`.bimopen` or `.rvt`) and writes `<out>`: `.ifc`, `.svg` or `.png` of a plan (the lowest level, or `--level`), `.png` of the 3D view (`--view 3d`), `.csv` of a schedule (`--schedule`), or `.bimopen`
`run`	Opens a file, the sample, or a new project, then runs the options in order: `--cmd` one line, `--script` a file of lines (empty lines and lines starting with `#` are skipped), `--save` and `--export` a file
`commands`	Every command, with its description and usage
`mcp`	The MCP server on standard input and output
`--version`	The version

The messages are in the language of the environment variable `BIMOPEN_LANG` (`it`, `en`, `de`, `fr`, `es`, `ar`), or of the system (`LANG`), or Italian. `bimopen-cli` ends with code 0 when everything worked, 1 when a command failed (the message says which), and 2 when the arguments are wrong (it prints the usage).

```
export BIMOPEN_LANG=en
bimopen-cli run --script house.txt --save house.bimopen --export house.ifc
bimopen-cli convert house.bimopen plan.svg
bimopen-cli convert house.bimopen doors.csv --schedule doors
bimopen-cli convert house.bimopen first.png --level "First Floor"
bimopen-cli convert house.bimopen house-3d.png --view 3d
bimopen-cli convert project.rvt project.bimopen
```

A script is a text file with one command per line, exactly as typed at the command line:

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```
# a small house
wall 0,0 10,0 10,6 0,6 c
door 5,0
window 10,3
floor walls
roof walls
```

The schedules are written as CSV with `;` between the columns, the headings in the language of the messages and a total row at the end.

MCP: BIMopen for AI agents

The Model Context Protocol (MCP) lets an AI agent use programs as tools. `bimopen-cli mcp` is an MCP server: it speaks JSON-RPC 2.0 on standard input and output, one message per line, and keeps one project open for the whole conversation. Without a file it starts from a new project; `--sample` starts from the sample house.

To add it to an agent, give the agent the command to start. For example, in Claude Code:

```
claude mcp add bimopen -- bimopen-cli mcp --sample
```

or, in a client configured with JSON:

```
{ "mcpServers": { "bimopen": { "command": "bimopen-cli", "args": ["mcp", "--sample"] } } }
```

The server offers seven tools:

Tool	Arguments	What it does
`command_line`	`line`	Types a line at the command line, exactly as a person would
`execute`	`command`, `args`	Runs a command with its arguments in JSON
`inspect_model`	–	A summary: levels, element counts, active view, selection
`query_elements`	`category` (optional)	The elements, with id, category and name: `walls`, `doors`, `windows`, `floors`, `rooms`, `roomseps`, `grids`, `levels`, `types`
`get_params`	`id`	Every parameter of an element
`schedule`	`kind`	A schedule: `walls`, `doors`, `windows`, `floors`, `rooms` or `materials`
`render`	`view`, `width`, `height`, `yaw`, `pitch`	A PNG image of a view: `3d`, the name of a level for its plan, of a section, elevation, callout or free 2D view, or the number or name of a sheet; `yaw` and `pitch` turn the 3D camera, in degrees

The answers carry the message of the command and, when there is one, its data in JSON, such as the ids of the elements created. An agent can check its own work by asking for a render.

> Tip
>
> Ask the agent to render the plan and the 3D view after each step: it sees what it built and corrects itself.

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Keyboard shortcuts

BIMopen has two kinds of shortcuts: **keys**, such as `Ctrl+Z`, which act at once, and **typed shortcuts**, such as `WA`, which you type letter by letter. At the command line a typed shortcut waits for `Enter`, like any command; typed over the drawing area, it acts as soon as it is complete when nothing longer begins with its letters (see below). Typed shortcuts are the same in every language.

On macOS, `Cmd` takes the place of `Ctrl`.

Keys

Key	What it does
`Ctrl+Z`	Undoes the last change
`Ctrl+Y` or `Ctrl+Shift+Z`	Redoes the change just undone
`Ctrl+S`	Saves the project; a project never saved asks for a file name
`Ctrl+P`	Opens **Print**
`Ctrl+A`	Selects the elements of the open plan's level, or of the whole model in a 3D view
`Del`	Deletes the selected elements, and the elements they host
`Home`	Fits the whole view in its window
`F1`	Opens the page of this manual about the open dialog, the tile under the mouse or the active tool
`Esc`	Cancels the step in progress; pressed again, ends the tool; with no tool, clears the selection
`Enter`	Runs the line typed at the command line; confirms a value in the element panel
`Up` / `Down`	At the command line, the lines typed before
`Space`	While drawing walls, swaps the exterior and interior side; while placing a component or a detail component, rotates it
`Shift`	While drawing, keeps segments horizontal or vertical (or frees them, when **Ortho** is on); with a click or a window, takes elements out of the selection; while dragging, frees a joined wall or keeps the move horizontal or vertical
`Ctrl` + click	Adds an element to the selection; held when a drag starts, places a copy
`Tab` / `Shift+Tab`	Highlights the next (previous) element under the cursor, or a chain of joined walls and lines
`Ctrl+Left`	Selects again what was selected before
`Left` `Right` `Up` `Down`	Move the selection a small step; with `Shift`, ten steps

`Ctrl+Z`, `Ctrl+Y`, `Ctrl+A`, `Del` and `Home` act on the model only while you are not typing in a field: in a field they act on the text.

In a floor, roof, stair or railing sketch, `Esc` (or a double-click) also finishes the sketch when it has enough points.

In the **Text Note** dialog, `Ctrl+Enter` inserts the note and `Esc` cancels it.

`Alt` pressed and released, or `F10`, shows the key tips of the rail and the tool panel. With no tool active and nothing typed, `Enter` repeats the last tool. After a click in the Project Index, the arrow keys and `Enter` move in its tree.

Mouse

	In plans and sections	In 3D views
Wheel	Zoom at the cursor	Zoom
Middle button, drag	Pan	Pan; with `Shift`, orbit
Left button, drag	Selection window (left to right: inside; right to left: touching), or moves the selected elements	The same
Right button	Context menu; with a tool active, the tool's menu (**Cancel**, **Snap Overrides** ▸…)	The same; dragging, orbit

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	In plans and sections	In 3D views
Double-click	Finishes a sketch	–

Typed shortcuts

Type the letters while no field is being edited, then press `Enter`. Without arguments, a shortcut starts the tool or opens the dialog; with arguments, it runs the command at once (see Command line and MCP).

With the cursor in the drawing area, the letters go to the command line (in the BIMopen style, to the search), and a shortcut acts as soon as it is typed, with no `Enter`, when no longer shortcut or command name begins with its letters: `VR` (cut heights), `WT` (tile), `ZE`, `ZF`, `ZA`, `ZP`, `ZR`, `ZO`, `ZS`, `HR`, `HH`, `HC`, `IC`, `DL` (detail line), `RT` (room label), `TG` (tag by category), `TX`, `CC` (copy), `MV`, `RM`, `LL` (level), `RC` (revision cloud), `PP` (shows or hides **Properties**), `PT`, `RP`, and the shortcuts you added. The others begin longer names and wait for the next letter, a space (arguments follow) or `Enter`: `VG` (`vgsub`…), `HI` (`hide`), `DI` (`dim`), `AL` (`align`), `WA` (`wall`), `DR` (`draworder`), `UN` (`undo`), `MA` (`mask`); so `dim`, `wall 0,0 5,0` or `WA 0,0 5,0` typed over the view run as typed. `MA` and `PT` say in the status bar that they are not available yet; `RP` starts the reference plane. Typed in the command line itself, a line waits for `Enter`.

While a tool waits for a point, in every view (plans, free 2D views, sections, elevations, the 3D view on its work plane, sketches), two letters override the snaps for that point only: `SE` (endpoint), `SM` (midpoint), `SN` (nearest), `SW` (work plane grid), `SQ` (quadrant), `SI` (intersection), `SC` (center), `SP` (perpendicular), `ST` (tangent), `SR` (remote objects), `SZ` (close), `SO` (no snap); `SS` ends the override in force. They are also chosen in the right button's **Snap Overrides ▸**; those a longer name begins with need `Enter`. The status bar says so, and `Esc` or the point taken ends it. With several snaps under the cursor, `Tab` goes to the next and `Shift+Tab` to the previous. Shortcuts added in **Keyboard Shortcuts** under a snap's name work as its two letters.

Modelling

Shortcut	Command	Starts
`WA` or `W`	`wall`	Wall
`DR`	`door`	Door
`WN`	`window`	Window
`SB`	`floor`	Floor (the **Flooring** button)
`RF`	`roof`	Roof
`CM`	`component`	Object
`ST`	`stair`	Stairs
`RA`	`railing`	Railing
`CG`	`curtaingrid`	Curtain grid
`MU`	`mullion`	Mullion
`CP`	`curtainpanel`	Curtain wall panel
`GR`	`grid`	Grid
`RM`	`room`	Room
`RS`	`roomsep`	Room separator
`PA`	`areaplan`	The **Area Plan** dialog
`CA`	`areaboundary`	Area outline
–	`area`	Area
`LG`	`legend`	Color fill legend
`SC`	`colorscheme`	The **Color Schemes** dialog

Modifying

Shortcut	Command	Starts
`MV` or `M`	`move`	Move

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Shortcut	Command	Starts
`CO`	`copy`	Copy
`RO`	`rotate`	Rotate
`MM`	`mirror`	Mirror on Axis
`DM`	`mirror`	Mirror on Line
`AR`	`array`	Repeat in Series
`OF`	`offset`	Offset
`AL`	`align`	Align
`TR`	`trim`	Shorten/Lengthen
`SL`	`split`	Split Element
`SG`	`split`	Split with Joint
`RE`	`resize`	Scale
`PN`	`pin`	Fix: pins the selection
`UP`	`unpin`	Unpins the selection
`SA`	`selectinstances`	Selects all instances of the selection's types visible in the view
`DEL`	`delete`	Deletes the selection
`U`	`undo`	Undoes the last change

Annotating and detailing

Shortcut	Command	Starts
`DI`	`dim`	Aligned dimension
`DL`	`lineardim`	Linear dimension
`DA`	`angulardim`	Angular dimension
`TX`	`text`	Text
`TG`	`tag`	Tags the untagged doors and windows of the level
–	`detailline`	Detail line
`RG`	`region`	Filled region
–	`mask`	Masking region
`CD`	`detailcomponent`	Detail component
`RD`	`repeatingdetail`	Repeating detail
`IS`	`insulation`	Insulation

> Note
>
> Typed alone, `DL` starts the detail line; followed by points at the command line, it draws a linear dimension (`lineardim`).

Views and settings

Shortcut	Command	Starts or opens
`SE`	`section`	Section
`EL`	`elevation`	Elevation
`DT`	`callout`	Callout (the **Detail** button)
`DS`	`splitsegment`	Split Segment of a section
`VD`	`draftingview`	The **Free 2D View** dialog
`PP`	`floorplan`	The **Floor Plan** dialog

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Shortcut	Command	Starts or opens
`DV`	`duplicateview`	Duplicates the active view
`VG` or `VV`	`vg`	**Visibility and Graphics**
`WF`, `HL`, `SD`	–	**Wireframe**, **Lines**, **Shaded**: the graphic style of the active view
`HH`, `HI`	–	**Hide Element**, **Isolate Element** for now
`HC`, `IC`	–	**Hide Category**, **Isolate Category** for now
`HR`	–	**Reset Isolate Temporarily**
`EH`, `VH`	–	**Hide in view**: the selected elements, their category
`RH`	–	**Reveal Hidden**, on and off
`TL`	`thinlines`	**Thin lines**, on and off
`SO`	`objectstyle`	**Styles by Category**
`MA`	`material`	**Materials**
`UN`	`units`	**Project units**
`PU`	`purge`	**Purge unused**
–	`filter`	**Filters**
–	`viewtemplate`	**View templates**
–	`applytemplate`	Apply a view template to the active view
–	`linestyle`	**Line styles**
–	`fillpattern`	**Fill patterns**
–	`saveview3d`	Saves the {3D} view with a name
–	`renameview`	Renames the active view

The commands with **–** have no short form, but their whole name works the same way. `help` lists every command and its short forms.

The zoom commands have typed shortcuts too: `ZR` zoom in region, `ZO` zoom out (2x), `ZF` zoom to fit, `ZA` zoom all to fit, `ZS` zoom sheet size, `ZP` previous pan/zoom. **Views ▸ Windows ▸ Panels ▸ Keyboard Shortcuts** gives any command shortcuts of your own (The interface).

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Troubleshooting

This chapter collects situations that can surprise you, and what to do about them. It grows with the manual: each area of the realignment adds its own.

Starting BIMopen

The web app shows «BIMopen non si è avviato» and an error

The browser could not give BIMopen a graphics surface: the message ends with `webgl2 not available` or with a WebGPU error. BIMopen in the browser needs WebGPU or WebGL2.

- Turn on hardware acceleration in the settings of the browser, then reload the page.
- Try another browser, or a newer version of the same one.
- Add `?webgl` to the address to use WebGL2 even where WebGPU exists but does not work.

The desktop app does not build on Linux

The interface needs the development files of xkbcommon, Wayland and GTK 3. On Debian and Ubuntu:

```
sudo apt-get install -y libxkbcommon-dev libwayland-dev libgtk-3-dev
```

The interface

The element panel or the Project Index has disappeared

Views ▸ Windows ▸ Panels shows or hides them: tick them again there. The **Properties** tile of the contextual card also shows the element panel again. A click on the rail entry already chosen folds the tool panel: click it again to open it.

What I type ends up at the command line

Whatever you type while no field is being edited goes to the command line, the search box of the top bar: this is how typed shortcuts work. To type in a field of the element panel or of a dialog, click the field first.

A tile is greyed

Some tools work only in the right view, and their tip says why. **Area Outline** and **Area** need an open area plan: the tip says **Open an area plan first**. **Save View** needs an open 3D view; **Document ▸ Sheets ▸ View** needs an open sheet; **Undo** and **Redo** are greyed when there is nothing to undo or redo.

`DL` starts a dimension instead of a detail line

`DL` is the short form of the linear dimension. To draw detail lines, use **Document ▸ Detail ▸ Detail line**, or type `detailline`.

Modelling

«no wall at that point»

A door or a window must be placed in a wall. Click on the wall itself, or give a point on its line at the command line.

A height set at the command line is enormous

At the command line a number without unit is in metres: `set Free Height 2700` makes the wall 2700 m high. Undo with `Ctrl+Z` and write the unit: `2700mm` or `2.7m`. In the element panel a number without unit is in the project's units.

Files

Unsaved work is gone after New, Open or Sample project

These commands replace the current project without asking. Save before using them: the asterisk **Unsaved changes** in the top bar shows that there are changes to save.

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Open… in the browser does not show a file dialog

In the browser, drag the `.bimopen` file into the page: the command line says **Drag a .bimopen file into the window to open it**.

bimopen-cli prints its usage and stops

The arguments are not the ones it expects: it ends with code 2 and shows the usage. A command that fails ends with code 1 and a message saying which line failed. See Command line and MCP.

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